APPLICATION DEVELOPMENT BLUEPRINT · WRITTEN FOR AN AI BUILD AGENT

OneDoc

The offline document studio for Android — write, structure, format, scan, dictate, listen. No account, no network, no ads.

From an empty repository to a fully tested, signed, installable APK: a Docs-class block editor, honest file fidelity, on-device AI behind SHA-verified packs, agent methodology, direct-APK QA, and release engineering — with machine-checkable acceptance criteria for every feature.

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OneDoc — Application Development Blueprint

Version 1.0 — the complete build contract for an autonomous AI developer.

OneDoc (`app.onedoc`) is an offline document studio for Android: write, structure, format, scan, dictate, listen, and export — with no account, no network, no ads, and no exceptions. This document is written to be executed, not admired: an AI agent given only this repository and this PDF must be able to produce a fully tested, release-quality APK without asking a single clarifying question.

Every requirement has an ID (F-xxx, UX-xxx, C-n, L-n). Every ID maps to at least one automated test case. Every chapter states not just *what* to build but *how to prove it works*. Verification is not a phase at the end; it is the shape of the whole program.

Reading rule for the agent: you may not write production code for a chapter you have not read. The reading order is in §29.1; the gates that enforce it are in Part X.

PART I

CHARTER, USERS AND NON-NEGOTIABLES

What OneDoc is, the five promises every requirement serves, the constraints that can never be weakened, and what is permanently out of scope. Everything in Parts II–XII is a consequence of this part.

Chapter 1 — Mission, promises and hard constraints

1.1 Mission statement

> **OneDoc gives one person, on one offline phone, a complete document studio — write, structure, format, scan to text, dictate, listen, and share — in one fast, private application that never requires an account, a network, or a subscription.**

The mission sentence is testable, word by word. "One application" means exactly one icon, one installer, one data store. "Offline phone" means the product delivers one hundred percent of its value in airplane mode, from first launch, with no configuration. "Never" means never — not "unless the user accepts a dialog", not "except for crash reporting", not "only on first run".

The reference bar is the writing experience of Google Docs and Microsoft Word: not their cloud features — their *editing* features. A user who writes letters, reports, minutes, essays, or a novel on OneDoc must not feel like they settled for a toy.

1.2 The five promises

Every requirement in this document serves exactly one of five promises. When two requirements conflict, the lower-ranked promise yields.

1. **P1 — Works everywhere, always.** Full functionality in airplane mode, on a low-end phone, on first launch, with no configuration. If a feature cannot honor this, the feature changes, not the promise.
2. **P2 — Private by architecture.** OneDoc never transmits user content because it *contains no transmit path*: no `INTERNET` permission, no WebView, no telemetry SDK, no updater. Privacy is structural, not a policy page.
3. **P3 — Zero-friction writing.** New document to first sentence in under two seconds and two taps. No modal walls, no wizards, no "sign in to continue" — there is nothing to sign in *to*.
4. **P4 — Fast and gentle.** Cold start under 1.2 s on Tier B hardware; typing latency P95 under 50 ms at 10,000 words; jank-free scrolling; battery-safe background behavior; no notifications except the reminders the user created.
5. **P5 — Built to be verified.** Every claim in this document maps to an automated check or a reviewable artifact. The agent proves, it does not assert. "It should work" is not a state; PASS, FAIL and BLOCKED are states.

1.3 Hard constraints

The constraints below are hard. They are repeated or refined in later chapters but never weakened. A task that violates a constraint is invalid, no matter how much work it embeds.

ID	Constraint	Rationale	Verified by
C1	No `INTERNET` and no `ACCESS_NETWORK_STATE` permission in the merged manifest of any shipped APK	P2; the owner's explicit requirement	`apkanalyzer` manifest check as a CI gate (§26.4)
C2	No sign-in, no account, no user identity of any kind	P1, P2	Dependency + code scan: no auth classes, no account APIs (§26.4)

ID	Constraint	Rationale	Verified by
C3	No ads, no ad SDKs, no tracking SDKs, no analytics, no purchase flows	Owner's explicit requirement	Deny-list dependency audit in CI (§26.4); direct-APK QA case TC-SEC-P-03
C4	All user data stays in app-private storage or user-granted SAF locations	P2	Security suite TC-SEC-* (§20.6)
C5	AI models are side-loaded packs verified by SHA-256; the app never downloads anything	P1, P2; AI must work on offline devices	Pack-manager tests TC-AI-PACK-* (§19.6)
C6	Android 8.0 (API 26) minimum; phones and tablets; portrait and landscape	Compatibility promise	Emulator matrix API 26/33/35/36 (§27.2)
C7	Editor laws L1–L10 (§13) are release blockers, exactly like the calculator laws were in the predecessor program	The editor *is* the product	Catalogue TC-EDIT-* + direct-APK QA (Part X)
C8	Max 3 parallel CI runners, ≤ 30 jobs per pipeline run	Runner quota discipline	Workflow configuration (§26.1)
C9	A single codebase ships both the full studio and, if the owner later asks, a lean "writer-only" flavor via Gradle flavor — zero duplicated editor code	Optionality without cost	Flavor build tasks (§26.2); not exercised at v1.0 unless the owner opts in
C10	Every shipped artifact carries provenance: commit SHA, workflow run ID, APK SHA-256	The owner verifies, not trusts	Release evidence bundle (§28.2)

1.4 Permanently out of scope

The following are rejected for the lifetime of this program, and any task proposing them is invalid: cloud sync of user content; accounts or profiles; social features; comments-from-others or real-time co-editing (single-user device product); push notifications from external services; in-app purchases; advertising of any kind; third-party content feeds; WebView-based document rendering; root access; device administration; accessibility-service abuse (OneDoc *serves* accessibility, it never *consumes* it); background location; contact, call-log or SMS access; cryptocurrency or payment features; executing arbitrary macros or scripts inside documents (a deliberate security stance — macro malware has no place here).

This list exists because autonomous agents drift toward "helpful" additions. Scope creep is a defect class (§25.3, category S-CREW) and is triaged as such: a feature nobody asked for is a bug somebody must now maintain.

1.5 Defaults where this document is silent

1. **Platform convention wins** over invention when they conflict: Material 3 behaviors, back-gesture semantics, share-sheet patterns, system back predictiveness.
2. **Simplest testable behavior wins** when two designs both satisfy a requirement.
3. **Battery and RAM win** over feature richness at equal user value.
4. **Determinism wins** over cleverness: same input, same output, same logs.
5. When genuinely ambiguous, the agent chooses the option closest to **Google Docs / Google Keep / Google Files** interaction patterns for the corresponding surface — because the owner's feedback history anchors on those references — then records the

choice as a D-nnn entry in the decision ledger (`docs/plan/decisions.md`).

Chapter 2 — Product rationale

2.1 Why a document studio

Documents are the most universal personal-computing artifact: a shopping list, a lease summary, a school essay, meeting minutes, a resume, the first chapter of a novel. The owner's requirement is a **fresh app** — not a continuation of any prior product — whose core is creating documents of Word/Google-Docs quality on a phone, fully offline.

Phones are where thought happens in stolen moments. Yet mobile writing tools mostly come in two broken flavors: crippled "note apps" that collapse the moment a document grows structure (headings, lists, tables), and office suites that are desktop UIs squeezed onto a screen, complete with sign-in walls and cloud nagging. OneDoc takes the third path: a native Android editor with a real block-and-mark document model (Part V), on-device AI that earns its place (Part VIII), and file handling honest about fidelity (Part VII).

2.2 Why offline-first, on-device AI

Three independent reasons, in priority order:

1. **The owner's requirement.** Work locally, on device, no sign-in, no internet. This is a product-definition fact, not a preference to optimize away.
2. **Trust asymmetry.** An offline-first app has nothing to leak. There is no telemetry pipe to abuse, no server to breach, no terms to erode. The privacy story fits in one sentence: *your words never leave your phone because there is no path out.*
3. **Reliability and latency.** On-device AI in 2026 covers the needed capability set — OCR (scan a page, get text), dictation (speak, see words), summarization and rewriting (small LLM), semantic search (embeddings) — at 40–800 ms latencies with zero availability risk. A feature that needs the network is a feature that dies in airplane mode, on a metro, in a basement classroom: exactly the moments people write.

2.3 Risks and their engineered answers

Risk	Consequence if ignored	Engineered answer
Editor complexity balloons; text-input bugs ship	Lost characters, broken undo — reputation-ending for a writing app	Strict block-and-mark model with property-based tests (§16); editor laws L1–L10 as release blockers (§13); IME composition test suite from day one (§15.3)
Document model is a dead end (no tables, no images)	Product hits a wall at G4 and needs a rewrite	Closed block-kind set designed up front, schema-versioned with migrations (§18.2); tables/images are block kinds v1, not afterthoughts
Format support over-promises ("opens .docx!") then corrupts	User loses trust and data	Declared fidelity levels F0–F3 per format (§18.2); importers refuse with reasons rather than guess; round-trip tests are release blockers
On-device AI drains battery or RAM on cheap phones	The promise P1 breaks exactly where it matters most	Device tiers A/B/C with fallback ladders; packs are opt-in side-loads; assistant controls hide (with reason) when hardware cannot run them (§19.4)
APK bloat toward office-suite gigantism	Slow installs, cheap-phone exclusion	Size budget 25 MB without packs (§26.4); per-PR APK-size check
Autonomous agent drifts, invents UI, invents scope	Incoherent product, unverifiable claims	Screen registry + token contract (Part III); gates with DoD (Part X); decision ledger reviewed at every gate

2.4 Users and the top journeys

OneDoc serves one user per device, and designs for three of them:

- **The correspondent** — letters, applications, filled forms; cares about clean print/export (PDF) and templates.
- **The compiler** — students, analysts, writers; long documents, headings, tables, versions, search across everything.
- **The capturer** — photographs pages and whiteboards, dictates on the move; cares about scan-to-text and dictation accuracy.

The five journeys that shape acceptance (full step-by-step versions with case ids in §27):

J-id	Journey	Success definition
J1	Blank page → formatted letter → PDF export	Under 3 minutes on first launch, no settings touched, PDF paginates cleanly
J2	Photograph 3 printed pages → one document with corrected OCR text	Under 2 minutes; OCR text lands in the editor as editable paragraphs
J3	Dictate meeting minutes → tidy → share as .docx	Speaker's words appear live; tidy-up pass is one tap; export keeps lists/headings
J4	20-document library → find "the paragraph about the warranty"	Full-text hit in <300 ms; semantic hit surfaced with pack installed
J5	Accidental deletion of half a document → recover	Version restore in under 30 s losing at most the last 2 s of typing (Law L3)

If a release candidate fails any journey end-to-end on the emulator matrix, the release does not ship. Journeys are tested on the **distributed APK**, not on source (§24).

2.5 What "done" means for v1.0

OneDoc v1.0 is done when: (a) all gates G0–G8 are closed with evidence (Part X); (b) the release APK passes every P0 case in the test catalogue on the emulator matrix; (c) the defect tracker has no open S1/S2; (d) the owner holds a signed artifact plus an evidence PDF whose every claim traces to a test id. Status without evidence does not exist in this program.

2.6 Reference teardown: what we take, what we reject

The bar is "writing quality of Google Docs / Word on a phone". That claim is only actionable when broken into specific, stolen-or-rejected behaviors. The teardown below is the design diligence behind Parts III and V; "TAKE" entries are requirements in Chapter 4 by reference.

App	TAKE (and where it lands here)	REJECT (and why)
Google Docs	selection toolbar that flips around the selection (UX-19); Enter-continues-list semantics (UX-23); outline/TOC coupling (F-035); live word count in context (F-027); find bar with counts (F-026)	account wall and "connect to network" states (C1/C2); offline as a degraded mode — OneDoc has no other mode; comment threads (single-user product)
Microsoft Word	real page metaphor with margins and break guides (F-037/038); print-fidelity PDF (F-051); table header repeat (§17.1); footnote handling (F-039)	ribbon-density UI (one-handed phones); sign-in-first flow (C2); file-format lock-in via fidelity FUD — OneDoc declares fidelity honestly (§18.2)
Google Keep	instant capture: FAB → editor with zero dialogs (UX-01); checklists as first-class (F-019); calm empty states (UX-36)	notes-not-documents ceiling (no structure, no tables) — the exact gap OneDoc exists to fill; color-as-organization only (UX-41)
Obsidian	plain-file honesty, fast full-text search (F-067), local-first durability mindset (Law L3)	markdown-source-as-truth (OneDoc's source is structured, exports markdown); plugin free-for-all (§1.4: no code execution in documents); link-graph UI complexity
LibreOffice mobile	format breadth ambition (F-054/055)	desktop-file dialogs on touch (F-031); fidelity overpromising without notes — the anti-pattern §18.1 outlaws; heavyweight APK (budget §26.4)

2.7 Why each promise earns its rank

The five promises (§1.2) are ranked for one reason: autonomous construction needs a tiebreaker more than a philosophy. The practical resolutions recorded here are citable in decisions:

- **P1 beats P3.** If a lovely two-tap flow requires a permission pre-prompt or a network check, the flow changes, not the rule (e.g., dictation requests mic at first use, not at launch — UX-32).
- **P2 beats P4.** The security suite runs on every RC even when it costs a day of schedule; there is no "ship first, encrypt later" (§20.3).
- **P4 beats breadth.** When G4 runs late, features drop before budgets do: a fast editor with fewer features is OneDoc; a slow editor with all features is a rejection (§21.4).
- **P5 outranks everything moment-to-moment.** Any work that cannot be evidenced (a "quick hack", an untested fix) is, by definition, not part of this program (§29.5).

2.9 Naming conventions (ids, files, branches, artifacts)

Feature ids `F-nnn`, UX `UX-nnn`, laws `L-n`, constraints `C-n`, cases `TC-<AREA>-<nnn>` (areas per Appendix C), journeys `J<n>`, gates `G<n>`, decisions `D-nnn`, sessions `S-nnn`, defects `OD-nnn`, risks `R-n`, screens `OD-<AREA>-<nnn>`, pack names `<cap>-<lang>`. Code: modules per §10.1; resources prefixed `od_`; a11y ids mirror screen ids (`od_edit_001_toolbar`). Branches: trunk-based short-lived `agent/<task-id>` branches merged to `arena/…` per session; tags `v<X.Y.Z>`; artifacts `OneDoc-v<X.Y.Z>-<code>-<sha7>.apk`. One naming style per thing, everywhere, forever — the traceability tooling greps these patterns, and ambiguity in names is ambiguity in evidence.

PART II

REQUIREMENTS AND PRODUCT SHAPE

The complete feature catalogue F-001…F-090 with priorities and test bindings, the acceptance UX criteria, the navigation graph, and the screen inventory. If a behavior is not listed here or in Part V–VIII, it does not exist in v1.0.

Chapter 3 — Product map and navigation

3.1 The product in one picture

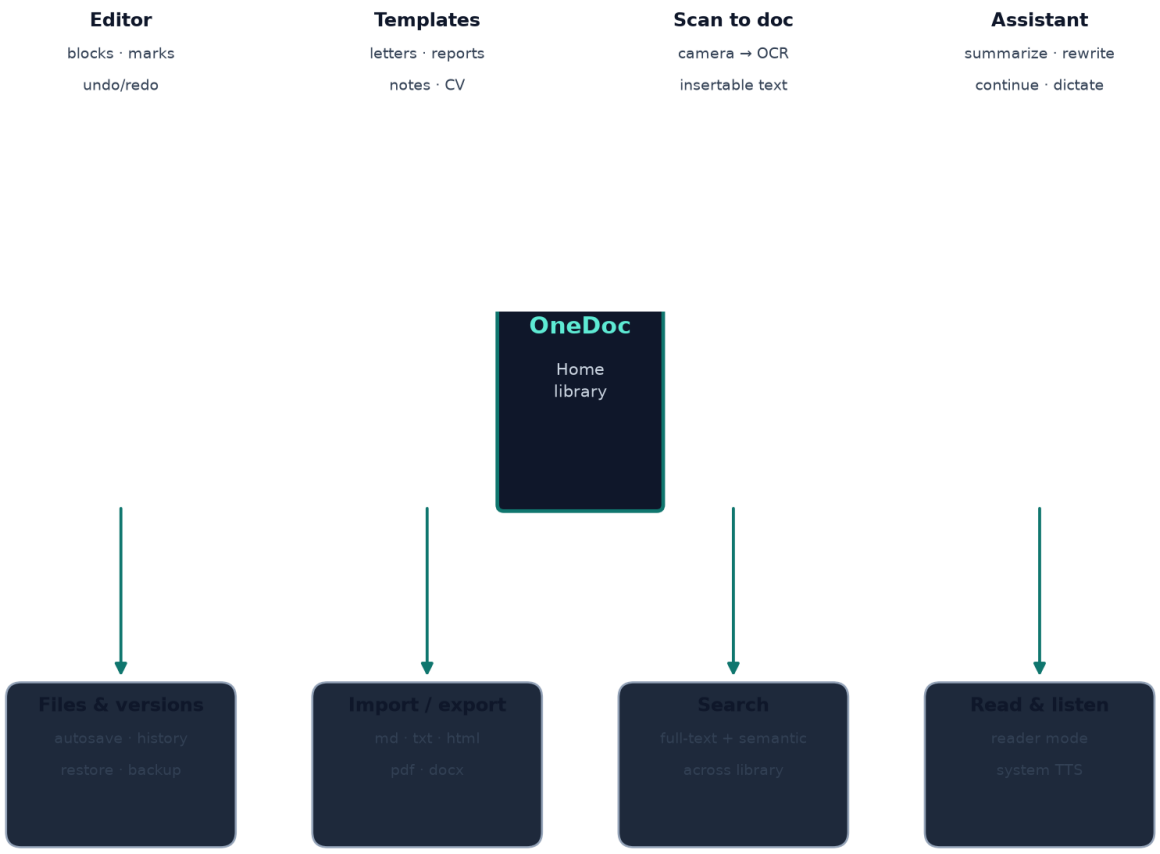


Figure 1. *OneDoc product map: the home library at the center; editor, templates, scan, assistant, files, formats, search and reading around it — one offline process.*

OneDoc has exactly four top-level destinations — **Docs** (home), **Templates**, **Assistant** (pack manager), **Settings** — plus the editor, which opens full-screen over anything. Everything else is a state or a sheet inside those five surfaces. This flatness is deliberate: a writing app must never make the user navigate.

3.2 Navigation graph (normative)

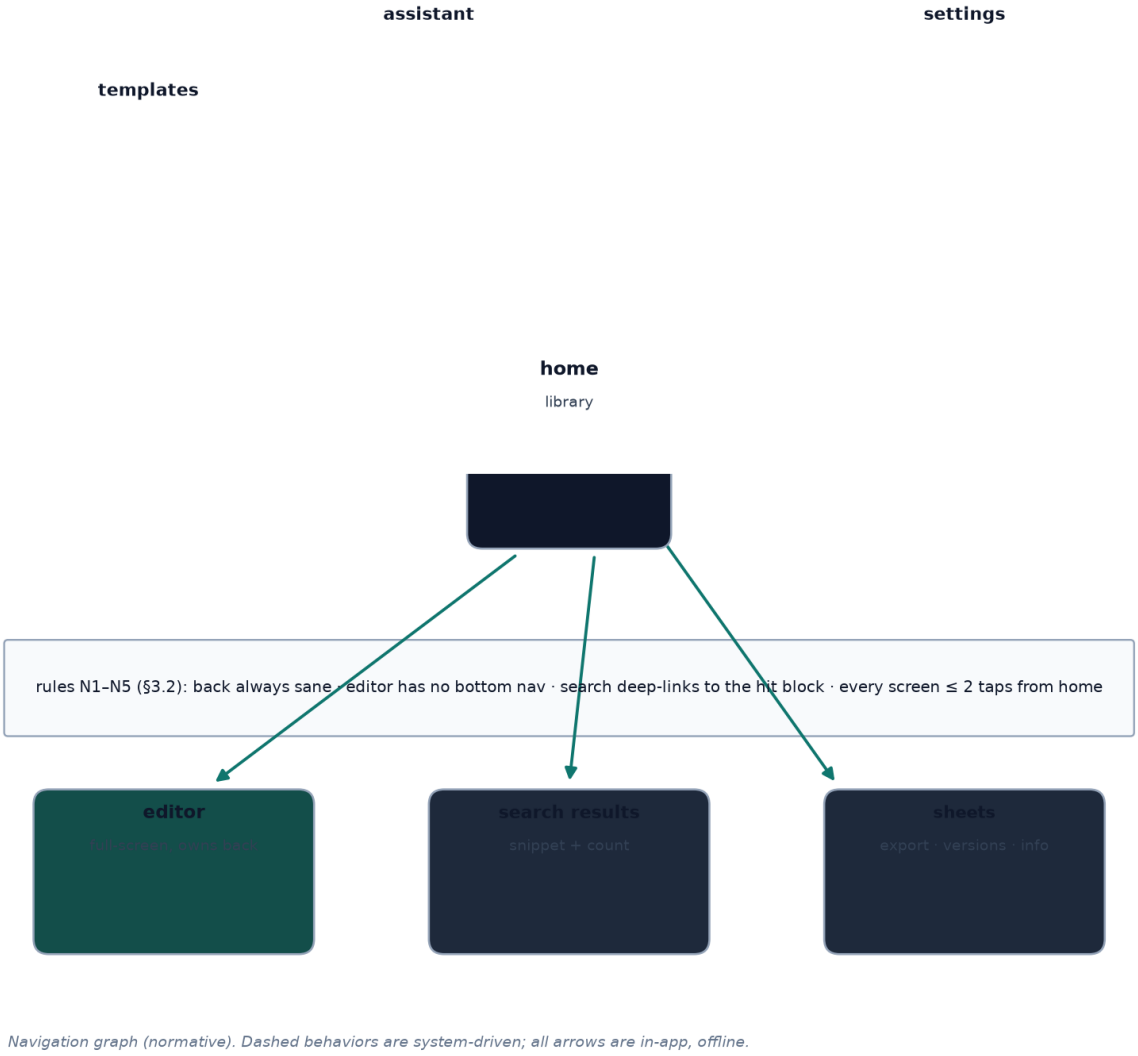
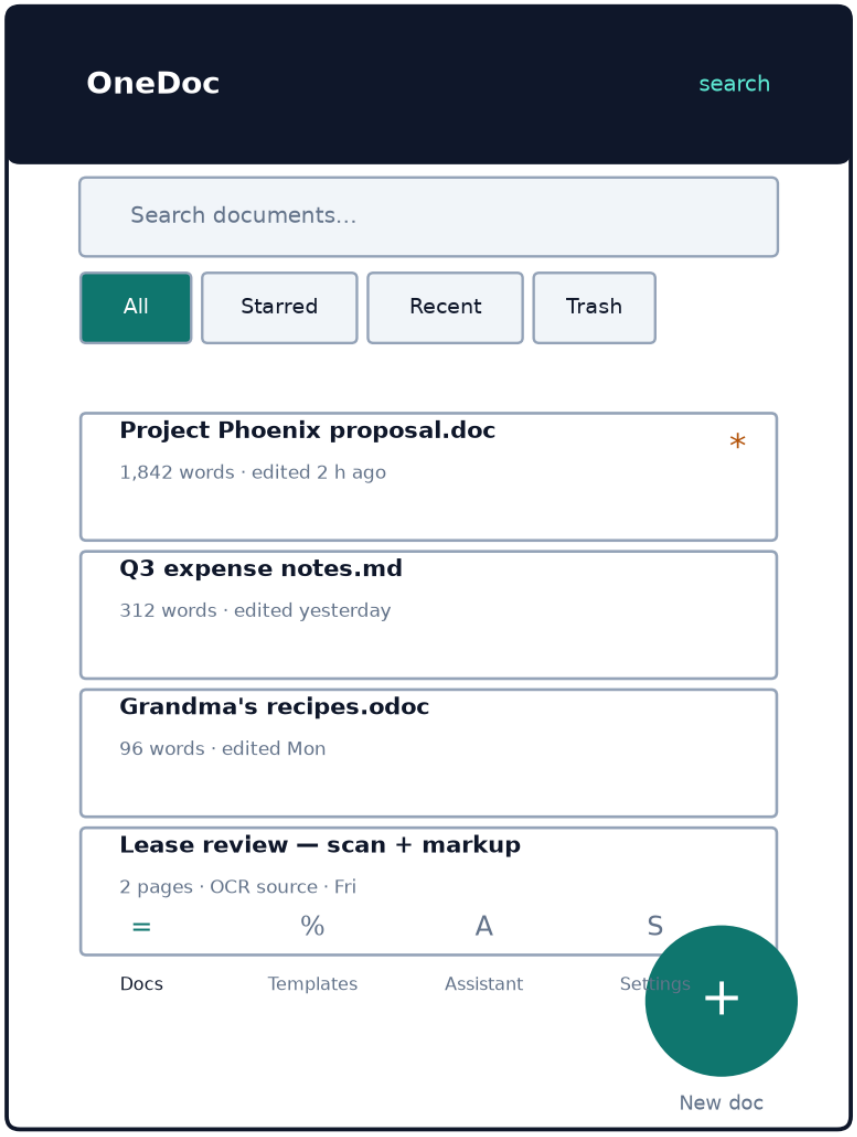


Figure 2. *Normative navigation graph. Solid arrows are user taps; dashed arrows are system-driven (deep links from search results into editor selections).*

Navigation rules:

1. **N1.** Back from the editor always lands on the surface that launched it (home, search result, template picker, assistant preview).
2. **N2.** System back inside the editor: first dismiss selection/toolbar, then dismiss keyboard, then confirm-or-autosave exit (Law L7: exit never loses more than 2 s of typing — in practice zero).
3. **N3.** Every screen is reachable in ≤ 2 taps from home; no screen requires scroll-then-tap-then-scroll.
4. **N4.** The editor has no bottom navigation bar; it owns the whole screen except system insets.
5. **N5.** Search results deep-link into the editor with the hit block scrolled into view and the term highlighted.

3.3 Home screen (library)



Documents library (screen OD-HOME-001): search, filter chips, doc cards, FAB, bottom nav.

Figure 3. *Documents library (OD-HOME-001): app bar with search, filter chips, document cards with title/meta/star, new-document FAB, bottom navigation.*

The library is a single scrollable list of document cards (title, word count, relative edit time, starred pin, format badge for imported files), with filter chips All / Starred / Recent / Trash, a search entry in the app bar, and a FAB that creates a document **immediately** — no naming dialog first (UX-01). Sorting: recent edit descending; starred documents pin to the top section. Trash holds deletions for 30 days with a restore action and a permanent-delete with confirmation. Long-press enters multi-select with actions: star, rename, duplicate, export, delete.

3.4 Screen inventory

The complete normative registry — one row per screen, with Figma frame ids, entry/exit, and states — lives in `docs/onedoc/blueprint/ui-screens.csv` (generated with this blueprint; §7.5 explains its contract). Summary:

Area	Screens	Registry prefix
Home / library	OD-HOME-001 library, OD-HOME-002 multi-select, OD-HOME-003 trash	OD-HOME
Editor	OD-EDIT-001 compose, OD-EDIT-002 selection state, OD-EDIT-003 insert sheet, OD-EDIT-004 find bar	OD-EDIT
Templates	OD-TPL-001 gallery, OD-TPL-002 preview	OD-TPL
Scan	OD-SCAN-001 camera, OD-SCAN-002 review/crop, OD-SCAN-003 OCR confirm	OD-SCAN
Assistant	OD-AI-001 pack manager, OD-AI-002 action sheet, OD-AI-003 generation preview	OD-AI
Files/data	OD-FILE-001 versions, OD-FILE-002 import, OD-FILE-003 export sheet	OD-FILE
Settings	OD-SET-001 root, OD-SET-002 editor, OD-SET-003 appearance, OD-SET-004 about/diagnostics	OD-SET

Chapter 4 — Feature catalogue (F-001 … F-090)

4.1 How to read the catalogue

Priority: **P0** = release-blocking (the product is broken without it), **P1** = must ship in v1.0, **P2** = committed stretch, may slip to v1.1 only with a logged decision. Every feature names its binding: the case ids (Chapter 5 and the catalogue in Appendix A) that prove it. The traceability tool (`tools/onedoc/traceability.py`) enforces the invariant **every F has ≥ 1 test binding**; `--check --strict` additionally fails if any P0 lacks a binding.

4.2 Documents and the library (F-001 … F-012)

ID	Feature	Priority	Bound to
F-001	Create blank document in one tap; opens editor with untitled doc; first character focuses body	P0	TC-LIB-001
F-002	Rename inline from card long-press → rename sheet; title mirrors first heading when untitled	P0	TC-LIB-002
F-003	Autosave: every edit persisted via journal + debounced model save (Law L3)	P0	TC-STATE-001
F-004	Star/unstar; starred section pinned on All chip	P1	TC-LIB-003
F-005	Duplicate document (full copy incl. assets, new id)	P1	TC-LIB-004
F-006	Move to trash with undo snackbar (7 s window)	P0	TC-LIB-005
F-007	Trash retention 30 days with per-item remaining time; restore; permanent delete with confirm	P1	TC-LIB-006
F-008	Sort: recent (default), title A–Z, word count; persisted per device	P2	TC-LIB-007
F-009	Empty states with one-action CTA (new doc / open templates)	P1	TC-LIB-008
F-010	Document info sheet: created/modified, words, chars, reading time, size, version count	P2	TC-LIB-009
F-011	Multi-select actions: star, duplicate, export, delete	P1	TC-LIB-010
F-012	Cold start to first interaction < 1.2 s on Tier B emulator with 200 docs	P0	TC-PERF-001

4.3 Editor core (F-013 … F-030)

ID	Feature	Priority	Bound to
F-013	Block text editing with per-block IME composition (no mid-composite breaks); Gboard, Samsung Keyboard, Swiftkey behaviors in scope	P0	TC-EDIT-001, TC-EDIT-IME-01
F-014	Enter splits block; Backspace at start merges with previous; both preserve marks sanely	P0	TC-EDIT-002
F-015	Undo/redo: single global stack, coalesced typing bursts (600 ms), grouped composite ops (paste, AI insert)	P0	TC-EDIT-003, TC-PROP-P01
F-016	Selection: handles, floating toolbar (bold/italic/underline/strike, H1/H2/H3, list, quote, align, link)	P0	TC-EDIT-004
F-017	Marks: bold, italic, underline, strike, code, highlight, link — non-overlapping per type (Law L2)	P0	TC-EDIT-005
F-018	Heading levels H1–H3 applied per block; outline drives TOC and document map	P0	TC-EDIT-006
F-019	Lists: bulleted, numbered (auto-continue, renumber on edit), checklists (tap box to check)	P0	TC-EDIT-007
F-020	Blockquote and code block; code block uses monospace, disables inline marks	P1	TC-EDIT-008
F-021	Alignment left/center/right/justify per block (non-heading)	P1	TC-EDIT-009
F-022	Horizontal divider insert; deleting content around it behaves as one block	P2	TC-EDIT-010
F-023	Paste: plain text, HTML subset (marks, lists, simple tables), image; clipboard sanitized	P0	TC-EDIT-011
F-024	Cut/copy of selection including multi-block range; paste round-trips exactly	P0	TC-EDIT-012
F-025	Find & replace: case toggle, whole-word, regex off by default; replace-all creates a pre-edit version first	P0	TC-EDIT-013
F-026	Find bar shows match count and navigates hits; highlight non-destructive	P0	TC-EDIT-014
F-027	Word/char count live in editor status chip; tap opens info sheet	P1	TC-EDIT-015
F-028	Hardware keyboard: Ctrl+B/I/U, Ctrl+Z/Y, Ctrl+F, arrow-key block traversal, Tab in lists	P1	TC-EDIT-016
F-029	Typing latency P95 < 50 ms at 10k words on Tier B emulator; frames never drop > 2 during IME burst	P0	TC-PERF-002
F-030	Orientation change and process death restore full cursor, scroll and selection state	P0	TC-STATE-002

4.4 Structure and objects (F-031 … F-044)

ID	Feature	Priority	Bound to
F-031	Tables: insert N×M, add/remove row/col, header row toggle, marks inside cells, swipe-safe cell editing	P0	TC-TBL-001
F-032	Table column width presets (equal / content); long text wraps; no horizontal doc scrolling from tables	P0	TC-TBL-002
F-033	Images: insert from gallery/camera; stored as assets; caption block; tap to view full-screen	P0	TC-IMG-001
F-034	Image resize presets (S/M/L/full-width) and alt-text editor (a11y, §9)	P1	TC-IMG-002
F-035	Table of contents block: generated from H1–H3, tap scrolls to target, refreshes on change	P1	TC-STRUCT-001
F-036	Document map sheet: outline list, current position indicator	P2	TC-STRUCT-002
F-037	Page setup: paper size (A4/Letter), margins (normal/narrow/wide), affecting PDF/print pagination	P1	TC-STRUCT-003
F-038	Page-break guides in compose mode (dashed) matching PDF pagination within ±1 line	P1	TC-STRUCT-004
F-039	Footnote blocks rendered at page bottom in PDF, inline sup markers in editor	P2	TC-STRUCT-005
F-040	Cover block: title/subtitle/date styling for reports and letters	P2	TC-STRUCT-006
F-041	Drag-to-reorder blocks via handle (with a11y equivalent: move up/down actions)	P1	TC-STRUCT-007
F-042	Duplicate/delete block from handle menu	P1	TC-STRUCT-008
F-043	Code block language label (stored, used for future highlighting; no execution ever)	P2	TC-STRUCT-009
F-044	Asset manager: list of images in doc, remove unused (with confirm)	P2	TC-IMG-003

4.5 Templates (F-045 … F-050)

ID	Feature	Priority	Bound to
F-045	12 built-in templates: blank, letter, meeting notes, essay/report, invoice, resume, weekly plan, project brief, minutes, recipe, reading list, thesis outline	P0	TC-TPL-001
F-046	Template preview (read-only render) before create	P1	TC-TPL-002
F-047	Creating from template instantiates a **copy**; template never mutates	P0	TC-TPL-003
F-048	Placeholders (Title, Date, [Your name]) tapped to fill; date placeholder auto-fills	P1	TC-TPL-004
F-049	User templates: "Save as template" from any document	P1	TC-TPL-005
F-050	Template gallery states: empty (never — built-ins), user section with delete	P2	TC-TPL-006

4.6 Files, formats, sharing (F-051 … F-060)

ID	Feature	Priority	Bound to
F-051	Export PDF via Android Print framework; fonts embedded; selectable text; page setup honored	P0	TC-IO-001
F-052	Export .odoc (native zip: document.json + assets + meta) — lossless F3	P0	TC-IO-002
F-053	Export Markdown F2; plain text F1; HTML F2 — per fidelity table §18.2	P0	TC-IO-003
F-054	Export .docx (OOXML subset writer): headings, paragraphs, marks, lists, tables, images	P1	TC-IO-004
F-055	Import .odoc F3, .md/.txt F2/F1, .html F2, .docx best-effort F1 with conversion note	P0	TC-IO-005
F-056	Import refuses unsupported formats with a specific reason (never a silent empty doc)	P0	TC-IO-006
F-057	Share sheet integration: export-then-share; never a raw private-path share	P0	TC-IO-007
F-058	Open-in from file managers for .odoc/.md/.txt/.html (intent filters, no internet mime traps)	P1	TC-IO-008
F-059	SAF: user-readable locations honored for import; export via create-document dialog	P0	TC-IO-009
F-060	Auto-backup: weekly zip of library (excluding trash) to user-chosen SAF folder; off by default	P2	TC-IO-010

4.7 Versions and state (F-061 … F-066)

ID	Feature	Priority	Bound to
F-061	Versions screen: auto snapshots (every 5 min of activity or 50 ops), manual "Save version", pre-edit snapshots before replace-all / AI insert / import-merge	P0	TC-STATE-003
F-062	Version preview read-only; Restore creates a pre-edit snapshot of current, then swaps — history never destroyed	P0	TC-STATE-004
F-063	Journal durability: kill -9 within 2 s of last keystroke loses nothing (Law L3)	P0	TC-STATE-005
F-064	Storage honest accounting: doc size + assets shown; unused assets cleaned on compaction	P2	TC-STATE-006
F-065	Migration: schema version bump path tested v1→vN with synthetic old documents	P0	TC-STATE-007
F-066	Multi-window/split-screen: editor remains single-instance, saves on focus loss	P1	TC-STATE-008

4.8 Search (F-067 … F-070)

ID	Feature	Priority	Bound to
F-067	Library search: title + full text (FTS5), < 300 ms on 200 docs / 1 MB text, results with snippet + hit count	P0	TC-SRCH-001
F-068	Result tap opens editor at hit block, term highlighted (N5)	P0	TC-SRCH-002
F-069	Search within document via find bar (F-025) shares the same highlight system	P1	TC-SRCH-003
F-070	Semantic search (MiniLM embeddings) when pack installed: "meaning" matches merged below FTS hits, labeled	P2	TC-SRCH-004

4.9 On-device AI (F-071 … F-082)

ID	Feature	Priority	Bound to
F-071	Scan-to-doc: camera capture → perspective crop → OCR (DBNet-lite + CRNN) → editable paragraphs with confidence coloring	P0	TC-SCAN-001, TC-AI-001
F-072	Multi-page scan: pages queue into one document, reorderable before insert	P1	TC-SCAN-002
F-073	OCR on-device only; pack required (18 MB); without pack the camera flow ends at "save image asset" with reason	P0	TC-AI-002
F-074	Dictation: whisper.cpp int8 streaming or system SpeechRecognizer per ladder §19.4; live partial results into a preview strip; Insert on stop	P0	TC-AI-003
F-075	Assistant actions on selection or whole doc: Summarize, Rewrite (polish/formal/casual), Continue writing, Fix grammar — each produces a **preview card**, insertion is explicit	P0	TC-AI-004
F-076	Assistant runtime: gemma-2B 4-bit, Tier A devices only; Tier B/C get extractive fallbacks per ladder	P0	TC-AI-005
F-077	Generation is streamed, cancellable, and never edits the document silently; a pre-edit version is snapshotted before any insert	P0	TC-AI-006
F-078	Pack manager: list packs (OCR/ASR/LLM/embeddings), size, license, install via SAF from side-loaded .odpack, SHA-256 verify at import and load; delete frees space	P0	TC-AI-PACK-01
F-079	Corrupt/wrong-SHA pack: refused with reason; capability degrades per ladder; no crash, no retry loop	P0	TC-AI-PACK-02
F-080	Read aloud: system TTS play/pause/stop from reader mode; sentence highlight follows speech	P1	TC-AI-007
F-081	Read aloud honors document range (from cursor); voice/rate from system settings only	P2	TC-AI-008
F-082	Every assistant control hides (not grays) when unavailable, with a one-line reason on long-press (tier/pack/ram)	P0	TC-AI-009

4.10 Reading, a11y, settings, platform (F-083 … F-090)

ID	Feature	Priority	Bound to
F-083	Reader mode: distraction-free render (no toolbars), font size control, TTS entry	P1	TC-READ-001
F-084	TalkBack: full traversal of editor by block; marks announced; actions (move block, format) via local context menu	P0	TC-A11Y-001
F-085	Contrast ≥ 4.5:1 body text in all 8 themes incl. high-contrast black/white	P0	TC-A11Y-002, DQ-4
F-086	Dynamic type: editor text scales to 200% system setting without clipping or truncation	P0	TC-A11Y-003
F-087	Themes: light, dark, sepia, paper, high-contrast ×2, plus accent choices; "match system" default	P1	TC-SET-001
F-088	Editor settings: default font (Inter/system/serif/mono), size 12–24 sp, autosave interval display, spell-check toggle (system service)	P1	TC-SET-002
F-089	About + diagnostics screen: version, commit, storage usage, pack states, self-test button (runs integrity suite locally)	P1	TC-SET-003
F-090	Locales: en (default), hi, mr string coverage; RTL layout mirroring correct	P2	TC-SET-004

Count check: 90 features — P0 = 48, P1 = 28, P2 = 14 (machine-verified by `traceability.py --check --strict`, which additionally enforces a binding on every P0). The machine-readable list with bindings lives in Appendix A and is parsed by `tools/onedoc/traceability.py`.

4.11 Design notes behind the catalogue (library and editor)

Library (F-001…012). The library's only job is to get out of the way. Cards show title, meta, star — never previews: preview tiles cost scroll speed and leak content in public places (a deliberate privacy choice, consistent with §20). Rename is a sheet, not an inline text war with card layout. Trash uses a retention period rather than "are you sure" chains: recovery exists, so deletion can stay one tap. Sorting is persisted per device and never asked about — default recency matches how returning users actually work. Cold start (F-012) is a feature, not an implementation detail: the library renders from the Room index in the first frame, which is why documents store their index fields (title, wordCount, updatedAt) denormalized in Room despite document.json being authoritative (§11.2).

Editor core (F-013…030). The editor's design center is the **single op stream**: everything a user does becomes ops in one log, and every surface (undo, versions, search-highlight, autosave) is a consumer of that log. This is why undo covers AI inserts and replace-all with zero special cases (they are composite op groups, §17.10). Selection is a first-class state machine (§15.7) because selection bugs are the most user-visible editor bugs. Marks get one chapter-level rule set (§14.5) instead of per-feature ad hoc behavior — the "why does bold disappear here" bug class is engineered out, not patched. The typing budget (F-029) is stated in the catalogue rather than a perf chapter because it is a *requirement*: if a feature breaks it, the feature waits.

Paste (F-023). Paste is the highest-risk input path (untrusted content, arbitrary structure). Design: sanitize-to-model in one step (no intermediate HTML DOM retained), image sizes capped, HTML subset per §18.2 with everything else dropped *and counted* (the conversion note can say "paste dropped 3 unsupported constructs").

4.12 Design notes: structure, templates, files

Tables (F-031/032). Tables on touch fail when they try to be spreadsheet grids: free column drag, per-cell borders, horizontal scroll. OneDoc's preset-width, wrap-always, popup-cell-editor design trades spreadsheet feel for calm document behavior — and that trade is a feature, stated here so it survives implementation pressure.

Images (F-033/034). Presets beat free-resize on touch (fat-finger precision); captions are blocks (they export everywhere); alt text is prompted at insert because retrofitting alt text never happens. EXIF stripping is privacy, not housekeeping (§20.4).

Templates (F-045…050). Templates are data, deliberately dumb: no logic, no computed fields beyond date fill, no code execution (§1.4). Twelve built-ins cover the correspondent and compiler personas (§2.4); "save as template" is how the capturer persona freezes recurring structures (meeting notes with fixed headings).

Files (F-051…060). The fidelity ladder (§18.2) is the honesty contract: refuse > guess. PDF via the platform Print framework is a dependency decision: it guarantees printer-parity pagination and selectable text for free, at the cost of style control — acceptable for v1.0, revisit only if a golden diff demands it.

Versions (F-061…066). Version kinds are semantic (auto/manual/pre_edit) because restore UX depends on knowing *why* a snapshot exists ("Before change" reads as safety; "Auto" reads as background). Restore never destroys (F-062) — the version graph is append-only, which also makes the restore implementation trivially crash-safe (§12.3 rules apply to versions as to everything).

4.13 Design notes: search, AI, settings

Search (F-067…070). Search is split by job: "find the doc" (library FTS, title-weighted) vs "find the place" (in-doc find bar) vs "find the meaning" (semantic, pack-gated, labeled "related" and ranked *below* lexical hits — meaning never overrides exact words). Deep-linking with highlight (N5) closes the loop that search opens; the shared highlight system (F-069) exists so the two surfaces cannot drift.

AI (F-071…082). Three principles produce every AI requirement: (1) *explicit over ambient* — preview cards, no ghost text (Law L10); (2) *ladder over error* — every unavailability is a graceful descent with a reason (§19.4), never a dead button; (3) *packs over bundling* — the APK stays small and the owner chooses what fits the device (C5). The LLM is Tier A-only in v1.0 because extractive fallbacks cover 80% of summarize/rewrite value at 0% of the RAM cost; the ladder makes that trade invisible except in the method label (UX-47), which is exactly as honest as P5 demands.

A11y & settings (F-083…090). TalkBack-complete is scoped to the letter flow (UX-37) because "accessibility everywhere" without a verified flow produces checkbox compliance; the letter flow exercises every mechanism (creation, selection, marks, export). Settings are few and instant-apply; themes are token-remaps (§6.3) not skins; diagnostics is a power tool with an off switch (it shows nothing unless opened).

3.5 Feature coverage sanity (by the numbers)

90 features over 10 areas — library 12, editor core 18, structure 14, templates 6, files 10, state 6, search 4, AI 12, read/a11y 4, settings/platform 4. P0 mass: editor core (13) + AI (9) + files (7) + state (4) = 33 of 48 P0s — the product *is* the editor and its durability; everything else earns its P1/P2 by the same testability bar. The count table regenerates with the catalogue (`traceability.py --stats`) so drift shows in CI, not at review.

2.8 What OneDoc tells users it is NOT (positioning honesty)

The welcome card and About say it plainly: OneDoc is not a cloud suite (nothing syncs, nothing signs in); not a PDF editor (PDFs export, they don't import — Scan is the path for paper); not a spreadsheet (tables are document furniture, not cells-with-formulas); not an AI assistant with a personality (the assistant edits text, it doesn't chat). Saying what a tool is not, in the product, prevents the one-star reviews that come from mismatched expectations — and keeps every promise in this document falsifiable.

PART II

REQUIREMENTS AND PRODUCT SHAPE

Continuation: the fifty acceptance criteria that define "easy UX". These are pass/fail behaviors, not vibes; each is bound to test cases in the catalogue.

Chapter 5 — UX acceptance criteria (UX-01 … UX-50)

5.1 How these were chosen

Each criterion below comes from a real friction pattern observed in mobile writing tools: the tap that should not exist, the dialog that interrupts thought, the state that silently vanishes. They are written as **verifiable behaviors** with exact counts (taps, seconds, milliseconds) because an autonomous builder needs numbers, not adjectives. Every criterion has at least one case in the catalogue; UX ids appear in test names as `UX-<nn>`.

5.2 Speed and directness

ID	Criterion	Verified by
UX-01	New-document FAB opens the editor directly: no template dialog, no naming dialog; a doc is created on first keystroke, not on tap	TC-UX-001
UX-02	Blank page to first typed character: ≤ 2 taps and ≤ 2 s on Tier B (cold)	TC-UX-001, TC-PERF-001
UX-03	Export PDF: ≤ 3 taps from editor (menu → Export PDF → share/save)	TC-UX-002
UX-04	Import a file from any app: share-to-OneDoc or open-in lands in import preview ≤ 2 taps	TC-UX-003
UX-05	Find & replace opens with the last find term preselected; Esc/back dismisses without leaving residue	TC-UX-004
UX-06	Undo of an AI insert is a single undo (grouped op), returning the document byte-exact to the pre-insert state	TC-UX-005, TC-EDIT-003
UX-07	Version restore completes in ≤ 30 s for a 50-page document including preview	TC-STATE-004
UX-08	Search-as-you-type: results start appearing after ≤ 2 characters, never a submit button	TC-SRCH-001
UX-09	Scan-to-doc: from camera open to editable text for one page ≤ 60 s on Tier B including OCR	TC-SCAN-001
UX-10	Dictation: first partial text on screen ≤ 1.5 s after speech starts	TC-AI-003

5.3 Never lose a keystroke

ID	Criterion	Verified by
UX-11	App killed (adb shell am kill / process death) mid-typing: zero loss on relaunch (journal replay)	TC-STATE-005
UX-12	Device rotation or split-screen mid-sentence: cursor, selection, IME composition state survive	TC-STATE-002
UX-13	Back from editor autosaves silently; no "Save?" dialogs anywhere in OneDoc, ever	TC-STATE-001
UX-14	Trash undo snackbar gives 7 s; after that, trash retains the doc for 30 days	TC-LIB-005/006
UX-15	A failed import (corrupt file) never creates an empty document and never destroys the clipboard	TC-IO-006
UX-16	Replace-all requires one confirm and creates a pre-edit version; a second tap of undo returns the original	TC-EDIT-013, TC-STATE-004
UX-17	Storage full during autosave: editor keeps running in memory, warning chip appears, retry is automatic — no data loss dialog storm	TC-STATE-009
UX-18	Export failure (SAF denied mid-write) leaves no partial file and shows the specific reason	TC-IO-011

5.4 The editor feels native

ID	Criterion	Verified by
UX-19	Selection toolbar never covers the selection; flips above/below by keyboard and screen edges	TC-UX-006
UX-20	Handles are ≥ 48 dp effective touch targets; toolbar buttons ≥ 44 dp (DQ-2)	TC-UX-006, DQ-2
UX-21	Long-press word → select shows system-standard behavior; drag handles work with stylus and mouse	TC-EDIT-004
UX-22	IME composition (pinyin, glide, Hindi transliteration): no flicker, no duplicate commit, no mark loss mid-composition	TC-EDIT-IME-01
UX-23	Enter on a list item continues the list; Enter on empty list item exits the list (Docs behavior)	TC-EDIT-007
UX-24	Heading style switch preserves inline marks; list↔paragraph switch preserves marks	TC-EDIT-006
UX-25	Scroll position anchors to the block being edited during IME keyboard show/hide (no jump)	TC-UX-007
UX-26	Typing never reflows the whole document: only the edited block and pagination guide update (measured via frame stats)	TC-PERF-002
UX-27	Two-finger scroll during IME composition does not corrupt text (composition is cancelled cleanly)	TC-EDIT-IME-02
UX-28	Hardware-keyboard shortcuts match §5.2 of the Android platform conventions; Ctrl+Z steps are identical to soft undo	TC-EDIT-016

5.5 Trust and calm

ID	Criterion	Verified by
UX-29	Every AI output appears in a preview card with an explicit Insert/Replace/Discard; nothing enters the doc implicitly	TC-AI-006
UX-30	Assistant unavailable states say *why* in one line ("gemma needs 8 GB RAM — this device has 4 GB")	TC-AI-009
UX-31	No notification is ever shown unless the user created a reminder or starts an export that needs progress	TC-SEC-P-02
UX-32	First launch shows a single welcome card (3 lines) with one button "Start writing"; no permission pre-prompt for camera/mic until the feature is used	TC-UX-008
UX-33	Deleting a document with versions warns about version count; deleting from trash is the only permanent path, double-confirm	TC-LIB-006
UX-34	Diagnostics screen shows only on-device facts; the "self-test" runs offline and takes ≤ 20 s	TC-SET-003
UX-35	All destructive actions have haptic confirm (tick) and are undoable or double-confirmed — never both silent and permanent	TC-UX-009
UX-36	Empty states teach one action, one line, one button; never a wall of tips	TC-LIB-008

5.6 Inclusive by default

ID	Criterion	Verified by
UX-37	TalkBack can create, edit, format and export a complete letter without sighted help	TC-A11Y-001
UX-38	Every image requires an alt-text prompt at insert (can be deferred, never silently skipped)	TC-IMG-001
UX-39	200% font scaling leaves every control usable; no text clipped by containers (DQ-5)	TC-A11Y-003
UX-40	Themes: dark theme is a first-class citizen; high-contrast themes pass 7:1 for body text	TC-SET-001, DQ-4
UX-41	Color is never the only carrier of meaning (OCR confidence uses color + underline style)	TC-AI-001, DQ-4
UX-42	One-handed reach: primary actions (FAB, toolbar, assistant) live in the bottom 40% of the screen	TC-UX-010
UX-43	RTL mirroring: layout, toolbar order and handle behavior mirror in RTL locales	TC-SET-004
UX-44	Reduced-motion setting removes non-essential animation	TC-UX-011

5.7 Honest machine behavior

ID	Criterion	Verified by
UX-45	Importers state the fidelity level achieved ("Imported as HTML — F2: lists and marks kept; tracked changes dropped")	TC-IO-005
UX-46	OCR low-confidence words are visually marked and tappable to fix; export never silently ships guessed words in a scan-sourced doc	TC-AI-001
UX-47	Summaries and rewrites state their method ("on-device gemma-2B" / "extractive fallback") in the preview card	TC-AI-004/005
UX-48	Pack install shows SHA-256 and asks for explicit confirm; verification failure shows both SHAs	TC-AI-PACK-01/02
UX-49	Word count matches the export (PDF text extraction within ±1%)	TC-IO-001
UX-50	The app never shows marketing, ratings prompts, "upgrade" nudges, or any network-dependent content — verifiable by string scan + traffic probe	TC-SEC-P-03

Count: UX-01…UX-50. The catalogue's ENV/A11Y/DIAG rows re-verify the platform-level ones; the direct-APK QA pipeline (Part X) runs the full UX set on every release candidate.

5.8 Why fifty criteria, in seven groups

The groups are ordered by the cost of getting them wrong. Speed (5.2) and durability (5.3) come first because they are the two ways a writing app loses users forever: wasted seconds and lost words. Native-feel (5.4) is where competitors leak trust daily — every toolbar-over-selection bug reads as "not a real app". Trust (5.5) is the AI contract: preview-explicit insertion is what makes on-device AI feel like a tool rather than an autonomous co-author. Inclusivity (5.6) is scope, not charity: TalkBack-complete flows find real bugs (traversal exposes states eyes skip). Honest machines (5.7) exist because fidelity lies and silent OCR guesses are the two fastest ways to corrupt someone's document without ever crashing.

Counts per group: speed 10 · durability 8 · native feel 10 · trust 8 · inclusivity 8 · honesty 6. The distribution is deliberate: no group may dominate, and any new criterion joins the group whose *failure* it prevents, which keeps the set from drifting into feature-list territory (S-CREW, §25.3).

5.9 Per-criterion rationale (why each exists, compressed)

UX-01/02 exist because dialog-gated creation is the #1 abandonment cause in note apps; the keystroke-not-tap doc creation makes "new document" free. UX-03/04 measure export/import reachability because file features hidden three menus deep don't get used. UX-05/06 are undo-dignity rules (AI must feel reversible or users stop trusting it). UX-07/08 put numbers on the two async surfaces (restore, search) so "slow" has a definition. UX-09/10 bound the AI capture paths — scan and dictate live or die by latency, not accuracy alone. UX-11…18 are the durability eight: every silent-loss scenario from the predecessor program's defect history became a criterion. UX-19…28 codify native-feel details competitors get wrong daily. UX-29…36 define the AI trust contract and calm-UI rules. UX-37…44 are inclusion as scope (each is testable, none is aspirational). UX-45…50 make the machine honest in writing — fidelity notes, confidence marks, method labels, hash transparency, count parity, and the absolute absence of marketing.

PART III

DESIGN LANGUAGE, FIGMA WORKFLOW AND SCREENS

The token system, typography, themes, motion; the Figma file contract and the screen registry the agent must obey; wireframes for the editor; accessibility annotations and the design-QA gate that blocks bad UI.

Chapter 6 — Design language

6.1 The contract in one line

OneDoc looks like a calm piece of paper with great tools. The design system optimizes for long-form reading and writing: high legibility, low chrome, generous white space, and controls that appear when needed and disappear when not. Nothing in the UI blinks, bounces, or begs.

6.2 Tokens (single source of truth)

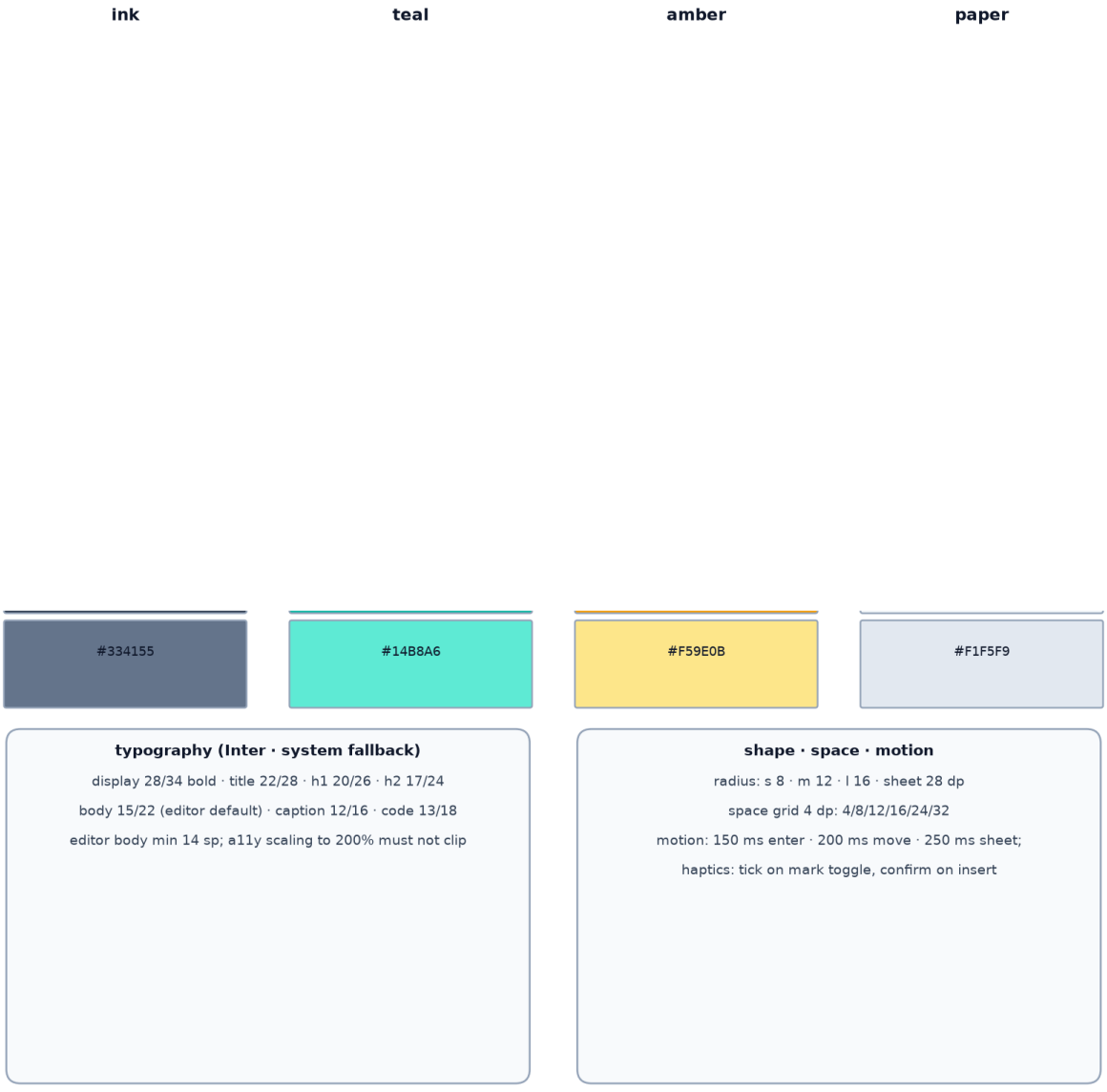


Figure 4. *OneDoc tokens: four ramps (ink, teal, amber, paper), typography scale, shape and motion values.*

All colors, type sizes, radii, spacing and durations are declared in `design/figma/tokens.json` and consumed two ways: the Figma library publishes them as styles/variables, and the Android build generates `OdTokens` from the same file (one-way, repo → Figma). Hand-picking a color in code or in Figma is a design-QA failure (DQ-1).

Token group	Values
`color.ink.*`	#0F172A · #1E293B · #334155 · #64748B
`color.teal.*`	#134E4A · #0F766E (accent) · #14B8A6 · #5EEAD4
`color.amber.*`	#92400E · #D97706 · #F59E0B · #FDE68A (highlights, warnings)
`color.paper.*`	#FFFFFF · #F8FAFC · #F1F5F9 · #E2E8F0 (surfaces)
`font.*`	display 28/34 · title 22/28 · h1 20/26 · h2 17/24 · body 15/22 · small 12/16 · code 13/18 (Inter, system fallback; editor body min 14 sp)
`radius.*`	s 8 · m 12 · l 16 · sheet 28 dp
`space.*`	4 dp grid: 4 · 8 · 12 · 16 · 24 · 32
`motion.*`	enter 150 ms · move 200 ms · sheet 250 ms; easing standard (decelerate 0.6)

6.3 The eight themes

Light · Dark · Sepia · Paper (warm white, low blue) · High-contrast white · High-contrast black, each with a teal or amber accent choice — 8 effective combinations. "Match system" is the default. Themes restyle *surfaces and text only*; the teal accent and semantic colors keep their relative luminance contract so contrast guarantees hold per theme (DQ-4 verifies per theme, not just in light).

6.4 Typography rules

1. Editor body text defaults to 15 sp Inter (system serif and mono selectable in settings); minimum 14 sp anywhere.
2. Heading scale is fixed relative to body (H1 1.33×, H2 1.13×, H3 1.0× semibold) so user font-size scaling flows through everything.
3. Line length in compose mode targets 60–75 characters at default width on phones; margins adapt on tablets to keep the measure.
4. Dynamic type to 200% must not clip: every container either grows or scrolls; `fixedSize()` is a banned modifier in editor UI (DQ-5).

6.5 Iconography, imagery, empty states

Icons: Material Symbols (outlined), 24 dp, 2 dp stroke optical weight, single color from `ink`. No illustrations in tool surfaces; at most one restrained spot illustration per empty state, drawn from the paper metaphor. Empty states (UX-36): one line, one action.

6.6 Motion and haptics

Motion exists to explain continuity, never to decorate: sheet enter 250 ms, toolbar appear 150 ms fade+8 dp slide, block drag 200 ms settle. `Animator` durations scale by 0.5× under battery saver and vanish under reduced-motion (UX-44). Haptics: tick on mark toggle, double-tick on insert (AI/asset), single confirm on destructive press — calibrated via `HapticFeedbackConstants`, no vibration SDK.

Chapter 7 — Figma workflow and the screen registry

7.1 Why Figma is in this document

The owner required the blueprint to specify how Figma links to development and how the file is used. The working model is **repo-first**: the repository is the source of truth for tokens; Figma is the mirror where the owner reviews and edits screens visually; changes flow back as token/registry pull requests. The agent never invents UI, and never waits on design: if a screen has a registry row, it is buildable; if it doesn't, the task is BLOCKED until a row exists.

7.2 The Figma file structure

File: `OneDoc Design System` (single file, two pages — "□ Foundations" and "□ Screens"), mirroring the registry exactly:

Figma page	Contents	Mirrors
Foundations	Color styles per token (`od/ink/900`…), text styles per token (`od/body/15`), effect styles, grid styles (4 dp), component set: buttons, chips, cards, toolbar, sheet, list item, FAB, app bar	`design/figma/tokens.json`
Screens	One frame per registry row, named `OD-<AREA>-<nnn> <name>`, organized in sections by area; every state in the registry's `states` column exists as a named variant or a marked frame	`docs/onedoc/blueprint/ui-screens.csv`

Frame naming is the join key: `OD-EDIT-002` in Figma ⇄ `OD-EDIT-002` in the registry ⇄ `odEdit002` screen id in code. Redlines: spacing/type specs are attached as Figma dev-mode annotations; the agent reads them via the registry's exported CSV mirror rather than screen-scraping Figma.

7.3 The workflow (one-way sync, review loop)



Figure 5. *Figma workflow: tokens.json is the source; the Figma library and screens mirror it; owner edits return as reviewed token/registry PRs; the design-QA gate blocks merges.*

1. **Agent** generates/updates tokens → runs the export script → commits both the JSON and the regenerated CSV mirror.
2. **Owner** (optional, no account needed by the agent) edits in Figma; at sync time the owner (or the agent with a read-only token, if the owner supplies one) exports the frame-name + style list and diffs it against the registry.
3. **Deltas** land as PRs titled `design: <frame-or-token>`; the design-QA checklist (§9.2) must pass before merge.
4. **The agent implements screens only from registry rows**; a frame without a row, or a row without a frame, fails DQ-3.

7.4 How the agent uses the file (no account required)

- **Default path:** the agent uses `ui-screens.csv` (screen id, frame, entry, exit, states, redline summary) as the buildable spec — it is committed to the repo and complete for v1.0.
- **Token path:** `design/figma/tokens.json` is authoritative for values; the CSV carries no values, only names.
- **Optional owner token:** if the owner provides a Figma personal access token in repo settings, the agent may run a nightly "registry drift" check (list frames, compare to CSV). Without a token, this check is skipped and drift is caught at the design-QA gate by the owner's review. The agent never requires Figma access to make progress — the CSV

is sufficient.

- **Redlines:** for every interactive component, the CSV `redline` column names the Figma component and the key measurements (heights, insets, states). This is what the agent implements against.

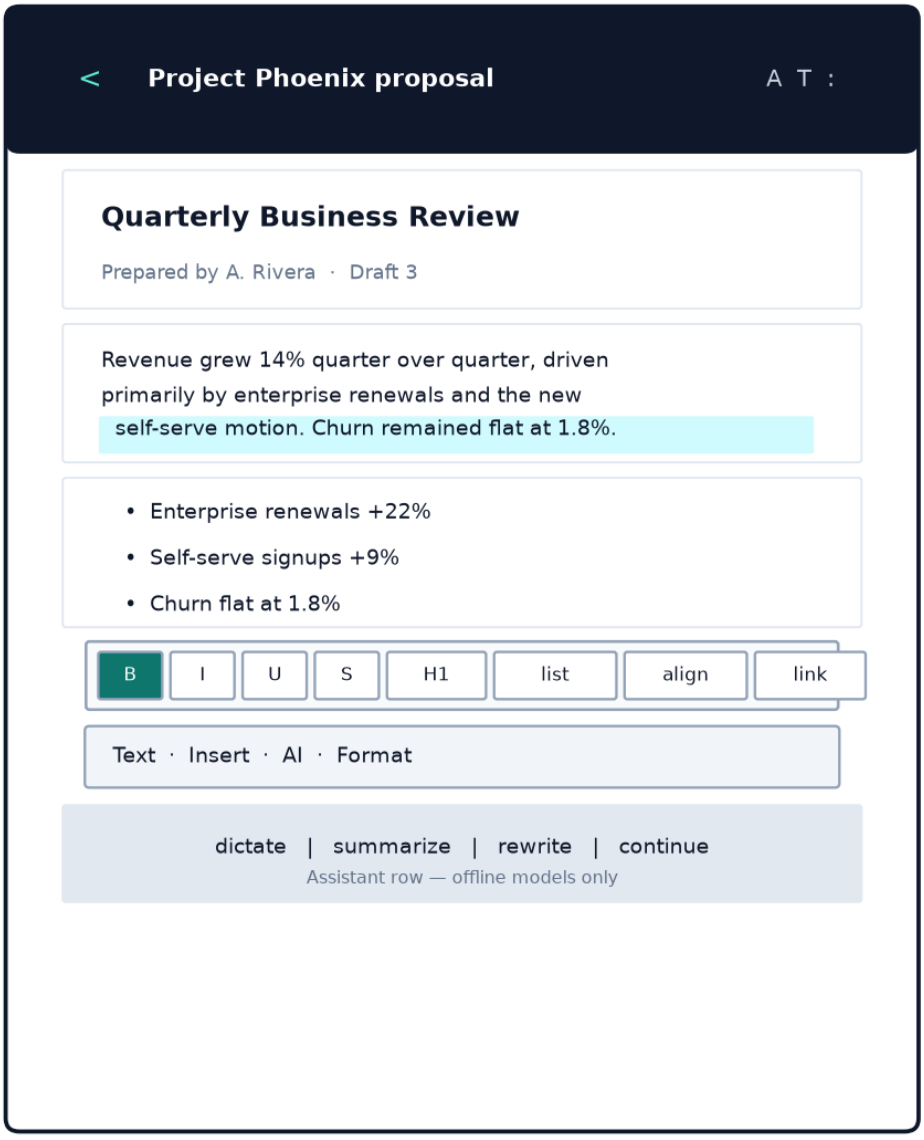
7.5 The screen registry contract (normative)

`docs/onedoc/blueprint/ui-screens.csv` columns: `screen_id, name, figma_frame, area, entry, exit, states, redline, test_ids`. Rules:

1. Every screen in §3.4 has exactly one row; frame ids match Figma exactly.
2. `states` lists all reachable states (loading/empty/error/content + feature states); the implementation must render each one; the catalogue tests the P0 ones.
3. A PR that adds UI without a registry row is rejected by the design-QA gate (DQ-3).
4. The registry is regenerated from the blueprint appendix by `tools/onedoc/traceability.py --emit-registry`, so PDF, CSV and Figma names can never drift silently.

Chapter 8 — Screen wireframes (editor)

8.1 The editor surface



Editor (OD-EDIT-001): page canvas, floating selection toolbar, tab row, assistant row.

Figure 6. *Editor (OD-EDIT-001/002): app bar with doc title and undo/redo; page canvas; floating selection toolbar; category tab row; assistant row with offline-only actions.*

The editor is a page metaphor, not a notes-list metaphor: text sits on a white page with margins and (in compose mode) dashed page-break guides (F-038). Chrome inventory is fixed: app bar (title, undo/redo, overflow), selection toolbar (appears on selection, §9 redlines), format tab row (Text · Insert · AI · Format) shown when the cursor is active and the keyboard is up, assistant row (only when relevant packs/tiers allow). Everything else — versions, export, find — lives in sheets so the page stays clean.

8.2 Behavior highlights tied to wireframe regions

Region	Rules			
App bar	Title shows doc title (editable via rename sheet); back is N1/N2; overflow holds Export, Find, Versions, Info, Settings			
Page canvas	16 dp side margins on phones; blocks are the touch units; tap empty page bottom = new paragraph at end			
Selection toolbar	Layout: B I U S	H1 H2 H3	list quote	align link; horizontal scroll beyond 8 actions; never covers selection (UX-19)
Tab row	Categories not commands: tapping Insert opens the insert sheet (image, table, divider, TOC, link); tapping AI opens the assistant sheet with availability reasons			
Assistant row	Quick actions for the four assistant verbs when a selection or whole-doc context exists; hidden entirely when no AI capability is active (F-082)			

Chapter 9 — Accessibility annotations and the design-QA gate

9.1 Accessibility is design, not a patch

Every screen ships with a traversal order matching the visual order; every interactive element has a role, a label and (where useful) a state description; every custom gesture has a menu equivalent. The editor exposes blocks as traversable units (F-084) with per-block actions: format, move up/down, delete, duplicate. Marks are announced as "bold, italic" prefixes on the affected text. Tables expose row/column headers. The full a11y acceptance script is catalogue section A11Y; the design-QA gate enforces the design half.

9.2 Design-QA gate (DQ-1 … DQ-8)

The gate runs as a PR checklist plus emulator checks; any DQ failure blocks merge of UI changes:

ID	Check	Method
DQ-1	No hardcoded colors/sizes in UI code: everything resolves to a token	Lint rule + code scan
DQ-2	Touch targets ≥ 48 dp (44 dp inside toolbars)	Lint + Compose/View inspection on emulator
DQ-3	Screen exists in registry with matching Figma frame; states match `states` column	Registry check in CI
DQ-4	Contrast: body 4.5:1, large 3:1, high-contrast themes 7:1 — all 8 themes	Theme snapshot tests
DQ-5	200% font scale: no clipped/truncated text on all screens	Emulator `font_scale=2.0` screenshot diff
DQ-6	Traversal order matches visual order; all actions reachable via TalkBack menu	Instrumented a11y audit
DQ-7	Motion respects reduced-motion and battery saver	Emulator setting toggle test
DQ-8	Empty/error states present per registry (no blank screens)	State-walk script per screen

Gate evidence lands in `reports/gates/` alongside the build gates; DQ failures are defects (severity S3 unless they hide a functional break).

8.3 Scan surfaces (OD-SCAN-001 … 003)

Screen	Purpose	States (registry `states` column)	Key redlines
OD-SCAN-001 camera	viewfinder with auto-capture on focus; page edges hint overlay	idle · focusing · captured-flash · no-camera-permission (rung-6 explanation + settings link)	shutter ≥ 56 dp bottom-center; permission state is a full-screen card, never a toast
OD-SCAN-002 review/crop	per-page perspective crop (4 corner handles, rotate ±90°), thumbnail queue with drag reorder (F-072)	single-page · queue · dragging · empty-queue	handles ≥ 48 dp; reorder uses long-press + drag with a11y "move page up/down" actions
OD-SCAN-003 OCR confirm	recognized text beside page image, confidence marks (UX-41), per-word fix field, insert CTA	processing (progress + cancel) · ready · no-pack (ends at save-image with reason, F-073) · low-yield (< 20 chars: warn, offer retry)	confidence uses color + underline style; fix field ≥ 44 dp; insert button bottom-anchored, always visible

Rules: capture resolution fixed at 2048 px long edge; processing happens on the AI executor with a cancel that reaches the native runtime within one frame; the review screen is the only place OCR errors are fixable before insert — after insert, confidence marks are gone (document is plain text + optional source-image asset).

8.4 Assistant surfaces (OD-AI-001 … 003)

Screen	Purpose	States	Key redlines
OD-AI-001 pack manager	list packs (OCR/ASR/LLM/embed), size, license, tier requirement, verify state; install via SAF; delete	no-packs (education card) · installed · verifying · failed (both SHAs, UX-48) · deleting-confirm	rows ≥ 64 dp; install is two explicit confirms (pick + SHA shown); no silent background verification loops
OD-AI-002 action sheet	the four verbs with availability per selection/doc context; unavailable rows hidden (F-082) with long-press reason	ready · partial (some verbs hidden) · all-hidden (sheet offers pack manager link)	sheet max-height 60%; verbs listed in fixed order Summarize/Rewrite/Continue/Fix
OD-AI-003 generation preview	streamed preview card, method label (UX-47), Cancel, Insert/Replace/Discard	generating (streaming, cancel enabled) · done · cancelled (no residue) · error (ladder descent notice, rung-4 offer)	card never covers cursor line; Insert = primary button; Esc/back = Discard with confirm only if text non-empty

Rules: only one generation at a time (second request cancels the first with a logged D-note); the preview card renders into a fixed-height scroll area (never grows into the toolbar); insertion applies as one grouped op — undo once returns exactly the pre-insert state (UX-06).

8.5 Files surfaces (OD-FILE-001 … 003)

Screen	Purpose	States	Key redlines
OD-FILE-001 versions	snapshot list (kind, time, size), preview (read-only), restore	empty (education line) · list · previewing · restoring (progress ≤ 30 s, UX-07) · restore-confirm (warns current will be snapshotted first)	kind chips color+label (UX-41); restore primary action disabled during preview render
OD-FILE-002 import	file pick (SAF) → sniff → preview (fidelity note, UX-45) → open	picking · sniffing · preview-ready · refused (reason, F-056) · too-large warn (> 50 MB)	refusal card names format + missing feature; never imports without preview step
OD-FILE-003 export sheet	format list with fidelity badges (§18.2), filename edit, SAF create, progress	ready · writing · done (share/open actions) · failed (reason, UX-18)	fidelity badges same visual system as conversion notes; filename sanitized (no path chars)

8.6 Settings surfaces (OD-SET-001 … 004)

Settings root lists four groups — Appearance (themes §6.3, accent), Editor (font family/size, autosave interval display, spell-check toggle), Files (backup toggle + SAF folder, storage accounting F-064), About/diagnostics (F-089). Rules: every setting persists instantly (no save button anywhere); diagnostics self-test (UX-34) runs the integrity suite (manifest self-check, pack SHAs, journal integrity, storage stats) and finishes with a copyable local report — nothing leaves the device; "About" shows version, commit, and the repo URL as plain text (never link-fetched, §20.2).

6.7 Component inventory (the closed set of UI parts)

Every screen composes from exactly these parts — a new custom widget requires a D-nnn entry and this table's update:

Component	Variants	States	Size contract
Button	filled · tonal · text	enabled · pressed · disabled (rare)	height 40/48 dp; label 15 sp medium
Icon button	bare · contained	same + selected (toggled)	48×48 dp target, 24 dp glyph
Chip (filter/assist)	filter · assist · input	unselected · selected · error	height 32 dp; label 14 sp
Card (doc card)	standard · compact	default · pressed · selected (multi-select)	min-height 72 dp; radius l
List item	one-line · two-line	default · selected · drag-anchor	height 56/72 dp; leading 40 dp slot
App bar	standard · editor	default · scrolled (elevation)	height 56 dp (+insets)
Toolbar (selection)	horizontal scroll	visible · flipped · expanded (overflow sheet)	height 44 dp; buttons ≥ 44 dp
Sheet (modal)	standard · full (editor overflow)	hidden · half · expanded · dragging	radius sheet; grab handle 32×4 dp
FAB	single	default · extended (New doc)	56 dp; bottom-right, above nav
Snackbar	action	visible · queued (one at a time)	bottom; 7 s default (trash undo contract)
Text field	single · multiline	default · focused · error (helper text)	min-height 48 dp; helper 12 sp
Tab row (editor categories)	fixed 4	selected · unselected · disabled	height 48 dp; indicator 3 dp
Dialog	confirm · destructive	—	used only where sheets are too heavy (permanent delete, pack delete)
Preview card (AI)	streaming · done	generating · done · cancelled · error	max 60% height; method label line mandatory

Do/don't (applies to every part): do respect token values only; don't add color outside ramps; do announce state changes to accessibility; don't animate anything over 250 ms; do support text scale 200%; don't introduce a second shadow elevation scale.

9.3 Focus-order specification method

Focus order is declared per screen in the registry (`a11y` column) and asserted by the audit (DQ-6). Declaration format: ordered node ids; Wildcard rows (list items) declare the pattern ("cards by recency"). The assertion walks with TalkBack semantics (not just `focusOrder`) on API 33/36: traversal must match visual order in both LTR and RTL, and no actionable node may be unreachable (orphan check). Custom views must register their virtual structure (`virtualViewCount`, child descriptions) — a custom view without a node contract fails review (§17.12).

9.4 Live-region and announcement rules

Announcements (polite): insert/restore done, export finished, generation completed, verification results. Never announced: typing echoes (TalkBack reads composition itself), autosave (silent by law), decorative state. Assertive is reserved for data-safety events: storage-full chip, journal-repair notice. The announce list is enumerated in strings tests like errors are (no ad hoc `announceForAccessibility` in feature code — key-based only).

9.5 The design-review checklist (PR template, UI path)

1. Registry row exists; states match; frame named (DQ-3).
2. Tokens only — no literals (DQ-1); sizes per component table (§6.7).
3. Targets ≥ 48/44 dp (DQ-2); primary actions in bottom 40% (UX-42).
4. Contrast per theme verified in snapshot tests (DQ-4); 200% walk clean (DQ-5).
5. Focus order declared + walk clean (DQ-6, W).
6. Motion within budget; reduced-motion honored (DQ-7).
7. Empty/error/loading states implemented per registry (DQ-8).
8. Strings via keys; new strings en+hi+mr (P).
9. Announcements per §9.4 only.
10. Screenshots attached at default and 200% scale.

6.8 Theme contrast engineering (how the promise holds per theme)

Body text pairs surface/ink with a maintained luminance-distance contract: each theme defines `surface`, `surfaceAlt`, `ink`, `inkMuted` such that ink/surface ≥ 7:1, inkMuted/surface ≥ 4.5:1, and accent-on-surface ≥ 3:1 at glyph weights used. The snapshot gate (DQ-4) computes WCAG ratios per theme per pair from the *token values* (not screenshots of text) and fails CI on regression — theme additions must extend the table, not eyeball colors. High-contrast themes push to ≥ 10:1/7:1 and forbid inkMuted on action labels (all controls full-ink). Sepia/paper trade blue-light for warmth while keeping the same numeric contract (§21.4 budgets unchanged).

8.7 Motion spec per transition (the complete list)

Transition	Spec
sheet enter/exit	250 ms, slide+fade, radius sheet; exit 200 ms
toolbar appear/dismiss	150 ms fade + 8 dp vertical slide
block drag settle	200 ms; neighbors reflow with 180 ms stagger ≤ 3 items
page guide appear	150 ms fade (compose mode only)
FAB → editor	shared-axis nothing: instant editor open (speed > flourish, UX-02)
search results stream	100 ms item fade-in, top-staggered
theme change	250 ms color cross-fade (activity recreation avoided)
AI streaming text	no animation (text appears as generated; card frame animates once)
reduced motion / battery saver	all of the above → 0 ms except opacity 100 ms

PART IV

ARCHITECTURE AND DATA

Module boundaries, the Android stack, the Room schema, the journal and version system that make keystroke loss impossible, and the concurrency rules that keep the main thread boring.

Chapter 10 — Architecture

10.1 Shape



Figure 7. *OneDoc architecture: feature modules over six core libraries over the Android platform, with the verification pyramid underneath.*

Single process, Kotlin-first, View-based UI (no experimental-UI dependency risk on API 26), modularized by Gradle:

Module	Owns	Never contains
`app`	Navigation, DI graph, themes, home, settings	Business logic
`feature-editor`	Editor screen, toolbars, sheets	Persistence, IO
`feature-home`, `feature-templates`, `feature-scan`, `feature-assistant`, `feature-search`	Their screens	Each other
`core-model`	Document, Block, Mark, Op — pure Kotlin, no Android imports	Anything impure

Module	Owns	Never contains
`core-editor`	Operation engine, undo, IME bridge, layout metrics	UI classes
`core-storage`	Room DB, .odoc container, journal, versions	UI, formats
`core-io`	md/txt/html/pdf/docx readers+writers	DB access
`core-ai`	Pack manager, OCR, ASR, LLM runtime, embeddings	UI, DB writes
`core-search`	FTS5 indexing, vector index, ranking	Editor

Dependency rule: arrows point downward only (`feature-*` → `core-*` → platform). `core-model` depends on nothing. Tests exist at every level; the highest proof is the direct-APK pipeline (Part X).

10.2 Threading model (§10.2 normative)

1. The **main thread** does UI and nothing else; any work over 4 ms measured at P95 moves off it.
2. One **storage executor** serializes all Room writes; the journal append is synchronous from the editor's perspective (write-ahead, §12.3) but executes on this executor with a bounded queue of 1 pending op batch.
3. **IO executors** (2 threads) run imports/exports/backups; cancellation is cooperative and checked at every file boundary.
4. **AI executors** (1 per active capability, max 2 concurrent) run OCR/ASR/LLM; priority: user-initiated action > background indexing.
5. Main-thread disk IO is a defect (category S-PERF, §25.3); strict-mode is enabled in debug builds and CI instruments it in tests.

10.3 Dependency and build hygiene

Version catalog for all deps; no dynamic versions (`+`); no annotation-processor beyond Room compiler and Hilt; R8 full-mode with keep rules owned by this repo; dependency deny-list enforced in CI (C1–C3): any artifact under `com.google.ads`, `com.facebook.ads`, `com.applovin`, `io.branch`, `com.mixpanel`, `com.amplitude`, `com.bugsnag`, `io.sentry`, or any `*analytics*` coordinate fails the build. New dependencies require a D-nnn ledger entry with an offline/no-telemetry justification.

Chapter 11 — Data layer

11.1 Entity model



Room schema v3. Full text in FTS5; embeddings only for docs where AI pack is installed (graceful absence).

Figure 8. *Room schema v3: documents, versions, assets, FTS index, vectors, meta.*

Table	Purpose	Notable columns
`documents`	One row per document	`id` (text, `d_` + 8 hex), `title`, `updatedAt`, `wordCount`, `starred`, `deletedAt?`, `schemaVersion`
`versions`	Snapshot blobs (zip of document.json + asset manifest)	`docId`, `createdAt`, `kind` (auto/manual/pre_edit), `sizeBytes`
`assets`	Images etc. stored app-private	`docId`, `kind`, `path`, `sha256`
`docs_fts`	FTS5 over title+plain text	external-content mapped to a `doc_blocks` shadow table (block-level hits for §3.2 N5 deep-links)
`vectors`	MiniLM embeddings per block	`docId`, `blockId`, `embedding` (f32 blob), `model`
`meta`	Schema version, settings, pack registry	key-value

11.2 The document payload

Authoritative content is a single `document.json` per document (schema in §14.2), stored zipped in app-private files, with the Room row as the index — never the other way. The database can be rebuilt from files + journal (the reverse is not true), which is what makes backup/export honest and crash recovery provable.

11.3 Journal

Every committed operation batch appends a framed, CRC-checked record to `<docId>.journal` (length-prefixed, fsync'd batch) **before** the debounced model save. On open, if the model file is newer than the last checkpoint, the journal replays forward. Compaction: at 512 records or 2 MB, fold into a new checkpoint file. The journal is the reason UX-11 can promise zero loss under `kill -9`.

11.4 Schema migrations

Room schema exports versioned JSON; each migration ships with (a) an up-test using a synthetic v(n-1) database from `core-storage/src/test/resources/`, (b) a dry-run against the seed corpus (§23.2), (c) a rollback *plan* (restore from journal + backup) documented in the migration notes. TC-STATE-007 runs the full v1→vN chain on emulators.

11.5 Encryption and privacy

Android Keystore-generated key (AES-256-GCM) encrypts the document store directory on devices with a lock-screen credential (StrongBox when available); on credential-less devices files stay app-private (sandbox) with a clear statement in About. The backup zip (F-060) is created in app cache and handed to SAF; it is encrypted with a user-passphrase-derived key (PBKDF2, 200k iterations) — the passphrase prompt is the only dialog in OneDoc that exists for security, and it explains itself in one line.

Chapter 12 — State, versions and the autosave pipeline

12.1 One document's life

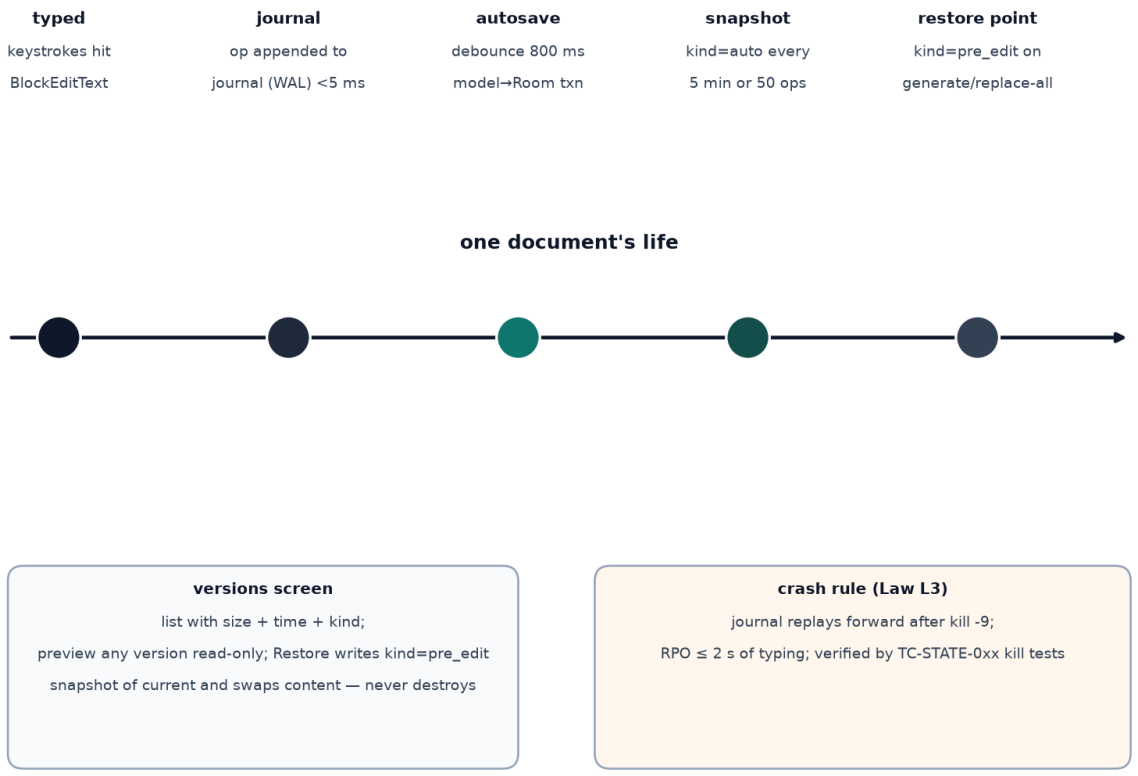


Figure 9. *From keystroke to journal to debounced save to snapshots to restore points.*

12.2 Snapshot policy (F-061 normative)

Kind	When	Kept
`auto`	5 min of activity or 50 committed op batches, whichever first	last 20 per doc
`manual`	"Save version" action	unlimited until user deletes
`pre_edit`	Before replace-all, AI insert/replace-all, import-merge, version restore	last 5 per doc, never auto-pruned

12.3 Crash-consistency rules (Law L3 machinery)

1. Journal record written+fsynced **before** ack of the edit batch in memory.
2. Model save (checkpoint + Room index update) is debounced 800 ms and always follows a checkpoint fold or journal replay boundary — never mid-batch.
3. On any open: `replay(journal) → validate CRC → repair or quarantine`. A quarantined journal is preserved byte-exact and the document opens at its last good state with a diagnostic note; data is never silently dropped.
4. `TC-STATE-005` proves the chain with instrumented kills at randomized points (100 trials); UX-11 is its user-facing promise.

12.4 Multi-window and process death

The editor is `singleTask`; a second launch intent brings the existing instance forward. Configuration changes (rotation, split, resize) never recreate the editor activity: state lives in a retained editor core; the view layer re-binds (TC-STATE-002). Process death restores via journal replay + saved scroll/cursor snapshot (F-030).

10.4 The DI graph (Hilt)

Components: `SingletonComponent` holds storage (DB, container store, journal service), pack registry, executors, token store (themes); `ActivityRetainedComponent` holds editor cores (one per open document, max 2); `ViewModelComponent` holds screen state. Rules: no `@Singleton` may hold a `Context` beyond `ApplicationContext`; no DI scope may reference UI classes; the editor core is created by an explicit `EditorCoreController` (not inject-on-demand) so its lifecycle (open → active → idle-saved → closed) is testable without UI.

10.5 Error-handling strategy (one taxonomy app-wide)

Every failure is a sealed `OdError` with three user-facing verbs: **retryable** (transient IO — auto-retry ≤ 3 with backoff, then calm banner), **actionable** (needs the user — specific reason + exactly one action button), **reportable** (engine invariant violated — quarantine + diagnostics entry + one-line apology, never a stack trace dialog). User strings live in the strings catalog; no error text is composed ad hoc in feature code (lint-enforced). The mapping error→verb is table-driven in `core-model/errors` so tests can enumerate it exhaustively (every error kind has a string and a verb — asserted within the TC-SET-004 strings tests).

11.6 On-disk layout (normative)

```
files/
  docs/<docId>/document.json(.zip)     # authoritative content
  docs/<docId>/journal                 # CRC-framed op log (§11.3)
  docs/<docId>/checkpoints/<n>.json.zip
  assets/<sha256>.<ext>                # content-addressed images etc.
  packs/<packName>/                    # verified side-loaded packs (§19.6)
  tmp/export|import|backup/            # lifecycle-managed temp (§20.3)
databases/onedoc.db (+ -wal, -shm)     # Room index (rebuildable, §11.2)
```

Invariant: `databases/` can be deleted and rebuilt from `files/docs` (the integrity self-test F-089 verifies this rebuildability on a sample document). Nothing under `files/` is world-readable; `tmp/` is cleared on start and after each operation completes or cancels.

12.6 Clocks and time

Documents store UTC epoch millis; display uses relative formatting ("2 h ago", locale-aware absolute after 7 days). Snapshot timing uses `SystemClock.elapsedRealtime` deltas (immune to wall-clock jumps); exported files embed wall time in metadata only (excluded from golden comparison, Law L6). Time-zone and clock-change torture is part of TC-STATE-006 (set clock forward/back across a save → no snapshot storm, no ordering flip).

11.7 Indexes and query plans

Room indices: `documents(deletedAt, updatedAt)` (library list), `documents(starred)` partial where starred=1, `versions(docId, createdAt)`, `assets(docId)`, `vectors(docId, blockId)` unique. FTS5 external-content with prefix index; `doc_blocks` shadow carries `(docId, blockIdx, plainText)`. The library list query is a single indexed scan + counts (EXPLAIN QUERY PLAN checked in a unit test — "no full scans on the hot path" is a test, not a hope). Search query plan: FTS MATCH → join to documents for title/updated → snippet materialization limited to top 50 hits. Any schema change re-runs the plan tests (schema checklist §29.5).

12.7 Compaction and vacuum runbook

Journal compaction (§11.3) triggers at 512 records/2 MB during open or idle (never during active composition — Law L5 respects the queue). Compaction = fold journal onto checkpoint → fsync new checkpoint → atomically swap pointer → delete old files. Vacuum (Room) runs at app update detection only (schema version change), wrapped in a pre-edit-style backup of the db file (cheap, local). Both procedures are idempotent and crash-safe: power loss mid-compaction leaves either old or new state, verified by TC-STATE-005's kill trials hitting the compaction window deliberately (1 of every 10 trials targets it).

10.6 The dependency ledger (allowed per module, with weight budgets)

Module	Allowed (summary)	Compiled weight budget
core-model	kotlin-stdlib only	< 100 KB
core-editor	+ androidx.core-ktx, text/ICU via platform	< 300 KB
core-storage	+ Room runtime, security-crypto-free (JCE)	< 2 MB
core-io	+ none (platform XML/zip/print)	< 500 KB
core-ai	+ onnxruntime OR llama.cpp binaries (pack-gated)	< 8 MB total both
core-search	+ Room (shared)	< 200 KB
feature-* / app	+ material, hilt, camerax (scan), recyclerview	< 10 MB combined
qa-driver	+ espresso, uiautomator, truth	test-only

Dex budget total ≤ 25 MB APK (§26.4); a dependency exceeding its budget needs a D-nnn with an alternative analysis. No module may add: image loaders, network stacks, JSON mappers beyond org.json platform, DI beyond Hilt, or any Kotlin multiplatform library (support burden).

12.8 The save pipeline state machine

`EDITING` →(batch committed)→ `JOURNAL_PENDING` →(fsync ack)→ `EDITING`; `EDITING` →(800 ms idle)→ `CHECKPOINTING` →(fold done)→ `SAVING_INDEX` →(Room txn ack)→ `IDLE_SAVED`; any state →(kill)→ recovery replays per §12.3. Invariants: exactly one `CHECKPOINTING` at a time per doc; `JOURNAL_PENDING` holds a bounded single pending batch (older batches are already fsynced); `SAVING_INDEX` failure retries with backoff and never blocks further editing (journal already has the truth — the index catches up). The machine is a sealed class with exhaustive transitions tested (TC-STATE-001 harness).

11.8 Corruption scenarios and drills (each has a test)

Scenario	Drill	Expected
truncated journal (power loss mid-write)	TC-STATE-005 kill window variant	last good frame replayed; partial frame discarded by CRC; no user loss
bit-rot in checkpoint zip	harness flips a byte	CRC/zip failure → rebuild from journal; self-test flags doc
corrupted document.json (invalid model)	harness mutates kind/offsets	validator refuses; doc opens read-only from last good checkpoint with repair banner
deleted asset file (image gone)	harness removes asset	missing-asset placeholder in doc; asset manager offers re-link/remove; export notes it
db wiped (§11.2 rebuild)	harness deletes databases/	index rebuilds from files+journal; library identical (sample verified)
partially-copied backup restore	harness truncates restore zip	restore aborts; existing library untouched; report explains

Drills live in `core-storage/src/test/drills/` and run in G2/G7 pipelines. Rule: corruption is an engineering input, not an error message — every scenario above ends with *user data safe or honestly reported*.

12.9 Uninstall and data lifecycle

Uninstall removes everything app-private (docs, packs, keys) by platform rule — the About screen says so in one line, because "where do my words live?" deserves a straight answer. SAF exports/backups persist by design (user-chosen locations); the backup passphrase is required to read them — OneDoc cannot recover it (no escrow, no account; the only unrecoverable secret in the product, stated at set-up). Clear-data (without uninstall) wipes the same stores; the next launch re-seeds schema and offers import/restore via the backup flow (TC-STATE-010 covers the clear-data → restore path).

10.7 The editor core class map (who does what)

Class (core-editor)	Responsibility
`EditorCore`	the retained non-UI brain: owns document, op log, undo stack, journal client
`OpEngine`	applies/normalizes ops; the only code allowed to mutate the model
`UndoStack`	batches + inverses + coalescing timers (§14.4)
`ImeBridge`	InputConnection implementation; composition queue (§15.8)
`Spanner`	block ↔ Spanned mapping (pure, deterministic, §14.6)
`SelectionState`	the §15.6 five-tuple + snapping + handle math
`ToolbarStateMachine`	§15.7 sealed transitions
`PagerMetrics`	pagination guides + PDF shared metrics (§17.4)
`BlockTextView`	the one custom text view; renders Spanned, hosts IME
`JournalClient`	CRC frames, fsync, compaction triggers (§11.3)

Everything UI-side (feature-editor) is rendering and gesture wiring only; a UI-free Robolectric test drives `EditorCore` through the full IME script set — the same tests CI runs, the same laws it checks.

PART V

THE DOCUMENT MODEL AND EDITOR ENGINE

The ten editor laws that are release blockers, the block-and-mark data model, the operation/undo engine, IME coexistence, and the property-based test system that makes correctness provable rather than hoped for.

Chapter 13 — The ten editor laws (release blockers)

13.1 Why laws

The predecessor program proved that a small set of absolute, test-bound behaviors — treated as laws — is what makes an autonomous build shippable: laws are few, precise, and each has a case id. OneDoc's editor has ten. A release candidate that fails any law does not ship, no matter what else is green.

Law	Statement	Proven by
L1 — What you see is the file	After autosave (≤ 800 ms after last edit), the persisted document equals the on-screen content byte-for-byte in text and marks	TC-LAW-001
L2 — Marks are sane	At any state, marks of the same type never improperly overlap; `normalize()` restores that invariant after every op, idempotently	TC-LAW-002, property P2
L3 — Zero-loss durability	A process kill at any instant loses at most 2 s of typing; after relaunch the document is valid and cursor-consistent	TC-STATE-005 (100 kill trials)
L4 — Undo is honest	Undo after any op sequence returns a state previously on screen (not merely "shorter text"); redo mirrors it; no user-visible op escapes the stack	TC-EDIT-003, property P1
L5 — IME composition is sacred	No op (autosave, snapshot, AI, find-highlight) ever mutates text inside an active composition; composition text is inviolable until committed	TC-EDIT-IME-01/02
L6 — Determinism	The same op list applied to the same document yields the identical result, always — including snapshot ids and export bytes (excluding timestamps)	TC-LAW-003 (golden files)
L7 — Exit never asks, never loses	Leaving the editor (back, home, app-switch kill) silently persists everything committed; no save dialogs exist	TC-STATE-001/002
L8 — Structure is closed	Only block kinds in the §14.3 set exist; unknown kinds from imports/opening new files are preserved verbatim (round-trip) with a conversion note, never dropped	TC-IO-002/005, TC-LAW-004
L9 — Selection survives	Cursor/selection survive: mark toggles, style changes, snapshot/restore of *other* docs, sheet open/close, split-screen resize	TC-EDIT-004, TC-STATE-002
L10 — No silent mutations	No code path (AI, importer, backup restore, migration) may modify document content without a visible, undoable, versioned operation	TC-AI-006, TC-STATE-004

13.2 Law provenance and precedence

Laws derive from the owner's acceptance criteria for the predecessor program (calculator laws L1–L10, CAT parity): a small surface must be *perfectly* right before breadth matters. When a feature task and a law conflict, the law wins and the feature is redesigned. Law violations are S1 defects by definition (§25.3).

Chapter 14 — Document model

14.1 Model choice, argued

Rich-text on mobile is dominated by two architectures: span-over-TextView (native, fast, but undo/structure logic becomes spaghetti) and block-based CRDT-style models (web heritage, robust, heavier). OneDoc uses a **block-and-mark model with an operation log** — the Docs/Notion shape — because: (a) laws L2/L4/L6 need an explicit op log and normalizer; (b) tables/images/TOC are blocks, which spans cannot express; (c) versioning is trivially "this op prefix"; (d) it is implementable and testable in pure Kotlin (`core-model` has zero Android imports), so the JVM property tests run in seconds, not on emulators.

14.2 The document



Figure 10. *document.json: schema-versioned blocks with ids, inline marks, and a closed kind set.*

```
{
  "schema": 3,
  "docId": "d_9f2c1a7b",
  "meta": { "title": "Project Phoenix proposal",
            "created": 1761900000000, "wordCount": 1842,
            "pageSetup": { "paper": "A4", "margin": "normal" } },
```

```
  "blocks": [
    { "id": "b01", "type": "h1", "text": "Quarterly Business Review" },
    { "id": "b02", "type": "para", "text": "Revenue grew 14% quarter over quarter…",
      "align": "left",
      "marks": [ { "type": "strong", "start": 0, "end": 7 },
                 { "type": "link", "start": 22, "end": 34, "href": "…" } ] },
    { "id": "b03", "type": "list", "listType": "bullet",
      "items": [ { "id": "i1", "text": "Enterprise renewals +22%", "checked": null } ] },
    { "id": "b04", "type": "table", "header": true,
      "cols": [ { "width": "equal" } ],
      "rows": [ { "id": "r1", "cells": [ { "id": "c1", "blocks": [] } ] } ] }
  ]
}
```

Invariants (checked by `core-model` validators after every op; violation = engine bug, fail-fast in debug):

- **I1** every block/item/cell has a stable id unique in the document; ids are never reused.
- **I2** blocks are ordered, no holes; text offsets are UTF-16 code-unit indices (matching Java/Kotlin strings exactly — no emoji off-by-one class of bugs).
- **I3** marks within one type are non-overlapping and half-open `[start, end)`; after `normalize()` the invariant holds and a second `normalize()` is a no-op.
- **I4** a list item's `checked` is non-null only for `check` lists; a table cell contains blocks (recursion depth 1 — no tables in tables in v1.0, refused at insert with a reason).

14.3 The closed block-kind set (Law L8)

`para, h1, h2, h3, list(bullet|number|check), quote, code, table, image, divider, toc, footnote, cover`. Thirteen kinds, schema-versioned. Adding a kind is a schema migration (§11.4) with round-trip golden files; no other extension point exists — custom plugins and embedded HTML are out of scope (§1.4).

14.4 Operations and undo

The engine's op set is minimal and sufficient: `insertText(block, pos, text)`, `deleteText(range)`, `setMarks(range, mark, on)`, `setBlockType(block, kind, params)`, `splitBlock(block, pos)`, `mergeBlocks(a, b)`, `moveBlock(from, to)`, `insertBlock(at, block)`, `removeBlock(block)`, `tableOp(kind, params)`. Composite ops (paste, AI insert, replace-all) are groups that undo atomically (UX-06).

Undo correctness is by **inverse application, not state snapshots**: each op knows its inverse; the stack stores committed batches with inverses and (for redo) reapplications. Coalescing: consecutive `insertText`/`deleteText` on the same block with no intervening op merge into one batch after a 600 ms idle boundary; cursor movements never enter the stack. Property P1 (§16.2) proves `apply(ops) then undo-all == original` over randomized op lists.

Chapter 15 — Text input and the IME contract

15.1 The bridge

Android IMEs type through `InputConnection`; the editor implements it over the block model with three absolute rules (Law L5):

1. **Composition isolation:** an active composition (`composingText` region) pins its block; the engine queues all non-IME ops (autosave snapshot, find re-highlight, AI cancellation cleanup) until `finishComposing`. The queue is drained in order after commit.
2. **Batched commits:** `commitText` applies as one `insertText` op (undoable as one step); `deleteSurroundingText` maps to `deleteText` with the same batching as key events.
3. **No silent re-layout of composed text:** the block being composed re-renders only its own text region; neighbors do not reflow (UX-26).

15.2 The known hard cases (each has a test)

Case	Behavior	Test
Emoji / surrogate pairs at selection	offsets in code units; deletion removes the pair atomically	TC-EDIT-017
Gboard glide typing with autocorrect	full composition replace lands as one op; marks outside composition untouched	TC-EDIT-IME-01
Hindi/Devanagari transliteration (owner locale)	combining-mark clusters never split; cluster-aware boundary helpers used everywhere	TC-EDIT-IME-03
Two-finger scroll mid-composition	composition cancelled cleanly via `finishComposing` before scroll op	TC-EDIT-IME-02
Physical keyboard + IME simultaneously	same op stream; no double-commit	TC-EDIT-016
Samsung keyboard "rewrite" suggestions	treated as commit/replace stream; covered by IME fuzzer (§15.3)	TC-EDIT-IME-04

15.3 The IME fuzzer

A Robolectric+instrumentation harness replays recorded `InputConnection` call sequences from the API 26/33/35/36 emulators' default IMEs plus scripted Gboard patterns (recorded once, stored as JSON scripts). The fuzzer interleaves random valid call sequences with engine ops and asserts laws L2/L4/L5 hold after every step. It runs on every PR touching `core-editor` and nightly on emulators; a red fuzzer blocks release regardless of other greens.

Chapter 16 — Property-based testing of the engine

16.1 Why properties, not examples

Example tests encode the bugs you imagined; property tests encode the laws you promised. The engine's core properties run in JVM seconds on thousands of random documents and op lists — this is what lets an autonomous agent refactor the editor without fear.

16.2 The property suite (P1 … P10)

P	Property	Source law
P1	For any valid doc D and op list O: `undoAll(apply(D, O)) == D` (text + marks + ids)	L4
P2	`normalize` is idempotent and restores I3 for any doc reachable by ops	L2
P3	For any doc, `plainText()` contains no zero-width engine internals; length equals sum of block lengths + separators	L1
P4	Split/merge are exact inverses for any split point (marks redistributed, never dropped)	L1
P5	Move/remove preserve ids and the no-holes invariant	I1, I2
P6	Table ops never orphan cells; depth-1 invariant holds under any op sequence	I4
P7	Export→import round-trip (odoc, md subset, html subset) yields `same(normalize, markSet)` equality per fidelity level	L6, L8
P8	Journal replay of any op-batch sequence yields the same state as in-memory application	L3
P9	Snapshot restore then redo-stack clear leaves a valid cursor ≤ doc length	L9
P10	Generators: `insertText` with random Unicode planes (emoji, Devanagari, CJK) never throws and keeps offsets valid	L5

Generators live in `core-model/src/test/gen/`; seeds are logged on failure so any red property reproduces exactly (Law L6 for the tests themselves). Coverage gates: engine line coverage ≥ 90%, property suite green = G3 exit criterion (§21.5).

14.5 Mark semantics per type (the edge rules)

Marks look symmetric in the schema and asymmetric in behavior. The table is normative; property P2 tests the invariants, these tests pin the semantics.

Mark	Typed at boundary	Overlap rule	Copy/paste	Export note
strong	extends (new char inherits bold)	merges with adjacent same-type	keeps	md `**` / html `<strong>`
em	extends	merges	keeps	md `*` / `<em>`
u	extends	merges	keeps	html `<u>` only (md has none — note)
strike	extends	merges	keeps	md `~~` / html `<s>`
code	does NOT extend (typed char is plain)	never merges; normalized to abut, not overlap	keeps	backticks / `<code>`
highlight	extends	same-type overlap forbidden (I3)	keeps	html `<mark>` (md: `==` extension, noted)

Mark	Typed at boundary	Overlap rule	Copy/paste	Export note
link	typed text inside link range extends href	one href per range; adjacent different-href links abut	keeps href; empty selection + paste-with-link = plain text (link not applied)	md `[t](href)` / `<a>`

Boundary rule common to all: a mark is "at" a caret when `caret ∈ [start, end)` after normalization; toolbar buttons show active state from the single block containing the caret (multi-block selection shows active only if all non-empty ranges carry the mark — Docs behavior, TC-EDIT-004).

14.6 The rendering stack (model → pixels)

`core-editor` binds blocks to Android text infrastructure with a strict pipeline:

1. **Model → span state:** each block owns one `BlockTextView` (custom `AppCompatTextView` subclass with caret/handle drawing disabled — we draw our own) bound via a `MarkSpanner` that maps marks to framework spans (`StyleSpan(b)`, `UnderlineSpan`, `StrikethroughSpan`, `BackgroundColorSpan`, `ClickableSpan`-like custom link span, `TypefaceSpan(mono)`). The spanner is pure and unit-tested: `document block → Spanned` is deterministic (Law L6).
2. **BlockTextView rules:** no scroll inside blocks (the document scrolls); `includeFontPadding=false` with our own line metrics; text IS the source for IME (`InputConnection` reads/writes the `Spanned` window directly within the block, §15.1).
3. **List rendering:** the RecyclerView grid lays out list items as blocks whose left inset carries the marker (bullet/number/checkbox) drawn by the item decoration — markers are never part of the text (Law L1: what you see is the file, markers included in exports but derived, not typed).
4. **Tables:** RecyclerView-of-RecyclerViews is banned (scroll trap, §17.1). A table block renders one custom view: measured columns, per-cell `StaticLayout`, cell editor is a popup `BlockTextView` anchored to the cell (cell content edits in place visually; the popup is transparent to the user).
5. **Pagination guides:** a `PagerMetrics` engine computes break lines from the same `StaticLayout` objects the blocks render with — guides are drawn as decorations at zero extra measure cost (§17.4 parity guarantee).

14.7 Memory and performance playbook

- **Window the document:** blocks are realized lazily (RecyclerView); a 300-page document keeps ~60 realized blocks. PagerMetrics for guides computes over offscreen blocks with cached `StaticLayout`s capped at 400.
- **Precompute once:** `StaticLayout` per block is cached by (text hash, width, textcache key); typing invalidates only the edited block (UX-26).
- **No allocation in** `onDraw` (lint + perf test): pens, paint, path objects are pooled per view.
- **Large text guard:** blocks > 20k chars (pasted walls) split visually into measured tiles internally while remaining one block in the model (Law L8 — the model never mutates to please the renderer).
- **Startup:** home renders first frame from Room index only; document content loads on open; cold-start budget §21.4 verified by macrobenchmark (TC-PERF-001).

15.4 Physical keyboards and pointers

Input	Behavior	Test
Arrow keys	caret traversal across block boundaries in visual order; Up/Down preserve target x-column (column memory)	TC-EDIT-016
Tab / Shift-Tab	indent/outdent list items; in tables moves cell focus	TC-EDIT-016
Ctrl+B/I/U/S	mark toggles on selection or at caret (word-scope)	TC-EDIT-016
Ctrl+Z / Ctrl+Y (Ctrl+Shift+Z)	same undo stack as soft keys (UX-28)	TC-EDIT-016
Ctrl+F	opens find bar preselecting current word	TC-EDIT-016, TC-UX-004
Esc	dismiss selection → toolbar → keyboard in that order	TC-UX-004
Mouse/stylus selection	same handles logic; right-click opens local context menu (TalkBack-menu parity)	TC-EDIT-004
IME + keyboard simultaneously	single op stream; composition from IME, shortcuts from keyboard — serialized by the bridge	TC-EDIT-IME-04

15.5 IME script format (the fuzzer's data)

Recorded scripts are JSON: `{ "keyboard": "gboard-en-sim", "events": [ {"t": 12, "call": "setComposingText", "args": ["hel"]}, {"t": 40, "call": "setComposingText", "args": ["hello"]}, {"t": 60, "call": "finishComposingText"}, {"t": 62, "call": "commitText", "args": ["\n"]}, ... ], "invariants": ["L2","L4","L5"] }`. Scripts ship in `core-editor/src/main/assets/imescripts/` (committed, reviewable). The fuzzer interleaves scripted events with random valid engine ops at recorded boundaries and asserts the listed invariants after every step (§15.3). Adding a keyboard = recording its script once; no code change.

16.3 Generator recipes (how the fuzzers randomize)

Generator	Produces
`docs(n, kinds)`	valid documents: 1–400 blocks, kind mix honoring I4 (no nested tables), random marks satisfying I3
`ops(k)`	op sequences biased toward the hard 20%: 40% split/merge at mark boundaries, 20% mark toggles spanning blocks, 20% list indent/outdent, 10% table ops, 10% unicode insert (planes 0/1/2/14: emoji, Devanagari, CJK, tags)
`imeEvents(k)`	valid `InputConnection` sequences from grammar (composing regions well-formed by construction)
`journals(k)`	op batches with randomized kill points for replay tests (P8, TC-STATE-005)

Seeds print on failure and reproduce byte-identically in CI (Law L6 applied to the tests). The suite runs 5,000 cases per property per PR and 50,000 nightly; shrinking (minimal counterexamples) is required output for any red.

15.6 Selection engine internals

Selection state is a five-tuple: `(anchorBlock, anchorOffset, focusBlock, focusOffset, mode)` where mode ∈ {caret, word, paragraph, all, handle-drag}. Rules:

1. Selection may span blocks (unlike marks — marks never cross block boundaries; a cross-block selection applying bold splits into per-block ranges, TC-EDIT-005).

2. Word/paragraph snapping uses ICU `BreakIterator` (same as find, §17.9); double-tap = word, triple = paragraph (never "select all" on triple — one-handed accidents).
3. Handle-drag dispatches through the gesture arena with slop 8 dp and a 100 ms pre-drag delay on handles only; during drag, auto-scroll activates within 10% screen margins at velocity ∝ overshoot, and the selection preview updates per frame (no commit until release — undo sees one composite op).
4. The floating toolbar positions from the focus rect: above by default, flips below when clipped by app bar or IME top; x clamps to screen; it never overlays the selection band (UX-19) — verified by a geometry unit test with a matrix of rects.
5. "Select all" exists only in the overflow sheet and local context menu (not a gesture) to prevent accidental whole-doc operations; it confirms when doc > 5k words (destructive-paste risk).

15.7 The toolbar as a state machine

Toolbar visibility = f(selection, IME, sheets): visible iff selection non-caret ∧ no sheet open ∧ IME allowed visible. State transitions are explicit (sealed class), making the "toolbar stuck" bug class structurally impossible; the machine is exhaustively unit-tested (every event × every state). Buttons read active-state from the §14.5 boundary rule and disable nothing — unavailable actions hide (same philosophy as F-082, applied to formatting: e.g., "clear marks" hides when selection has none).

16.4 The failure museum (classic editor bugs and what kills each)

Every exhibit is a real bug class from a decade of mobile editors. The museum's label names the killing test — this is why the property suite exists.

#	Exhibit	Killing test
1	"Undo deletes the wrong word after IME autocorrect" (composition commits as several ops)	IME-01 + P1: composition = one op
2	"Emoji splits into boxes when bolded" (surrogate halves)	EDIT-017 + P10
3	"Bold bleeds into the next paragraph after merge" (mark ranges not remapped on merge)	EDIT-002 + P4
4	"Cursor jumps to document start after rotation" (state in Activity, not retained core)	STATE-002 + L9
5	"Typing lags after page 40" (whole-doc reflow per keystroke)	PERF-002 + §14.7 windowing
6	"Autosave wrote half a document during kill" (save not transactional)	STATE-005 + P8
7	"Select-all + paste lost the last paragraph" (range end-exclusive mixups)	EDIT-012 + P1 half-open invariants
8	"Checkbox state ignored in export" (kind-specific fields dropped by exporter)	P7 + goldens
9	"Toolbar stayed after keyboard closed" (state machine holes)	§15.7 exhaustive transition test
10	"Same doc, different export bytes on re-export" (timestamps/hash drift)	LAW-003 goldens minus timestamps
11	"Numbered list renumbered wrong after deleting item 2 of 5" (renumber not an op)	EDIT-007 + P5
12	"Link swallowed the space after it" (boundary rule §14.5 absent)	EDIT-005 boundary matrix

Rule: any bug found in the wild joins the museum with a test id — the museum is append-only, like everything else in this program.

15.8 The composition queue (exact semantics)

States: `idle` → `composing(block, range)` → `committing` → `idle`. While `composing`: (1) all non-IME mutation requests enqueue (never merge); (2) journal writes for *prior committed* batches continue (durability does not wait for composition); (3) snapshots may run (they read last committed state); (4) find-highlight refresh defers; (5) AI cancellation cleanup defers. On `committing`: apply queued IME delta as one op batch; drain queue in FIFO; each drained item re-validates its precondition (e.g., highlight term still present) and self-drops if stale. Death in `composing`: uncommitted composition text is lost *by platform design* (never entered the op log); loss ≤ the composition itself — documented in the LAW-003 evidence, counted inside the 2-second L3 bound. Queue depth is bounded at 64 with overflow = invariant alarm (E_IME_VIOLATION) — deeper than any real burst.

16.5 Property-suite economics

PR runs: 5,000 cases/property ≈ 40 s JVM parallel — cheap enough to never skip. Nightly: 50,000 + cross-property composition (P1×P2×P8 chains). Failure budget: a property red is a G3/G4 gate blocker regardless of other greens (laws outrank features, §13.2). Shrink results are cached per seed so triage reads a minimal counterexample, not a stack trace. The suite's runtime is itself budgeted (§21.4 table) — property runtime creep is a defect against developer velocity, treated like any budget breach.

PART VI

FEATURES: STRUCTURE AND OBJECTS

Implementation contracts for the features that make OneDoc Docs-class: tables, images, templates, search, reader mode, pagination. Part II said what; this part says how, with the cases that prove it.

Chapter 17 — Structure and objects: engineering contracts

17.1 Tables (F-031, F-032)

Tables are blocks containing one level of cells; each cell holds paragraphs/lists. Rendering uses a measured grid table with per-column max-width; the doc never scrolls horizontally because of a table — narrow columns wrap (UX contract, TC-TBL-002). Editing: tap cell → caret with cell-scoped selection; the table toolbar (row/col add-remove, header toggle) appears anchored to the table. Column presets: equal or content-weighted (recomputed on "re-fit", not live — live reflow during typing causes layout thrash and is banned). Marks inside cells round-trip everywhere (property P6). PDF export renders tables with header repeat across page breaks (F-051).

17.2 Images (F-033, F-034)

Insert paths: gallery picker, camera, and from-scan (§19.2). Images are copied into `assets/` (JPEG 85 quality, max edge 2048 px, EXIF rotated and stripped of location data — privacy, §20.4); stored hash-named; referenced by asset id. Resizing is preset-based (S 25% / M 50% / L 75% / full measure) — free-drag resize is explicitly rejected (fiddly on touch, breaks measure). Alt-text (F-034): the insert flow ends with the alt-text field focused (UX-38), deferrable but tracked; an image without alt text shows a dotted outline in the editor and is listed in the asset manager. Full-screen viewer respects theme background and has TalkBack description from alt text.

17.3 Table of contents and document map (F-035, F-036)

The TOC block renders from H1–H3 at layout time (never a stored copy — always current); entries scroll to their block with position animation 200 ms. In exports, TOC becomes a static list with page numbers (PDF) or links (HTML/odoc). The map sheet (portrait rails) lists the outline with the live position dot; entries are drag-reorderable via the same moveBlock engine op, giving instant outline-level reorganization.

17.4 Pagination (F-037, F-038)

Compose mode shows page-break guides computed by the same metrics engine that drives PDF export (same font metrics, same line-breaking config) — the ±1 line guarantee (TC-STRUCT-004). Paper A4/Letter and margin presets change measure and guides live. Footnotes (F-039) render inline as superscripts with a bottom-of-page sheet on tap in editor; in PDF they are true page footnotes (Print framework region per page). This "one metrics engine, two renderers" rule is what makes WYSIWYG honest without a full pagination layout engine in-loop (which would wreck typing performance).

17.5 Templates (F-045 … F-050)

Built-ins are bundled `document.json` assets with placeholders `{name}`, `{date}`, `{your_name}` (tapped → fill sheet; date auto-fills locale-aware). Templates never execute anything; they are data. "Save as template" copies the current document minus content-ids into the user template store (Room table `templates`, schema v3). Gallery ordering: built-ins in a fixed pedagogical order (blank, letter, meeting notes, …), user templates recency-first. Preview renders the template read-only in reader mode with placeholders visibly marked.

17.6 Search (F-067 … F-070)

Indexing: on document save (debounced), `core-search` extracts per-block plain text into the FTS5 external-content shadow table; indexing a 10k-word doc must take < 150 ms background (measured, TC-SRCH-005). Query path: FTS5 `MATCH` with prefix queries, snippet() for the hit context, per-doc hit counts; results ranked by (title match, recency, density). Semantic layer (F-070, pack-gated): per-block MiniLM embeddings computed during idle maintenance windows (JobScheduler-style periodic task, battery-constrained); query embeds once (23 ms on Tier B) and merges cosine hits below FTS hits with a "related" label. Without the pack, semantic search simply doesn't appear (F-082) — no empty section, no error.

17.7 Reader mode and read-aloud (F-083, F-080)

Reader mode renders the same `document.json` through the read-only renderer: no chrome beyond font-size stepper, theme follows reader theme; TTS entry starts `TextToSpeech` from the first visible block, highlighting the sentence currently spoken (utterance-range callbacks), pause/resume/stop. Ranges (F-081) start from cursor when present. TTS never reads mark punctuation noise (bullet markers are spoken as list items).

17.8 The features that deliberately do not exist in v1.0

Word count goals, distraction-free timers, markdown shortcuts while typing (a v1.1 candidate — D-nnn reserved), collaborative anything, macros, plugins, custom themes beyond the token system. Each exclusion has a reason logged in §1.4 or §30; the agent adds none of them without an owner decision.

17.9 Find & replace internals

Find shares the highlight system with search deep-links (F-069): highlights are `BackgroundColorSpan` ranges recomputed on `textChanged` for the visible window only (never the whole doc). Match engine: case-fold via `String.equalsIgnoreCase` semantics (locale-invariant root fold for the default toggle), whole-word via ICU `BreakIterator` word boundaries (never regex □ — that is ASCII-only). Replace-all applies as one grouped composite op and snapshots `pre_edit` first (F-025); replacement text containing block-separator markers is rejected with a hint (use per-match replace). Regex mode is deliberately absent in v1.0 (recorded shelf item §30.2) — the support cost of 100 user regexes was judged worse than the feature's value.

17.10 How features join the undo system

Feature	Undo shape
paste (any flavor)	one composite group
replace-all	one composite group + pre-edit version
AI insert/replace	one composite group + pre-edit version (Law L10)
import-merge (scan multi-page into existing doc)	one composite group + pre-edit version
checklist toggle	single op, coalesces with adjacent same-block toggles
drag-reorder	one moveBlock op; a11y move actions emit the same op
table row/col ops	composite of remove/insert ops, atomic undo
template instantiate	not an undo — it is a new document
version restore	clears redo stack, snapshots current first (F-062)

Rule: any new feature that mutates content MUST join this table (PR checklist §29.5) — an ununduable mutation is a Law L4 violation by definition.

17.11 Reader-mode rendering rules

Reader reuses the same spanner (§14.6) with a chrome-free viewport: no toolbars, no guides; measure widens to screen minus 20 dp; theme = reader theme (sepia default at night by system clock); font-size stepper persists per document. Lists render markers from decoration (same as editor); tables render static with horizontal-in-block scrolling only when exceeding measure (allowed here — no editing in reader); images render at measure width with alt text as caption-on-tap. TTS (§19.7) anchors to the first visible block and follows scroll by pausing on exit (never reading offscreen).

17.12 Accessibility of custom views (engineering contract)

Custom views (BlockTextView, table view, toolbar) expose: role (`AccessibilityNodeInfo` class names kept semantic), state (selected/marked), actions (format toggles, move up/down, delete, duplicate — mirroring §17 handle menu), and traversal order matching visual order via `screenReaderFocusOrder`. Table view exposes cell matrix with row/column headers in `collectionInfo`. The toolbar exposes each action with state (bold on/off). The a11y audit (TC-A11Y-001) walks these nodes; any custom view merged without node contract fails review (DQ-6).

17.13 Checklists (the quiet feature done right)

Check items are list items of kind `check`; the tap target is the box (48 dp) or the whole row toggle; checked state is a mark on the item (exports per §18.2: md `- [x]`, html `data-check`, docx glyph, pdf drawn box). Progress: a check-list block with ≥ 2 items shows a subtle "2/5" chip in its top-right that fades after first interaction — progress is information, not gamification (no bars, no colors beyond token ramps). Long lists (> 30 items) paginate guides as normal blocks; no virtualized special case (blocks already window, §14.7).

17.14 Statistics and the info sheet (counting contracts)

Words = ICU grapheme clusters split on line/space boundaries (same number everywhere: chip, info sheet, exports — UX-49's ±1% is for PDF extraction noise, not definitional drift). Characters = code points (what users think); the info sheet shows both chars and "chars with spaces". Reading time = words ÷ 200 wpm rounded up. Size = document.json.zip + assets on disk. Version count = rows in `versions`. Every number names its definition in the info sheet's fine print — honest counts are part of P5.

17.15 Export-presets memory (small kindness, defined precisely)

The export sheet remembers the last used format+paper per user (device-local, one row in `meta`), preselecting it next time — nothing cloud, nothing clever; it just stops asking the same question daily. Reset by long-press on the format list ("Always ask again"). Tested in TC-IO-001-b (preselect honored; long-press resets).

PART VII

FILES, FORMATS AND FIDELITY

The honest format matrix, the exporters and importers, sharing and SAF, backup. OneDoc never lies about fidelity: every conversion states what it kept and what it dropped.

Chapter 18 — Files and formats

18.1 Principles

1. **Refuse, don't guess.** An importer that cannot represent content returns a specific refusal (format X, feature Y), never an empty document (F-056, UX-15).
2. **Declare fidelity.** Every conversion result carries a fidelity level (§18.2) shown to the user (UX-45) and asserted by tests (property P7).
3. **Lossless native.** `.odoc` is the canonical form; every other format is a projection. Importers that cannot map something preserve it in a round-trip-safe extension when the target is `.odoc` (Law L8).
4. **No network, ever.** "Open with" handlers accept local files/streams only; URIs requiring remote fetch are refused with a reason (C1).

18.2 The fidelity matrix

Format matrix and declared fidelity levels

Markdown .md	in+out	**F2**	headings/lists/bold/italic/links/code; tables → pipe tables
Plain text .txt	in+out	**F1**	text only; block structure flattened
HTML .html	in+out	**F2**	semantic subset: h1-3, p, ul/ol, table, img, strong/em/u/a
PDF .pdf (export)	out	**F3**	paginated via Print framework; fonts embedded; selectable text
DOCX .docx (export)	out	**F2**	OOXML subset writer: styles, lists, tables, images
DOCX .docx (import)	in	**F1**	best-effort reader: text + basic marks; unsupported → F0 note
OneDoc .odoc	in+out	**F3**	native zip container: document.json + assets + meta — lossless

F3 lossless · F2 lossless for the covered subset, others degrade with a conversion note · F1 best-effort text · F0 unsupported (refused with reason)

Figure 11. *Format matrix with declared fidelity levels and covered subsets.*

Format	Direction	Level	Covered subset
OneDoc `.odoc`	in+out	**F3**	everything (zip: document.json + assets + meta)
Markdown `.md`	in+out	**F2**	h1–h3, para, bullet/number/check lists, quote, code, divider, strong/em/code/highlight/link marks, pipe tables
Plain text `.txt`	in+out	**F1**	text only; blank-line paragraph inference on import
HTML `.html`	in+out	**F2**	h1–h3, p, ul/ol/li, blockquote, pre/code, table (simple), img, strong/em/u/s/a, mark; scripts/styles stripped always
PDF (export)	out	**F3**	paginated via Print framework, fonts embedded, selectable text, TOC page numbers, table header repeat, footnotes
PDF (import)	—	**F0**	refused in v1.0 ("use Scan-to-doc for page images") — honest and better than garbage text
DOCX (export)	out	**F2**	styles for headings, para+marks, lists, tables, images, page setup
DOCX (import)	in	**F1**	best-effort text + basic marks; unsupported constructs listed per-paragraph in the conversion note

F0 = refused with reason · F1 = best-effort, stated · F2 = lossless for the covered subset, everything else degrades with a note · F3 = lossless. The conversion note (UX-45) is generated from a structured diff (elements dropped/transformed count), not a canned string.

18.3 Exporters

Export is a pure function in `core-io`: `document.json → bytes` with golden-file tests per format (Law L6). PDF export uses `PdfDocument`/Print framework with the pagination metrics from §17.4; embedded fonts are subset Inter + DejaVu fallback chain; exported bytes are deterministic except the embedded creation timestamp (excluded from golden comparison). DOCX export writes a minimal but valid OOXML package (validated against the OOXML strict subset in tests by re-opening with the importer — round-trip gate). Share sheet (F-057) exports to cache and hands a `FileProvider` URI; private paths are never shared raw.

18.4 Importers

Import order of operations: sniff magic bytes/MIME → choose reader → parse to a candidate `document.json` → validate (I1–I4) → produce conversion note → open editor. SAF sources stream; sizes over 50 MB warn (mobile RAM honesty) but proceed. HTML import sanitizes by whitelist parser (no regex parsing, ever — the parser is a real tokenizer with tests; scripts/styles/event attributes cannot survive it, proven by TC-SEC-IO-01). DOCX import reuses the shared OOXML subset reader; features beyond it become per-paragraph notes.

18.5 Sharing, open-in and SAF (F-057 … F-059)

Intent filters: `application/zip` view for `.odoc`, `text/markdown`, `text/plain`, `text/html` — `android.intent.action.VIEW` with content: URIs only. Export uses `ACTION_CREATE_DOCUMENT`; import reads via `ContentResolver` with size caps. The app registers no launcher handlers for URLs/links that imply network fetch. Round-trip case TC-IO-012 proves share→receive→import→export parity for md/html/odoc on the emulator.

18.6 Backup and restore (F-060)

Weekly (default off) background backup: zip of all non-trashed documents (odoc form) + manifest with SHAs → user-chosen SAF folder → encrypted (§11.5). Restore imports the manifest, diffs by docId+content hash, and imports as new versions on top of existing docs (never silently overwrites); the restore report lists added/updated/conflicts. TC-IO-013 runs backup→wipe→restore on the emulator and asserts the library is byte-identical at F3.

18.7 Charset and encoding policy

All text is UTF-8 at rest (document.json, exports except UTF-16 for DOCX XML which mandates UTF-8 too — compliant). Offsets in the model are UTF-16 code units to mirror JVM strings exactly (§14.2 I2); exporters that emit UTF-8 bytes recount line/char metrics in their own units (word count is grapheme-cluster based via ICU, the number users see everywhere — one definition, TC-EDIT-015). BOM policy: readers accept and strip; writers never emit. Legacy encodings (latin-1, UTF-16 txt): sniffed via BOM/validators; converted with a conversion note; binary-garbage detection (NUL bytes, < 90% valid decode) refuses with reason (F-056).

18.8 IO error taxonomy and user strings (normative examples)

Error	Verb	User string (en)
`E_IO_SAF_DENIED`	actionable	"OneDoc can't write to that folder. Pick a different location."
`E_IO_UNSUPPORTED {fmt}`	actionable	"{fmt} isn't supported. Supported: odoc, md, txt, html, docx."
`E_IO_CORRUPT {fmt, why}`	actionable	"This file is damaged ({why}). OneDoc won't risk opening it."
`E_IO_TOO_LARGE`	actionable	"That file is over 50 MB — too big for a phone to convert safely."
`E_IO_DISK_FULL`	retryable	"Storage is full. Free some space — OneDoc will retry."
`E_IO_PARTIAL_WRITE`	retryable	"Writing was interrupted. Retrying…" (auto; no partial file remains, UX-18)

Every error kind has exactly one string in every shipped locale; the strings catalog test enumerates the taxonomy (no orphan strings, no missing strings — TC-SET-004 extended).

18.9 Golden-file management

Golden corpus: `core-io/src/test/resources/golden/<format>/` — input fixtures + expected outputs, byte-exact for odoc/md/html/txt, normalized (timestamps stripped) for pdf/docx. Regeneration is an explicit task (`--update-goldens`) whose diff MUST be reviewed line-by-line in the PR (silent golden churn is how fidelity regressions sneak through). New exporter feature ⇒ new golden case in the same PR (checklist §29.5). Round-trip goldens pair with property P7; a golden change without a P7-observed behavior change is suspect by default.

18.10 Per-format implementation specs (writers)

Markdown writer rules. Headings `#`×level; blank line between blocks; bold `**`, italic `*` (nested → `***` at shared boundaries); code spans pick backtick runs (longest-run fencing, §13.4-style rule); lists `-` / `1.` / `- [x]`; tables GFM pipe style with escaped pipes (`\\|`); links `[t](href)` with URL-escaped parens/spaces; check state → `[ ]`/`[x]`; no reference links (inline only — deterministic output).

HTML writer rules. One `<article>` root; semantic tags only from the §18.2 subset; marks map strong/em/u/s/code/mark/a; `lang` on root; checklists render `<li data-check="done">` (documented extension); images `<img src="assets/<sha>.<ext>">` inside a zip-flavored payload when sharing, or base64 when single-file requested (size warn > 2 MB).

DOCX writer rules. Styles: Heading1–3 ↔ h1–h3; Normal ↔ para; ListParagraph+numbering.xml ↔ lists; bold/italic/underline/strike runs ↔ marks; `w:shd` highlight ↔ highlight; tables `w:tbl` with `w:tblW` pct; images inline with extent; core props (title, created) from meta. The writer emits a minimal, valid package — validated by reopening with the importer plus an XML-schema subset check (TC-IO-004).

18.11 Per-format implementation specs (readers)

Markdown reader. Line-based state machine: ATX headings, setext rejected (note), fences with language capture, GFM tables, task lists, nested lists by indent (2/4-space tolerated, normalized), emphasis parser (delimiter-run algorithm, CommonMark subset), links inline only, HTML passthrough **rejected** (security posture, §20.4) with note. Undecidable constructs are not guessed: a setext heading becomes a paragraph with a note line, never silently.

HTML reader. Real tokenizer (state machine per WHATWG scanning subset) → whitelist tree → model. Unknown-but-harmless tags unwrap (keep text); hostile constructs (script/style/iframe/object/embed/form, on* attributes, javascript: URLs) drop structurally and count into the conversion note. Entities decoded via a fixed table (no external fetches — ever, for any purpose).

DOCX reader. Zip → document.xml walk: paragraphs with runs (rPr → marks), numbering refs → list kinds, tables (nested tables → flattened with note), images → assets, headers/footers/footnotes → v1.0: footnotes mapped to footnote blocks; headers/footers dropped with note. `w:ins`/`w:del` (tracked changes) → accepted text with a "tracked changes were flattened" note (§30.2 shelf: preserve as suggestions).

18.12 Worked fidelity examples (what the note actually says)

1. **Docs-exported HTML with a comment thread** → import: "Imported as HTML — F2. Kept: headings, paragraphs, bold/italic, one list. Dropped: 2 comment threads, 1 embedded drawing (image-only frame kept as image)."
2. **Word file with tracked changes** → "Imported as Word — F1. Kept: text, headings, bold/italic/underline, one table (2 columns flattened to paragraphs: 1 of 9). Flattened: tracked changes (accepted text). Dropped: header/footer."
3. **Markdown with setext heading** → "Imported as Markdown — F2. Note: 1 underline-style heading read as a paragraph (OneDoc keeps # headings)."
4. **HTML with script** → "Imported as HTML — F2. Removed for safety: 1 script block, 1 form." (SEC-IO-01 pairs with this note in tests.)
5. **DOCX with SmartArt** → "Imported as Word — F1. Dropped: 1 SmartArt diagram (content listed below as text)."
6. **PDF** → refused: "PDF isn't imported in OneDoc. Use Scan to doc for page images — it's better for real-world PDFs."

Each example is a golden test input (§18.9) — the note text above is asserted verbatim (en locale) by TC-IO-005's dataset.

18.13 PDF export internals

Engine: `android.print` with a custom `PrintDocumentAdapter` rendering through the same PagerMetrics (§17.4) to `PdfDocument` pages (A4/Letter per F-037). Fonts: Inter regular/semibold subsets embedded per page-family; DejaVu fallback chain for glyphs Inter lacks (Devanagari matras render via system Noto — subset embedded likewise). Text layer: real glyph runs (selectable/searchable, UX-49 count source). TOC entries carry link annotations to target pages; internal links (F-035) preserved; table headers repeat per break via per-page re-render of the header row region; footnotes bottom-of-page regions (§17.4). Metadata: title/author from doc meta; creation timestamp only (excluded from goldens). Deterministic layout: same doc+settings+font set → identical page breaks (LAW-003 asserts).

PART VIII

ON-DEVICE AI

Scan-to-doc, dictation, the writing assistant, semantic search and read-aloud — every capability running locally, every model a side-loaded SHA-verified pack, every unavailable state explained, nothing ever downloaded.

Chapter 19 — On-device AI

19.1 The five capabilities and their stacks

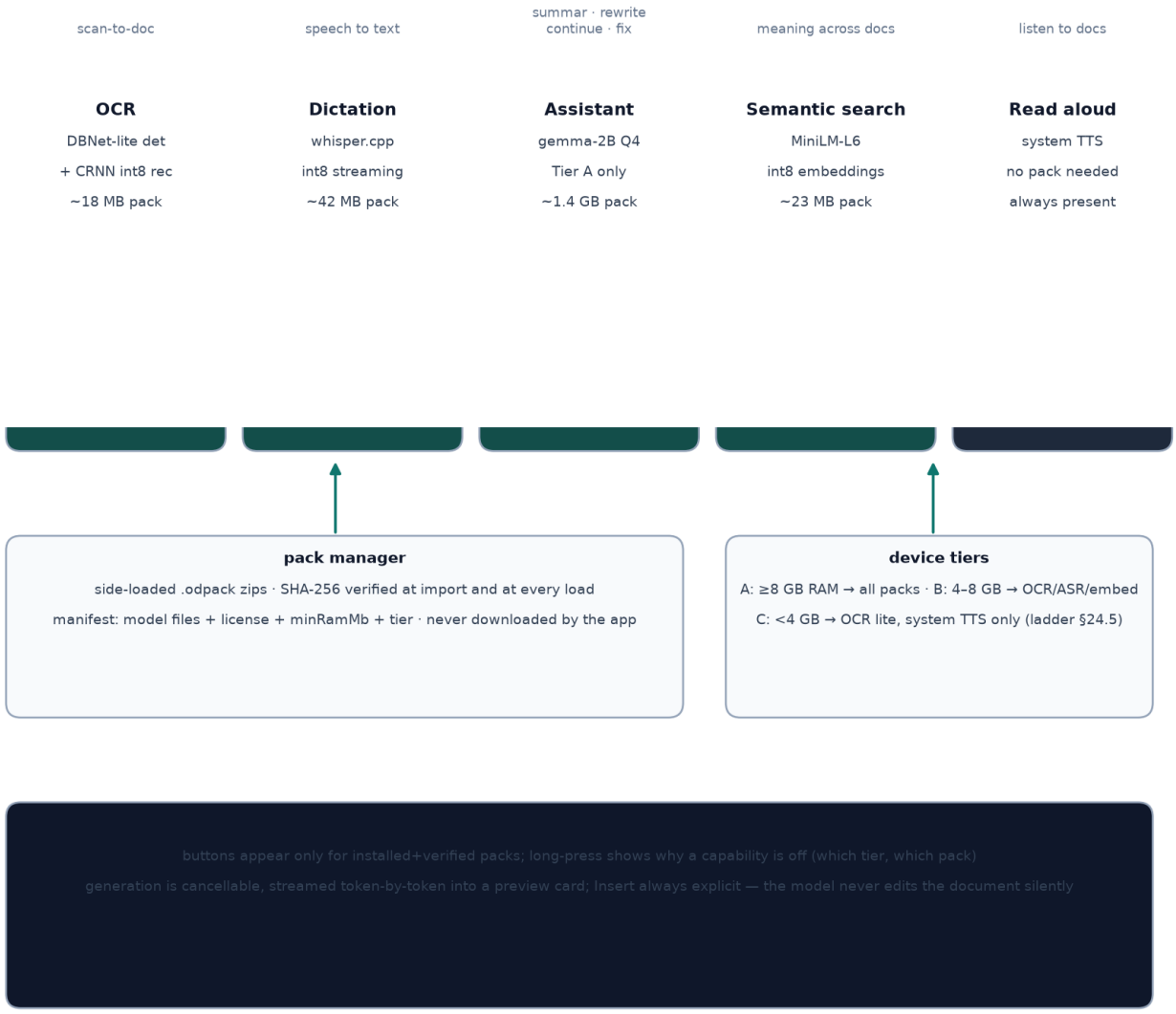


Figure 12. *The AI stack: five capability modules over the pack manager and device tiers; the assistant row is the single surface where AI meets the user.*

Capability	Engine	Pack	Latency budget (Tier B)
Scan-to-doc OCR	DBNet-lite detection + CRNN recognition, int8, NNAPI/GPU delegate	`ocr-en` ~18 MB (en; hi pack separate)	≤ 12 s/page incl. capture
Dictation	whisper.cpp int8 streaming (Tier A/B) or system `SpeechRecognizer` (Tier C ladder)	`asr-en` ~42 MB	first partial ≤ 1.5 s (UX-10)
Writing assistant	gemma-2B 4-bit (q4_0, llama.cpp-compatible runtime), Tier A only	`llm-gemma2b` ~1.4 GB	first token ≤ 2.5 s, then ≥ 8 tok/s
Semantic search	MiniLM-L6 int8 quantized embeddings	`embed-minilm` ~23 MB	embed block ≤ 25 ms; query ≤ 40 ms
Read aloud	Android system TTS	none	system-provided

Model runtimes are pinned (exact versions in Appendix F). Runtimes are compiled into the app as AAR/native libs; **models are never bundled** — they install as side-loaded `.odpack` files (F-078), keeping the APK ≤ 25 MB (§26.4) and letting owners choose what fits their device.

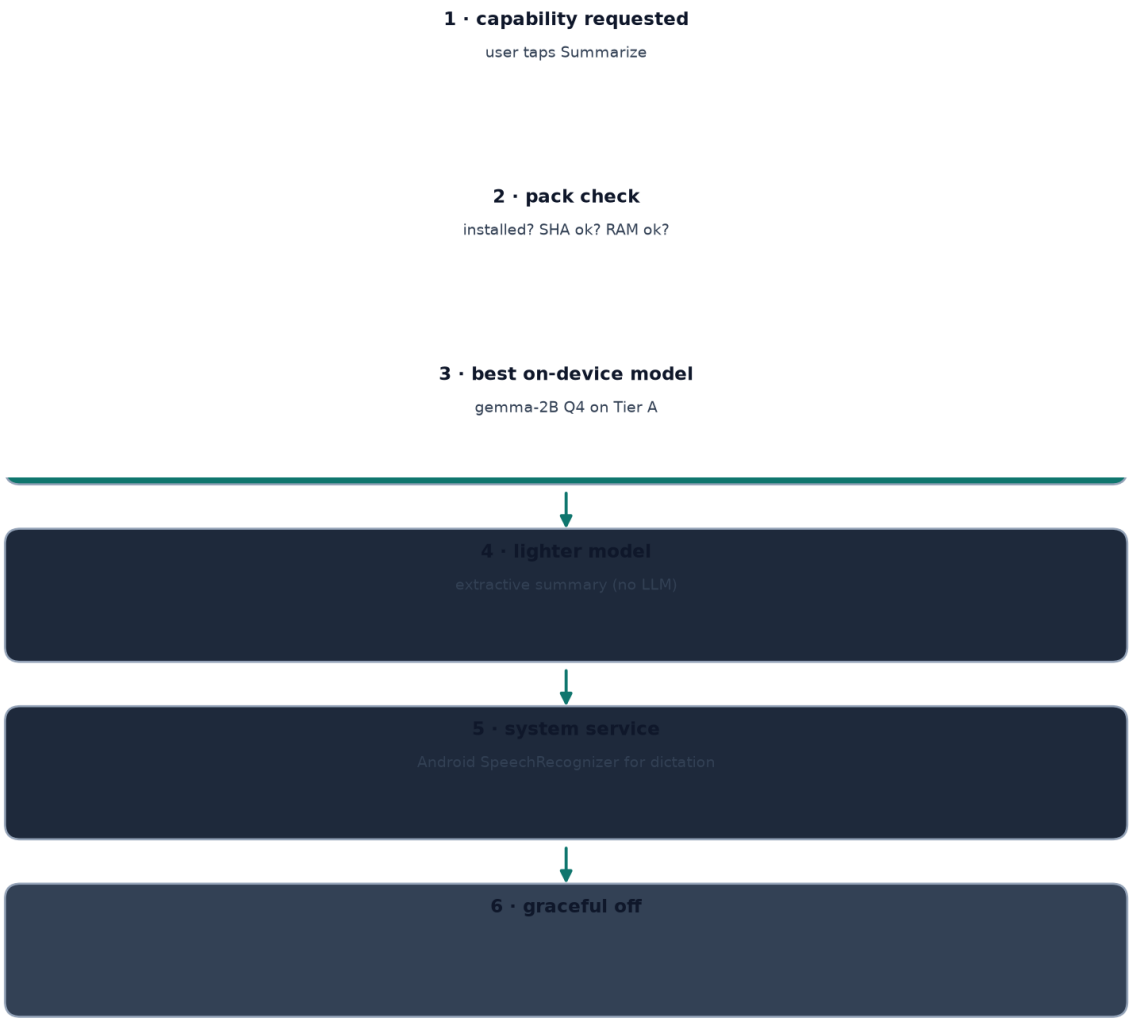
19.2 Scan-to-doc (F-071 … F-073)

Flow: camera screen (CameraX, no internet-capable extensions) → auto-capture on focus → perspective crop editor (drag 4 corners, rotate) → per-page OCR → review screen where recognized text sits beside the page image with **confidence coloring + underline style** (UX-41, DQ-4): low-confidence words (score < 0.82) are marked; tapping a word focuses an inline fix field. "Insert into document" appends pages as paragraph blocks (page headers as separators); the source images attach as assets linked in the conversion note. Without the OCR pack the flow ends after crop at "Save image to document" with a one-line reason (F-073, F-082). Multi-page (F-072): pages queue with thumbnails, reorder before OCR batch runs.

19.3 Dictation (F-074)

The dictation strip appears above the keyboard: live partial transcript, stop → preview card with the full text (editable) → Insert at cursor as one grouped op (undoable as one step, UX-06). Language: pack language or system locale for the SpeechRecognizer rung; never both at once. Mic permission is requested the first time dictation opens (UX-32: no pre-prompt), and audio is never buffered to disk — frames live in memory until transcription, then are dropped (§20.4).

19.4 The availability ladder (F-076, F-082 — normative)



Every AI feature descends exactly one ladder; the step taken is logged in the diagnostics screen (no network, no crash).

Figure 13. *Every AI capability descends exactly one ladder: best pack → lighter model → system service → hidden with reason.*

Rung	When	Example (assistant)
1	requested	user taps Summarize
2	pack present, SHA ok, RAM+thermal ok	gate check passes
3	best on-device model	gemma-2B 4-bit (Tier A)
4	lighter local method, no LLM	extractive summary (TextRank over block embeddings or sentence scoring)
5	system service	system SpeechRecognizer for dictation on Tier C
6	hidden with reason	control not shown; long-press explains: tier / pack missing / RAM

Rules: the ladder step taken is recorded in the preview card (UX-47) and diagnostics; rung 6 is a *hidden control*, never a disabled one (F-082) and never an error dialog. Thermal/RAM gates: before loading the LLM, `ActivityManager` memory class + a thermal-status check (PowerManager.OnThermalStatusChangedListener); at "severe" the runtime swaps to rung 4 mid-session and says so in the card.

19.5 The assistant surface (F-075, F-077)

Four verbs: **Summarize** (selection → bullet digest; doc → abstract), **Rewrite** (polish / formal / casual), **Continue writing** (two-sentence continuation from cursor, style-matched), **Fix grammar** (selection-level corrections presented as a diff list, each item individually accepted). Every output lands in a **preview card** with method label, token-stream rendering, Cancel, and explicit Insert/Replace — insertion snapshots `pre_edit` first (F-077, Law L10). The card is the only AI surface; there is no ambient/inline ghost-text anywhere in v1.0 (deliberate: Law L10 plus battery discipline).

Prompting: fixed system prompts stored in `core-ai/assets/prompts.json` (versioned, deterministic, temperature 0.3, max 512 new tokens); the document context is truncated by block boundaries to the model's window; prompts never contain metadata beyond the selected text. The prompts and decoding params are acceptance-tested: TC-AI-010 pins a set of summarization fixtures with quality rubric checks (extractive-faithfulness heuristics) and runtime-invariance checks (same input → same output).

19.6 Pack manager (F-078, F-079)

Packs are `.odpack` zips: `manifest.json` (name, version, minRamMb, tier, license, files[] with sha256), model files, license text. Install: SAF pick → streaming SHA-256 verify → copy into `files/packs/<name>/` → registry row in `meta` → capability appears. Verify runs again at every load (files can rot on disk); mismatch ⇒ refused with both hashes shown (UX-48) and ladder descent. Delete frees space with confirm. The manager shows per pack: size, tier requirement, license name, verification state. No pack ever ships inside the APK; no download path exists anywhere in the code (C5) — the direct-APK QA proves network silence during every AI flow (TC-SEC-P-03).

19.7 Read aloud (F-080, F-081)

System `TextToSpeech` only — no neural pack in v1.0 (recorded as a D-reserved decision for v1.1). Sentence highlight follows `onRangeStart`; the reader controls are play/pause/stop + rate via system settings. If no TTS engine exists (rare), the control hides with reason (ladder rung 6).

19.8 AI performance and memory budgets

Capability	Model	Tier B (3 GB AVD) budget	Tier A (8 GB) budget	Enforced by
OCR	DBNet-lite+CRNN int8	≤ 12 s/page; peak RSS +250 MB	≤ 6 s/page	TC-AI-001 timed
ASR	whisper int8 streaming	RTF ≤ 0.6; first partial ≤ 1.5 s	RTF ≤ 0.35	TC-AI-003 timed
LLM	gemma-2B q4_0	not loaded (Tier gate)	first token ≤ 2.5 s; ≥ 8 tok/s; peak RSS +1.9 GB	TC-AI-005 timed
Embeddings	MiniLM-L6 int8	≤ 25 ms/block; index 10k blocks ≤ 60 s idle	≤ 12 ms/block	TC-SRCH-005 timed
Read aloud	system TTS	system-provided	system-provided	TC-AI-007

Budget breach = S2 defect (§25.2). The AI executors run at `THREAD_PRIORITY_BACKGROUND` except during an active user action; embedding backfills yield to any user interaction (checked by a scheduler test).

19.9 AI privacy rules (all capabilities)

1. No audio, image, text, or embedding ever leaves the process (structural: no network path exists — C1; probes re-verify per RC: TC-SEC-P-03).
2. Audio frames are memory-only, dropped after insert (§20.4); scan images are written only when the user chooses "save image asset".
3. Embeddings are stored per-block in `vectors` with the pack's model id; deleting the pack deletes its vectors (space honesty, F-078).
4. The assistant's context window contains document text only — never metadata, never other documents.
5. Every AI surface labels its method and model (UX-47); diagnostics show last-run stats (duration, rung) but no content.

19.10 The prompt library (verbatim, versioned in `core-ai/assets/prompts.json`)

System prompt — SUMMARIZE:

```
You are a concise summarizer inside an offline writing app. Summarize the TEXT
for the document's author. Use at most 6 bullets or 2 sentences, whichever
fits the content. Quote key figures exactly. Never invent facts. Output only
the summary, no preamble.
```

System prompt — REWRITE (variant slot: polish / formal / casual):

```
You are a careful rewriter inside an offline writing app. Rewrite the TEXT in
a {polish|formal|casual} register. Preserve meaning, names, numbers and
formatting cues. Keep length within ±20%. Output only the rewritten text.
```

System prompt — CONTINUE:

```
You are a co-writer inside an offline writing app. Continue the TEXT with at
most two sentences that match its voice, tense and formatting. Do not repeat
existing phrases. Output only the continuation.
```

System prompt — FIX GRAMMAR:

```
You are a precise copy editor inside an offline writing app. List each grammar
or spelling fix in the TEXT as: ORIGINAL -> CORRECTED. One per line, in order.
If there are no issues output exactly: NO_ISSUES. Never change meaning.
```

Decoding: temperature 0.3, top-p 0.9, max 512 new tokens, seed pinned per run (invariance tested, TC-AI-010). Changing any prompt is a D-nnn decision + TC-AI-010 re-run.

19.11 The .odpack format (normative)

```
pack.odocpack (zip, store or deflate)
├─ manifest.json
├─ models/<files…>          # exact names per manifest
└─ LICENSE.txt
manifest.json:
{
  "format": 1,                        # pack format version (app accepts ≤ 2)
  "name": "ocr-en",                   # registry key
  "capability": "ocr|asr|llm|embed",
  "version": "1.3.0",
  "language": "en",
  "minRamMb": 2048,
  "tier": "A|B|C",                    # minimum tier for any use
  "files": [ {"path": "models/det.onnx", "sha256": "…", "bytes": 9123441} ],
  "license": "Apache-2.0",            # spdx id; full text in LICENSE.txt
  "runtime": {"engine": "onnxruntime", "min": "1.17.0"}   # app refuses older runtimes
}
```

Rules: total unpacked size capped at 8 GB; zip entries validated (count ≤ 64, no absolute/.. paths — TC-SEC-PK-02); SHA-256 verification streams (no full-load into RAM); verification state cached with file mtime+size re-check on every load. The manager UI never hides the license — install confirm shows spdx id + links (text) to LICENSE.txt inside the pack.

19.12 The OCR and ASR pipelines in detail

OCR preprocessing. Captured frame → grayscale → local adaptive binarization (Sauvola window 31, k 0.34 — chosen for receipt/whiteboard robustness) → deskew via Hough text-line angle (±15° clamp) → perspective rectify from crop quad → resize to detector input (long edge 960 px, aspect preserved, letterbox).

Detection (DBNet-lite). Input 1×960×N; output probability map → threshold 0.3 → connected components → polygon simplify (Douglas-Peucker ε=2 px) → rotated boxes. Page-text filter: drop boxes with height < 8 px source or fill ratio < 0.25 (noise, page edges).

Recognition (CRNN int8). Per box (batched ≤ 8): crop → height-32 resize, width ≤ 512 (long boxes tile with 16 px overlap, merged by CTC blank alignment); CTC beam (width 4) over the en charset (94 glyphs + case); output text + per-character confidence = frame posterior max.

Confidence aggregation. Word confidence = min of char confidences (pessimistic — marks the word if ANY glyph is unsure); threshold 0.82 (§19.2). Page confidence shown as chip; < 0.6 page-level → "low yield" state offers retake (OD-SCAN-003 states).

ASR (whisper.cpp) streaming. 16 kHz mono PCM frames (VAD-gated, 30 ms silence = segment hint); 10 s sliding window with 1 s overlap; partial = decode current window (temperature 0.0, no re-sampling); final on stop = full-buffer decode with condition-on-previous-text; language pinned to pack language; timestamps suppressed (we render live, not timed captions).

Both pipelines are CPU/GPU-delegate configurable with a benchmark at first pack load choosing the delegate (decision cached per device, diagnostics-visible). Numeric defaults above are the v1.0 pins; changing one is a D-nnn entry + TC-AI-001 re-run.

19.13 Model cards (what each pack is and is not)

ocr-en (DBNet-lite + CRNN int8). Trained for printed Latin text 8–40 pt on plain backgrounds. Strong: books, letters, screenshots of text. Weak: handwriting, dense tables (columns linearize), stylized fonts, low light — hence confidence marks and the fix pass (§19.2). Not a translator, not a layout parser.

asr-en (whisper int8). Single-channel near-field speech en-US/en-IN. Strong: dictation pace, meeting proximity. Weak: far-field, heavy accents beyond training, music. Verbatim by design (no silent cleanup) — tidy is explicit (§19.3).

llm-gemma2b (gemma-2B-it q4_0). Short-form assist: summaries ≤ 6 bullets, rewrites ±20%, two-sentence continuations, grammar lists. Not: long-form generation, citations, facts from outside the text (prompts forbid invention; TC-AI-010 rubric enforces). Tier A-only; thermal descends to extractive.

embed-minilm (MiniLM-L6 int8). English semantic similarity for "related" ranking. Not a search engine replacement — it ranks *below* lexical hits (§17.6), never above; wrong-language packs must not be installed over en corpora (language field checked at load).

Each card ships inside its pack manifest (`about` field) and renders in the pack manager detail sheet — the owner reads what they installed without leaving the app.

19.14 Dictation quality expectations per locale (set at v1.0, honestly)

en-US/en-IN (pack + system rungs): WER ≤ 15% near-field (fixture V.2). hi-IN (system rung only in v1.0 — no hi ASR pack; shelf item): WER targets are system-dependent, so the UI promises *not* accuracy but liveness (partials + explicit insert); the pack manager marks hi ASR "coming with pack v1.x" rather than shipping a weak model (§30.2). This is the honesty pattern for every AI capability: ship where the bar is provable, label where it isn't, never fake confidence.

19.15 Assistant edge-case matrix (each cell has a behavior + test)

Input	Behavior	Test
selection > 4,000 words	auto-chunk summarization (per-chunk bullets, merged, labeled "long text, chunked")	TC-AI-004-b
empty / 1-word document	verbs hidden; sheet explains "write a little first"	TC-AI-009-b
RTL (Hebrew/Arabic paste)	rewrite preserves direction; preview renders RTL	TC-AI-004-c
text inside tables	verbs operate on the selected cell's blocks only	TC-AI-004-d
code blocks	excluded from Summarize/Fix (not prose); Continue skips code	TC-AI-004-e
mixed-language selection	method label notes dominant language; no translation ever	TC-AI-004-f
offline-from-first-launch (never had packs)	every AI surface hidden with pack reason; no crashes, no empty sheets	TC-ENV-004-b
generation while typing elsewhere	generation continues in card; insertion only after explicit tap; no caret steal	TC-AI-006-b

PART IX

SECURITY AND PRIVACY

The threat model, structural no-network proofs, storage encryption, clipboard and asset hygiene, the security suite. OneDoc's privacy story is one sentence because its architecture makes it true.

Chapter 20 — Security and privacy

20.1 Threat model (one device, one user)

Adversaries: (a) network attackers — moot by architecture, proven anyway; (b) other apps on the device reading OneDoc's files — mitigated by app-private storage + encryption-at-rest; (c) malicious input files — the real threat: crafted HTML/DOCX/odoc payloads; (d) physical access to an unlocked phone — out of scope beyond app-private storage; (e) OneDoc itself exfiltrating — impossible by construction, proven by manifest + runtime probes. Not in threat model: device root, screen recording, system-level compromise (documented non-goals).

20.2 Structural no-network (C1, C2, C3)

Layered proofs, each independently tested:

1. **Manifest gate:** no `INTERNET`, no `ACCESS_NETWORK_STATE` in the merged release manifest — CI fails the build otherwise (§22.4). Note the consequence honored throughout the codebase: without `INTERNET`, *no* library can open a socket that succeeds; even a hostile dependency is sandboxed by the platform.
2. **Dependency deny-list** (§10.3) with resolution-graph audit on every build.
3. **Runtime probe:** direct-APK QA installs the release APK on an emulator with no network and exercises every flow (all journeys + all AI flows); TC-SEC-P-03 asserts zero socket syscalls (strace-backed logcat check) and zero `UnknownHost`-class attempts.
4. **String/asset scan:** release APK contains no URLs outside license files and the About screen's repo link (allow-listed, plain text, never fetched).

20.3 Storage

Documents, versions, assets: app-private internal storage; Keystore AES-256-GCM directory encryption where a lock-screen credential exists (§11.5). Backups: passphrase-encrypted zips in user-chosen SAF locations. Cache: export/share temp files are written to cache, handed via FileProvider (grant-per-URI, read-only), and deleted after share cancel/completion (`FileProvider` path hygiene tested by TC-SEC-IO-02). Logs: diagnostics are in-memory ring buffer, 200 lines, never written to world-readable storage, never leave the device.

20.4 Input hygiene

Clipboard: OneDoc reads clipboard only on explicit paste; never reads it in background; Android 13+ clipboard-access toast is therefore never triggered spuriously (TC-SEC-P-04). Images: EXIF GPS and serial stripped on import (§17.2). HTML import: whitelist tokenizer; event handlers, `javascript:` URIs, external resource references are structurally impossible to preserve (TC-SEC-IO-01). DOCX/odoc: zip-bomb guard (entry count + ratio caps), path-traversal guard on entry names, entity-expansion caps. Audio (dictation): in-memory only, dropped after insert; no dump even in debug builds.

20.5 Permissions ledger

The complete permission set of release `app.onedoc`: `CAMERA` (scan), `RECORD_AUDIO` (dictation), `READ_MEDIA_IMAGES`/`READ_MEDIA_VISUAL_USER_SELECTED` (image insert), `VIBRATE` (haptics), `POST_NOTIFICATIONS` (export progress only, F-031 guarantees, UX-31). Nothing else. Each permission is requested at feature first-use (UX-32), each maps to a catalogue row (SEC area), and the manifest test asserts this exact set — an added permission fails CI.

20.6 The security suite (TC-SEC-*)

Suite	Cases	Proof
TC-SEC-P-*	privacy probes: network silence, notification silence, clipboard hygiene	direct-APK runs
TC-SEC-IO-*	hostile files: script-laden HTML, zip-bomb odoc, traversal-name pack, oversized DOCX	JVM + instrumentation
TC-SEC-ST-*	storage: FileProvider grants, cache cleanup, encryption-at-rest smoke (keystore present)	instrumentation
TC-SEC-PK-*	packs: wrong SHA, tampered zip, path-traversal entry name	JVM + instrumentation

Security defects are S1 by definition (§25.3). The suite is part of G7 and re-runs on every release candidate.

20.7 Build and dependency integrity

Reproducibility posture: pinned Gradle wrapper + version catalog; dependency lockfiles committed (`gradle.lockfile` per configuration); `--refresh-dependencies` forbidden in CI (cache-only builds prove lock completeness). Checksum verification: Gradle verifies artifact hashes against lockfile entries; a mismatched or new transitive dependency fails the build rather than joining quietly. The dependency inventory (SBOM-lite: name/version/sha256 list generated at build) is attached to the provenance bundle (§28.2) so every APK can answer "what's inside you" without network lookups. R8 keep rules are owned in-repo and tested by the release smoke (reflection seams listed, not discovered).

PART X

QUALITY METHODOLOGY AND GATES

The gate plan G0–G8 with DoDs, the CI architecture and its budgets, the emulator matrix, the direct-APK QA pipeline that produces the evidence PDF, and the defect system. The rules here are inherited from a prior program's hard-won lessons and are not negotiable.

Chapter 21 — Quality methodology

21.1 The principle: prove on the artifact

Source-level tests (JVM, Robolectric, instrumentation) are development aids. The **proof** that the owner holds is always the same shape: a GitHub Actions run that built the **release APK**, installed it on emulators, drove the real UI, and produced an evidence PDF whose every PASS/FAIL/BLOCKED row cites a case id and carries screenshot provenance. A green publisher job is *not* a passing QA result — QA shards decide the verdict, and their failures must surface as FAIL in the report even when publishers succeed (`if: always()`).

21.2 Verdict vocabulary (normative)

Verdict	Meaning	May be counted as
PASS	the asserted behavior was exercised and held	PASS
FAIL	the asserted behavior was exercised and broken	FAIL — a defect row is opened
BLOCKED	the environment (emulator, accessibility tree, input synthesis) prevented a fair assertion	neither — re-triaged every cycle

BLOCKED doctrine (inherited, hard rule): a blocked gesture — e.g. a stationary hold-then-drag that `adb input` cannot reproduce — is never evidence of a product defect and never evidence of health. It is logged with reason + environment fingerprint and re-tried in later cycles with new tooling. BLOCKED counts never silently shrink without a defect id or a tooling-change decision (D-nnn).

21.3 The test pyramid, budgeted



Figure 14. *Four layers: JVM, Robolectric, instrumentation, and the direct-APK proof layer.*

Layer	Runtime	Scope	Budget
1 JVM	seconds	core-model/io/search: properties, goldens, parsers	≥ 60% of all cases
2 Robolectric	minutes	editor ops, journal replay, IME scripts	correctness harness
3 Instrumentation	per-PR	API 26/33/35/36: a11y, perf micro, IME fuzzer	smoke subset per PR
4 Direct-APK QA	release gate	full catalogue on the artifact	G7 + every RC

21.4 Performance budgets (enforced, not aspirational)

Metric	Budget	Measured by
Cold start → first interaction, 200 docs, Tier B	< 1.2 s	TC-PERF-001 macrobenchmark + APK run
Typing latency P95, 10k-word doc	< 50 ms	TC-PERF-002 frame metrics
Full-text search, 200 docs / 1 MB	< 300 ms	TC-SRCH-001
Library scroll, 200 docs	jank < 1% frames	macrobenchmark
APK size (no packs)	≤ 25 MB	CI size gate
Export 50-page PDF	< 10 s	TC-IO-001 timed

Regression = budget exceeded on the emulator matrix → S2 defect. Optimizations ship with before/after numbers in the decision ledger.

21.5 Gates G0–G8

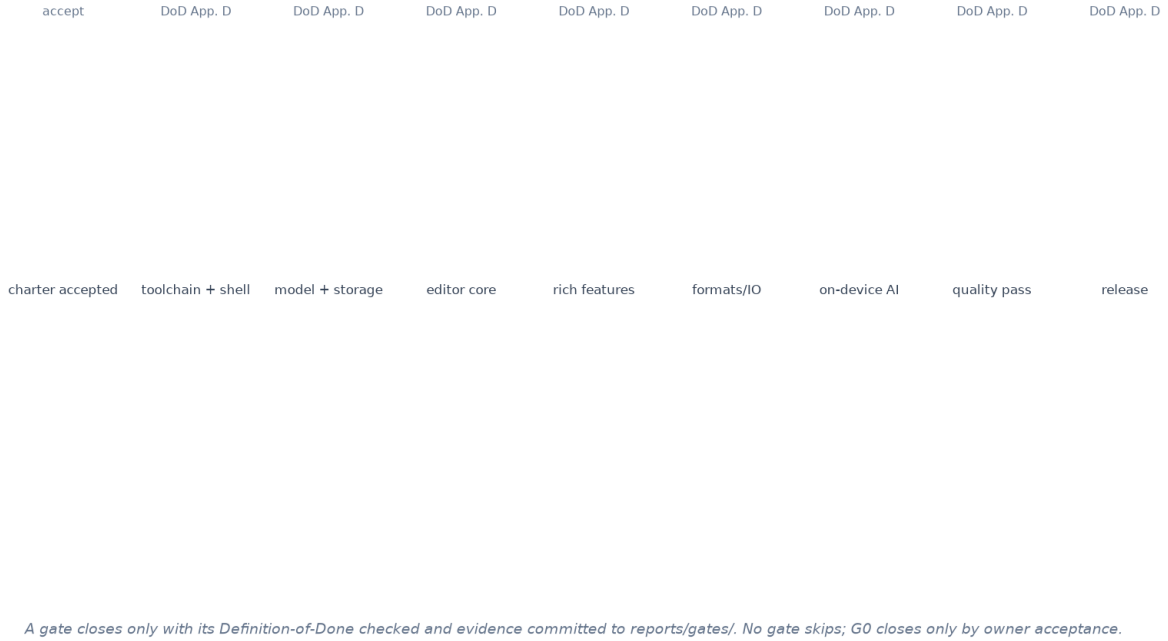


Figure 15. *The gate line: G0 charter acceptance through G8 release; each closes with its DoD evidence.*

Gate	Name	Definition of Done (abbreviated — full DoD in Appendix D)
G0	Charter accepted	owner accepts this blueprint; gate report updated; task graph seeded
G1	Toolchain & shell	SDK 36/emulators pinned; both build flavors of an empty shell assemble; CI pipeline skeleton green ≤ 30 jobs
G2	Model & storage	core-model + core-storage done; property suite green; schema v3 exported; kill-trial harness passes locally
G3	Editor core	block editing, IME contract, undo, selection; property P1–P10 green; typing budget met on Tier B emulator
G4	Rich features	tables, images, templates, find/replace, versions UI; laws L1–L10 suites green

Gate	Name	Definition of Done (abbreviated — full DoD in Appendix D)
G5	Formats & IO	all exporters/importers; round-trip properties; PDF export pagination parity; conversion notes
G6	On-device AI	scan/dictation/assistant/packs behind feature flags; ladder honored; pack tests green; network-silence probe green
G7	Quality pass	full catalogue on release APK across matrix; journeys J1–J5 end-to-end; security suite; performance budgets; zero open S1/S2
G8	Release	signed artifacts, release notes, evidence PDF, defect tracker final state, provenance bundle

Gate discipline: a gate closes only with its DoD checked and evidence in `reports/gates/G<k>.md`; failing DoD blocks later-gate tasks (no debt sneaking forward); any gate may reopen a closed earlier gate explicitly.

Chapter 22 — CI architecture

22.1 Budgets (C8, hard)

Maximum **3 parallel runners**, maximum **30 jobs per pipeline run**. Workflows achieve full-matrix coverage by sharding the matrix across stages (build once, reuse artifact) rather than multiplying jobs; the nightly pipeline may use all 30; per-PR pipelines target ≤ 8. Quota protection: concurrency groups cancel superseded runs (`concurrency: group=branch, cancel-in-progress`).

22.2 Pipelines

Pipeline	Trigger	Content
`pr.yml`	every PR	assemble debug + unit (JVM) + Robolectric + lint/token/manifest gates; ≤ 8 jobs
`instrumented.yml`	merge to main + nightly	emulator matrix smoke (API 26/33/35/36): editor smoke, IME fuzzer, a11y audit
`onedoc-apk-qa.yml`	release candidate / weekly / manual	the direct-APK QA (§24): build release → gates → install → drive → report PDF ≤ 30 jobs

22.3 Publisher separation

Report/publish jobs (PDF assembly, release notes) run `if: always()` so evidence exists even for red runs — and the report's verdict banner reflects the QA shards, never the publisher's exit code. This separation is inherited from the predecessor program where a green publisher once masked real failures; it is codified here so it can never regress.

22.4 The cheap gates that catch the expensive mistakes

Manifest permission gate (exact set, §20.5) · deny-list dependency audit · token lint (DQ-1) · APK size gate · schema-freeze check (room schema JSON diff requires a D-nnn entry) · swear-jar check for banned APIs (main-thread IO in debug strictmode tests).

Chapter 23 — Test environments

23.1 Emulator matrix

AVD	API	Device class	Role
`od-api26`	26 (Android 8.0)	Pixel 2, 2 GB	minimum-spec compatibility (C6)
`od-api33`	33	Pixel 4, 3 GB	Tier B baseline for budgets
`od-api35`	35	Pixel 6, 4 GB	Tier B+ / permissive UX checks
`od-api36`	36	Pixel 8, 8 GB	Tier A: full AI incl. LLM pack

Host pins: platform `platforms;android-36`, build-tools `36.0.0`, command-line tools current stable; emulator images google_apis (no play store — deterministic). Tier emulation uses AVD RAM profiles; the LLM pack test runs only on `od-api36`.

23.2 What runs where

P0 catalogue cases run on all four; PERF on 33 (Tier B baseline); AI pack flows on 36 + 26 (graceful degradation proof); IME fuzzer on 26/33/35/36; a11y audit on 33/36. Known-allowed BLOCKEDs (from the predecessor's doctrine) are pre-registered in Appendix G with their environment fingerprints.

23.3 Determinism and hygiene

Emulators boot from fixed snapshots with cleared app data per run; clocks pinned for golden tests; locale en-US default, hi-IN run for the owner-locale set; screenshots captured at deterministic view states, embedded into the evidence PDF with run/case metadata stamped on the image corner.

Chapter 24 — The direct-APK QA pipeline

24.1 Shape

`onedoc-apk-qa.yml`: (1) build signed release APK (provenance: commit, run id, SHA-256); (2) static gates (manifest, deny-list, size); (3) matrix jobs: install APK → grant nothing → run catalogue shards via instrumentation driving the real UI through `UiAutomation`/accessibility APIs; (4) per-case records PASS/FAIL/BLOCKED + screenshots + logs; (5) publisher job assembles `reports/onedoc-qa-<version>.pdf` with embedded screenshots and provenance; (6) defect-tracker diff job proposes new rows.

24.2 Evidence PDF (the owner's artifact)

Structure: cover (version, commit, run link, SHA-256) → verdict banner (counts by severity/verdict) → journey results with screenshot strips → per-case table (id, area, verdict, evidence link) → BLOCKED register → defect summary → methodology appendix. Every screenshot carries a corner stamp: build SHA + case id + timestamp. The PDF is committed to `reports/` and linked from the release.

24.3 Driving rules

Emulator driving uses the accessibility tree as the source of truth (resource ids are stable contract surfaces — `od_*` prefix); input synthesis accepts its own limits (§21.2). Flaky-once = rerun with same seed; flaky-twice = quarantine with defect id. Assertion failures include the accessibility snapshot dump — no "element not found" without the tree attached.

Chapter 25 — Defect management

25.1 Flow

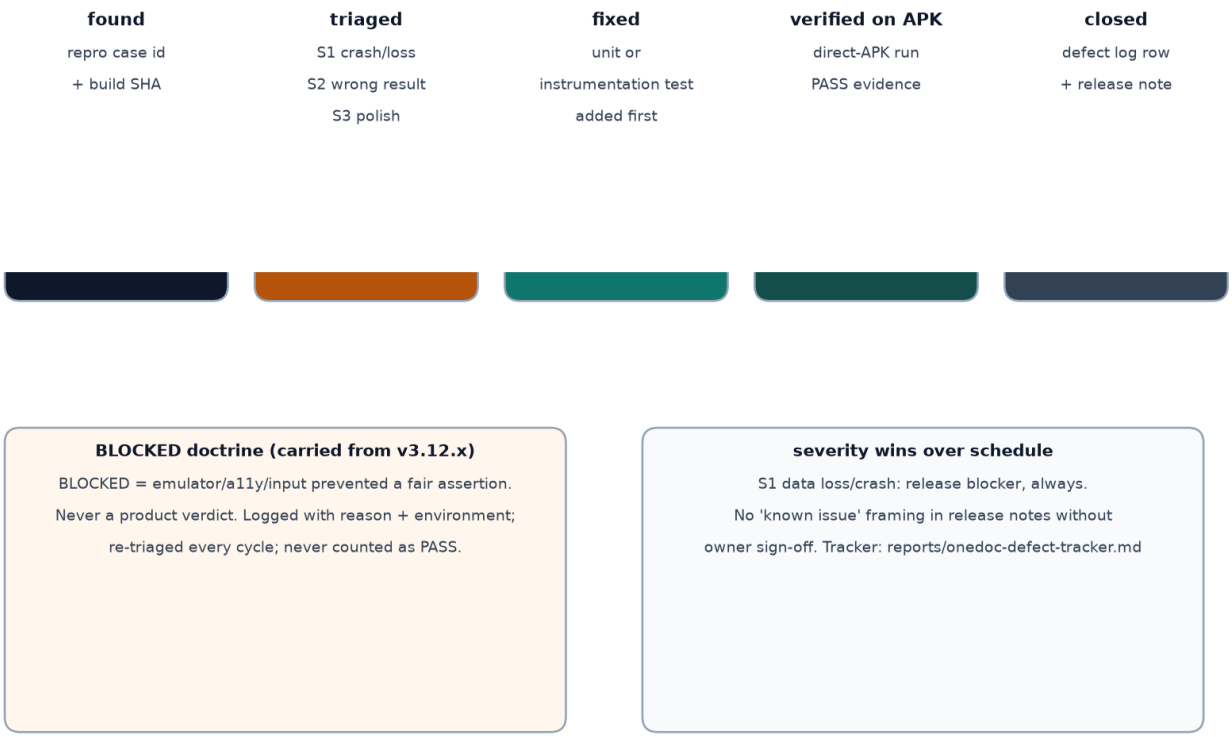


Figure 16. *Defect flow: found → triaged → fixed (test first) → verified on APK → closed.*

Defects live in `reports/onedoc-defect-tracker.md` (append-only rows; the predecessor's tracker format and history are preserved untouched in the same directory). Every row: id (OD-001…), source case id, build SHA, severity, repro, evidence link, status, fix commit, verified-on run.

25.2 Severity

S1 — data loss/corruption, crash, law violation, security; release blocker always. **S2** — wrong result, budget regression, broken P0 UX criterion. **S3** — polish, DQ design failures. Fix rule: a fix lands only with a test that failed first (regression test) — no test, no fix.

25.3 Triage categories

S-EDIT (engine), S-IME (input), S-IO (formats), S-AI (models/ladder), S-UI (design), S-PERF (budgets), S-A11Y, S-SEC (always S1), S-STATE (durability), S-CREW (scope creep — a feature nobody asked for; triaged by *removing* it, not fixing it, per §1.4).

24.4 Evidence-PDF schema (normative sections)

#	Section	Required fields
1	Cover	app, version, versionCode, commit SHA, workflow run id+URL, APK SHA-256, date, catalogue version
2	Verdict banner	counts by verdict (PASS/FAIL/BLOCKED), by area; overall GO/NO-GO with rule citation (§21.2)
3	Journeys	J1–J5 step tables with per-step screenshot strips
4	Case results	full table: case id, area, level, AVD, verdict, evidence ref (screenshot/log), duration
5	Performance	budget table actuals (§21.4) with run links
6	BLOCKED register	each block: case, reason, environment fingerprint, first-seen, retries
7	Defect summary	new/open/closed since last report, tracker link
8	Method appendix	matrix description, driving rules, tool versions

Layout rule: screenshots are embedded (not linked) at 2-up per row with corner stamps (§23.3); the PDF is self-contained — a printed copy proves as much as the run link.

24.5 Case-procedure format (how cases are written down)

Every catalogue case (Appendix C) expands in the QA harness to a procedure: `given / when / then` with exact app state, driver commands, and assertions. The harness sources procedures from `qa/onedoc/procedures/<case>.json` (generated skeleton at G3 from the catalogue; humans/agent refine assertions). A case without a procedure file cannot enter the run manifest — this is what stops "we tested it" claims that no one can replay.

25.4 Defect metrics (what the tracker reports)

Lead time to fix (median by severity), reopen rate (< 5%), BLOCKED aging (no block older than 2 cycles without a tooling-change decision), S1 count at any gate (must be 0 to pass), fix-with-test rate (100% by rule §25.2). Metrics come from tracker rows only — no side spreadsheets; the tracker is the single source (same discipline as the predecessor's tracker, which stays untouched for history).

24.6 Writing assertions that do not flake

Rules learned the hard way (predecessor cycles), applied to every procedure file:

1. **Assert on the accessibility tree, not pixels.** Text, content-desc, resource ids (`od_*`), and node states are deterministic; pixel colors are not (antialiasing, GPU quirks).
2. **Wait for states, never sleeps.** `waitUntil { nodeExists / propEquals }` with budgets; a `sleep` in a procedure is a defect in review.
3. **Pin the seed for anything random.** Content generators take seeds from the run manifest; failures replay byte-identical (§16.3).
4. **One logical assertion per wait cycle** — compound asserts report which conjunct failed (procedure format enforces assertion ids).
5. **Idempotent pre-conditions.** Procedures may assume: app installed, no prior data, permissions granted at feature use. They may not assume: time of day, locale beyond the run's pinned locale, battery level, network (there is none — assert its absence where relevant).

6. **Screenshots are evidence, not assertions** (except DQ visual checks, which diff against committed baselines with a 1% perceptual tolerance and human review on miss).

Flaky history is tracked per case in the run manifest; a case with 2 flakes in 30 days enters quarantine with a defect id (§24.3) — flakiness is a bug in the test or the app, never weather.

23.4 Device-farm notes (emulator tier fidelity)

AVD RAM profiles approximate real tiers but are optimistic on thermals and storage speed. Consequences: (1) LLM budgets measured on API 36 AVD carry a ×1.5 safety factor for physical Tier A devices before the capability is claimed in release notes; (2) thermal-descent tests (E_AI_THERMAL) are simulated via a test hook (diagnostics hidden toggle), never by cooking a core; (3) storage-full tests use a quota'd loopback path, not a filled disk. Where emulator truth is unknowable (cellular radios, real mics), the case is L5 and pre-BLOCKED (Appendix G doctrine) — honesty over theater.

23.5 Locale, timezone and layout runs

Nightly adds: hi-IN run (owner-locale strings + IME-03 fixtures), ar-RTL layout sweep (mirroring), timezone torture (clock jumps across save/snapshot boundaries — TC-STATE-006), font-scale 2.0 walk (DQ-5 evidence refresh). Tablet: 800dp/1280dp landscape sweep of home/editor/scan/settings (TC-ENV-003) — measure adaptation per §6.4(3), not stretched-phone layouts.

21.7 The gate evidence index (template every gate report follows)

```
# Gate G<k> report — <date>
Verdict: OPEN | CLOSED
DoD checklist: <App. D items with ✓ + evidence link each>
Evidence: runs <links>, artifacts <paths>, decisions <D-nnn touched>
Budgets: <table actual vs budget>
Open items: <list or "none">
Carry-over risks: <R-ids touched>
```

22.5 Quota math (how the matrix fits 30 jobs)

Full pipeline: 1 build+gates → 4 matrix QA shards (26/33/35/36) → 1 journeys shard → 1 perf shard → 1 security shard → 1 evidence publisher = 9 jobs (≤ 30 ✓) at ≤ 3 parallel by matrix-group scheduling (two waves). Nightly instrumented adds fuzzer×4 + a11y×2 + locale×2 = 8 jobs, reusing builds. PR pipeline: build+unit+lint+flavor-matrix = 6–8 jobs. The manifest publishes per-run job counts — budget breach is a visible number, not a vibe.

25.5 Defect row schema (tracker columns, exact)

`id | discovered (date, cycle) | source case | build SHA | severity (S1/S2/S3) | category (S-*) | title | repro (steps or procedure ref) | evidence (links) | status (open/in_progress/fixed_pending_verify/verified/closed/wontfix-owner) | fix commit | verified-on (run id, case) | notes`

Rules: rows append-only (status transitions edit the row in place with a dated note; never delete); `wontfix` requires owner sign-off (release note listing, §28.1); severity overrides are owner-only. The tracker renders to the release PDF via the same engine as the evidence report (one renderer, one style).

24.7 The qa-driver module (code layout)

```
qa-driver/src/main/
  pages/        # screen objects: Home, Editor, Sheets, Scan, Assistant, Settings (a11y-id bas
    ed)
  steps/        # one object per catalogue case group, methods = procedure steps
  verdict/      # VerdictRecorder: PASS/FAIL/BLOCKED + evidence paths + timings
  drivers/      # uiautomator wrappers, gesture helpers with honest limits (§21.2)
  rules/        # budgets, wait policies, screenshot corner-stamper
  manifest/     # reads Appendix U table → run manifest (shards, seeds)
```

Rules: pages never assert; steps assert via verdict recorder only; drivers expose capability queries (`canHoldDrag()` false on emulator → procedure auto-marks BLOCKED with fingerprint, §21.2) — the harness cannot lie about what it clicked.

21.8 The G7 audit script (the final acceptance walk)

The gate-7 evidence is presented in this exact order, and the owner (or a reviewer) can replay it in ~45 minutes:

1. **Cover math (5 min):** catalogue version, case count, verdict counts — every number links to the manifest.
2. **Journeys (15 min):** five strips left-to-right; each step's assertion visible; artifacts open.
3. **Laws & properties (5 min):** LAW suite green + property suite seeds/run links.
4. **Performance & size (5 min):** budget table actuals; size gate log.
5. **Security (5 min):** SEC suite + manifest permission diff (empty) + network-silence probe log.
6. **BLOCKED register (5 min):** every entry has reason + retry history + owner-relevant consequence.
7. **Defects (5 min):** tracker diff; zero open S1/S2 asserted; S3 list previewed for release notes.

A gate report not structured this way does not close G7 — the audit script is the DoD's reading order.

23.6 The IME fuzzer farm schedule

Per PR touching `core-editor`: Robolectric fuzzer (scripts ×10k steps) in CI. Nightly: full farm on 4 AVDs × script set (gboard-en-sim, samsung-sim, hindi-translit-sim, generic-AP26) × 50k steps with randomized op interleaving; seeds rotate daily; a red seed pins a defect with its replay JSON. Recording new keyboards: run the record harness on an AVD with the target IME attached, commit the JSON script — the farm picks it up next night (§15.5). The farm's verdicts roll into the nightly report next to the catalogue shards; L5 evidence always cites the farm run id.

22.6 CI security and secrets

Runners get no secrets: the RC keystore is a base64 secret injected only into the release-build job, never printed; owner keystore arrives out-of-band (§N.5) and never enters CI at all. Third-party actions are pinned by commit SHA (allow-list in the workflow lint); no `pull_request_target` on forks (this repo is single-party; the rule stays for hygiene). Logs are the leak surface to police: no APK paths with tokens, no environment dumps (a lint scans workflow logs post-run in the publisher job). Emulator snapshots carry no user data (§23.3) — they are boot-state only, rebuilt weekly from the pinned images.

24.8 Case-triage runbook (when a shard goes red)

1. Read the assertion id + attached a11y dump (never guess from the title).
2. Reproduce locally with the same seed (`dev.sh qa <case>`); if it passes locally ×3, suspect the environment → rerun shard once; still red → environment defect (S-PERF class against harness) with the shard log.
3. Real red: minimality (which step first fails), then §25 flow — defect row same day.
4. BLOCKED-path check: was the driver honest (`canHoldDrag()` etc.)? If the red is actually a blocked gesture misfiring as FAIL, fix the driver capability query — a false FAIL is as bad as a false PASS (§21.2 doctrine cuts both ways).

24.9 Reading a QA log line (the format every case prints)

```
[TC-EDIT-IME-01] avd=od-api33 build=abc1234 seed=88410
  step 3/7 wait node od_edit_body ... ok (1.2s)
  step 4/7 ime glide-replace script gboard-en-sim ... ok
  step 5/7 assert single-op-undo ... FAIL
    expected: undo count == 1 after commit batch
    actual:   undo count == 3 (ops: insert,delete,insert)
    tree: <dump attached: fails/TC-EDIT-IME-01/tree-step5.json>
  VERDICT: FAIL (defect candidate) evidence: fails/TC-EDIT-IME-01/
```

Format rules: one line per step (action + duration), assertions quote expected vs actual (never "assertion failed" bare), evidence paths relative to the run artifact root. The publisher parses this format directly into the evidence PDF's case table — the log *is* the source of the report, never retyped.

21.9 The three-question release review (the owner's shortcut)

Before reading anything else in a release's evidence, three questions with one-line answers:

1. **Is it the real artifact?** SHA in provenance = SHA of the downloaded APK.
2. **Does it work for a human?** J1–J5 strips green (the journeys are the product).
3. **Is anything hidden?** BLOCKED register + defect diff + fidelity notes — read these three before believing the green.

If all three satisfy, the release is good; the remaining 190 pages of evidence exist for the day one of them doesn't.

24.10 The evidence-PDF build script

The publisher job renders the report from structured run output (`verdict/*.json` + `journeys/*/steps.json` + screenshots) through the same PDF engine as this blueprint — one renderer, one style, zero retyping. Inputs are checksummed into the report footer (run manifest hash), so the PDF itself is verifiable against the run that produced it. A failed publisher rerun must produce byte-identical output modulo timestamps (Law L6 applied to our own tooling).

PART XI

BUILD, ACCEPTANCE AND DELIVERY

Gradle topology and flavors, the five acceptance journeys end-to-end, the release checklist, signing and the provenance bundle. This part is what "deliver the app" concretely means.

Chapter 26 — Build topology

26.1 Gradle

Gradle wrapper (pinned distribution), version catalog, AGP and Kotlin pinned per Appendix F. Modules per §10.1. Build types: `debug` (strictmode, leak-canary-free — no telemetry ever, leak canary is local-only, allowed), `release` (R8 full, resource shrinking, no debuggables). Signing: debug keystore for QA; release keystore provided by the owner at the gate-8 step (until then, release builds are signed with a CI-held keystore whose key is marked "release candidate").

26.2 Flavors (C9 — reserved)

`full` (the studio, default) and `writer` (library+editor+templates+formats only; feature modules scan/assistant/search-semantic excluded by flavor-aware DI). v1.0 ships `full`; the `writer` flavor must keep assembling in CI so the seam never rots, but is not distributed unless the owner opts in (D-reserved).

26.3 Versioning

`versionName MAJOR.MINOR.PATCH`, `versionCode` monotonic; build stamps commit SHA + run id into `BuildConfig` and the About screen (F-089). Every artifact name embeds version+code+SHA: `OneDoc-v1.0.0-<code>-<sha7>.apk`.

26.4 Size budget

APK ≤ 25 MB without packs (§21.4): enforced by a CI gate that fails the build when the release APK exceeds the budget; per-PR size delta reported in the job summary. Fonts subset; PNG assets quantized; no unused ABI splits shipped (single universal APK for direct distribution; ABI splits only if the owner requests them).

Chapter 27 — Acceptance journeys (J1 … J5, full scripts)

27.1 J1 — Blank page to formatted letter to PDF

Start: fresh install, no docs. Steps: launch (T0) → tap FAB → type salutation → select line → bold via toolbar → H1 the subject line → insert divider → type two paragraphs → menu → Export PDF → save via SAF → open PDF from files. Assert: ≤ 2 taps to editor (UX-02), formatting visible, PDF pages with correct measure, word count ±1% (UX-49), total median time under 3 min on API 33. Evidence: screenshot strip + the exported PDF attached. Case: TC-J1.

27.2 J2 — Photograph three pages into one document

Start: home. Steps: FAB → Insert → Scan → capture 3 pages (fixture images via emulator camera virtual scene) → reorder page 2 before page 1 → OCR → review screen shows low-confidence word marked → fix one word → insert. Assert: editable paragraphs in chosen order, confidence marks, images attached as assets, conversion note present, ≤ 60 s/page on API 33 (UX-09). Case: TC-J2 (OCR pack present) and TC-J2-degraded (no pack → ends at save-image with reason, ladder rung 6 UX).

27.3 J3 — Dictate minutes, tidy, share as DOCX

Steps: new doc → dictate fixture audio (emulator mic injection) → partials appear ≤ 1.5 s (UX-10) → stop → preview card labeled with method → insert → select first line → H1 → Summarize (extractive rung on API 33) → preview card labels method (UX-47) → insert below → Export DOCX → share via share sheet. Assert: docx opens in importer round-trip with headings/list parity (F2), all inserts are single-undo groups (UX-06). Case: TC-J3 (+ TC-J3-llm variant on API 36 with gemma rung labeled).

27.4 J4 — Find the paragraph about the warranty in 20 documents

Setup: seed corpus (20 docs incl. a 6-page "Appliance manual" mentioning warranty once obliquely). Steps: home search "warranty" → FTS results with snippets ≤ 300 ms (UX-08) → tap hit → editor opens at block with highlight (N5). Variant with `embed-minilm` pack: query "what covers repairs" surfaces the warranty paragraph under "related". Case: TC-J4 / TC-J4-semantic.

27.5 J5 — Recover half a deleted document

Steps: open seeded 3,000-word doc → select from mid-document to end → delete → type replacement sentence → wait 1 s → `adb shell am kill` → relaunch → assert full content minus zero words present (journal replay) → now Versions → open auto snapshot from before deletion → preview → Restore → assert pre_edit snapshot created, deletion undone, redo stack cleared. Assert: zero loss (UX-11), restore ≤ 30 s (UX-07). Case: TC-J5 (the UX-11 kill trial is TC-STATE-005's 100-trial version).

Chapter 28 — Delivery

28.1 Release checklist (G8)

1. G7 evidence green on the RC build (same SHA as release).
2. Defect tracker: no open S1/S2; S3 list accepted in release notes.
3. Journeys J1–J5 green on matrix; evidence PDF assembled and committed.
4. Manifest/deny-list/size gates green on the exact release artifact.
5. Artifacts: universal release APK (+ AAB optional), SHA-256, `.odpack` bundles documented (never shipped inside APK).
6. Release notes: features, fidelity notes, known BLOCKEDs with reasons, defect summary — no marketing language, no "known issues" framing of S1/S2 (they cannot exist at G8).
7. Tag `v<X.Y.Z>`; provenance bundle committed (§28.2).

28.2 The provenance bundle

`releases/<version>/provenance.json`: version, versionCode, commit SHA, workflow run id+URL, APK SHA-256 + size, pack manifest hashes referenced (not embedded), catalogue version (case count + commit of qa CSV), tool versions (Appendix F). The owner verifies by recomputing the APK hash — trust is a checksum away, as promised in P5/C10.

28.3 Post-release

Crash-free operation is *expected, not monitored* — there is no telemetry (C3). The owner's feedback loop is the diagnostics screen export (a local text file the owner can share manually if they choose) and the defect tracker. Version upgrades follow the same gate discipline with migration tests (§11.4) as the entry gate.

26.5 CI caching and speed hygiene

Gradle build cache + configuration cache on; wrapper distribution cached at runner level; emulator AVDs pre-booted from golden snapshots (§23.1) with app data cleared per run; artifact reuse: one release build fans out to all QA shards (no shard rebuilds). Budget: per-PR pipeline ≤ 12 min P95; nightly full matrix ≤ 75 min. Cache poisoning guards: caches keyed by lockfile hash; no cache writes from release branches' QA runs.

28.4 Owner handoff note (template, committed with the release)

```
OneDoc v<X.Y.Z> — handoff
What shipped: <features by area, 10 lines max>
Quality: <catalogue verdicts; journeys; budgets; BLOCKED register summary>
Known limits: <S3 list + fidelity notes worth remembering>
Artifacts: APK + SHA-256, evidence PDF, provenance.json, tracker link
How to verify in 5 minutes: install -> J1 walkthrough -> About self-test -> compare SHA
```

The note is deliberately owner-shaped: verification-first, jargon-light, and honest about limits — the same voice as §28.1's release notes rule.

27.6 Journey evidence layout (how J-cases document themselves)

Each journey produces a self-contained folder: `artifacts/journeys/<J-id>/` containing `steps.json` (step, action, assertion, verdict, duration) and `screens/NN-<step>.png` (corner-stamped). The evidence PDF renders journeys as a strip per step with the assertion text as caption — the owner walks the story left-to-right without reading logs. Journey steps double as the demo scripts (Appendix S) — one source, two uses, no drift.

28.5 Owner verification runbook (5 minutes, offline)

1. Download APK + `provenance.json` from the release.
2. `sha256sum OneDoc-…apk` — compare to provenance (match = artifact integrity).
3. Install; launch; run About → self-test (≤ 20 s, must pass).
4. Walk J1 (create → format → export PDF) — the letter lands in your files.
5. Optional deep check: `apkanalyzer manifest permissions <apk>` shows the §20.5 set exactly.

If any step fails, the release is not accepted — file the tracker id from the evidence PDF and stop. Trust arrives from checksums and tests, never from promises (P5, C10).

28.6 Release-day script (automated, idempotent)

1. `./scripts/release-verify.sh v<X.Y.Z>`: checks tag → build from tag → SHA match to provenance → static gates.
2. Re-run TC-J1..J5 + SEC-P-03 on the tag build (30-minute smoke; full matrix already done at G7 — the smoke proves *this* artifact).
3. `gh release create v<X.Y.Z>` with APK, evidence PDF, provenance.json, notes (Appendix M voice).
4. Post-release task graph update: gate G8 → closed; maintenance mode §30.4 armed (nightlies continue).
5. Rollback posture: a bad release is never "patched" — a new version ships through the same gates; the bad tag's release notes get an amber supersede line.

26.7 The local build and run script (dev loop)

```
./scripts/dev.sh build    # assembleFullDebug + install on first booted AVD
./scripts/dev.sh qa TC-J1 # build RC, install, run one journey, open screenshots
./scripts/dev.sh laws     # JVM law+property suite only (fast inner loop)
./scripts/dev.sh clean    # uninstall + clear data + reboot AVD state
```

The scripts are thin wrappers over the N.4 commands — same paths, same gates, so "works on my machine" and "works in CI" are the same sentence. CI invokes the identical scripts (workflow lint asserts it), keeping drift between local and CI at zero.

PART XII

WAY OF WORKING FOR THE AGENT

The operating loop the autonomous developer follows: reading order, the decision ledger and session log, the task graph, how to handle ambiguity and blockage, and the open risks the next builder should watch.

Chapter 29 — The operating loop

29.1 Reading order (first 30 minutes)

1. This Part XII (10 min): the loop, the ledgers, the blockage rules.
2. Part I (5 min): promises, constraints, out-of-scope — the constitution.
3. Part X §21–§22 (5 min): verdict vocabulary, gates, CI budgets.
4. Part V §13 (5 min): the ten laws.
5. Skim Part II catalogue + Appendix D gate DoDs (5 min): what "done" means.

Then work gate-by-gate. Do not write code for a chapter you have not read; do not open a gate whose predecessor is open.

29.2 The ledgers (append-only, always)

- **Decisions** — `docs/onedoc/plan/decisions.md`, format `D-nnn / date / context / decision / consequences`. Every non-trivial choice (dependency added, schema changed, fidelity altered, budget traded) gets one entry before the code lands. Entries are never edited; a reversed decision gets a new entry referencing the old.
- **Sessions** — `docs/onedoc/plan/sessions.md`, format `S-nnn / date / goal / done / failed / next`. One block per working session, appended at session end even when the goal failed — especially then.
- **Gates** — `reports/onedoc/gates/G<k>.md`: one report per gate, DoD checklist with evidence links, verdict, open items.

29.3 The task graph

`docs/onedoc/plan/task-graph.yaml` seeds every task with id, gate, depends-on, DoD, and case bindings. The agent expands each gate's slice just-in-time (details of G4+ are deliberately unplanned at G1 — plan-depth is a liability at 5 gates' distance), keeps statuses honest (`todo/doing/blocked/done` with evidence links), and never marks done without its DoD met.

29.4 Ambiguity and blockage

1. Blueprint first: §1.5 defaults resolve most ambiguity; record the chosen default as D-nnn.
2. Platform convention next (Material 3, Android patterns).
3. Still ambiguous and consequential (user-visible, irreversible, budget-affecting): choose the smallest reversible option, log it as D-nnn with the reopen condition, and surface it in the next gate report under "Decisions awaiting owner review".
4. Hard blockage (tool missing, quota exhausted, environment broken): mark tasks `blocked` with reason, close what remains closeable, and write the blockage into the gate report — a stalled agent that reports precisely is doing its job; a stalled agent that invents scope is not.

29.5 Quality self-checklist (every PR)

Laws touched? → property tests updated. UI touched? → registry row + DQ checklist. Format touched? → golden round-trip. Schema touched? → migration + D-nnn. Perf touched? → budget re-measured. Catalogue touched? → `traceability.py --check --strict` green. If none apply, say so in the PR body — the checklist's purpose is to force the question.

Chapter 30 — Open risks and the v1.1 shelf

30.1 Watchlist

Risk	Signal to watch	Pre-agreed response
IME diversity breaks composition invariants	fuzzer red on any keyboard	freeze features, IME-only gate (§15.3) until green
LLM RAM envelope too tight on real Tier A	gen latency > 2× budget on API 36 AVD	demote assistant to rung 4 default, keep rung 3 for ≥ 12 GB devices (D-nnn)
Pagination parity drift (guide vs PDF)	TC-STRUCT-004 off-by->1	single-metrics-engine audit; freeze font fallback changes
APK size creep	PR size delta > +0.5 MB	shrink pass before any feature merge
Figma drift	DQ-3 failures	registry re-sync task; drift report to owner
OCR accuracy below fixture bar	TC-AI-001 accuracy < 95% on fixture set	pack version bump; ladder gains "confirm each page" rung

30.2 The v1.1 shelf (recorded, not scheduled)

Markdown input shortcuts while typing · footnotes endnote mode · custom template import (.odoc as template) · neural TTS pack · additional OCR language packs (hi, mr) · writer-flavor distribution. Items graduate to the plan only by owner decision; the agent builds them only after a D-nnn says so.

29.6 Communication artifacts (what the owner sees, when)

Cadence	Artifact	Location
every session end	session log block (S-nnn)	docs/onedoc/plan/sessions.md
every decision	ledger entry (D-nnn)	docs/onedoc/plan/decisions.md
every PR	checklist-filled description	GitHub
every gate	gate report + evidence links	reports/onedoc/gates/G<k>.md
every RC	evidence PDF + tracker diff	reports/
release	release notes + handoff note + provenance bundle	GitHub Release + releases/<ver>/

Tone rule for all artifacts: facts with links, no adjectives ("green", "3 FAILs closed by OD-012" not "great progress"). The owner should be able to reconstruct the program's entire history from artifacts alone, without asking a single question.

29.7 The agent's daily loop (operating cadence)

1. **Morning:** read nightly results; red nightly = today's first task (§N.6). Check BLOCKED register for retry candidates with new tooling.
2. **Work block:** pull the top task from the graph whose dependencies are done; expand its slice just-in-time; write the test/procedure first where the type demands it (laws, formats, security).
3. **PR cadence:** small PRs (≤ 400 changed lines excluding generated), checklist-filled (§29.5), merged only green.

4. **Ledgers:** D-nnn at decision time (not end of day); S-nnn block at session end with the failed/next lines honest.
5. **Gate approach:** when a gate's DoD is within reach, freeze new scope, close evidence gaps first (evidence debt is the only debt that compounds).
6. **End of day:** task-graph statuses true; gate reports touched if evidence moved; nothing "done" without its DoD.

30.4 Post-v1.0 maintenance protocol

Between releases the program runs in "lighthouse mode": nightly pipelines keep running (emulator matrix + properties + fuzzer), dependency audit weekly (deny-list + lockfile refresh review), and every D-nnn-marked shelf item (§30.2) waits for an owner decision. A maintenance cycle re-runs the release checklist on any rebuilt artifact — no rebuild ships without evidence, including "just a dependency bump" (the SBOM and provenance regenerate; §28.2).

29.8 Asking the owner (the template for genuine blockers)

When §29.4's ladder exhausts and the matter is truly the owner's (keystore, Figma access, scope change, quota change), the agent posts ONE structured request, in the gate report under "Decisions awaiting owner review":

```
ASK-<n> (date, gate)
Need: <one sentence>
Options: A) <x — cost/risk> B) <y — cost/risk> C) do-nothing <consequence>
Recommendation: <option + why>
Workaround until answered: <what proceeds without this>
```

No repeated pings; the ask lives in the report until answered (answer becomes a D-nnn). This keeps autonomy honest: the agent never blocks silently and never nags.

29.9 What the agent must never do (the short list)

1. Never add a network permission, account, ad SDK, or telemetry — §1.3/§1.4 make such tasks invalid, not negotiable.
2. Never edit history: no rebase/force-push on program branches, no edits to closed gate reports, ledger rows, or the predecessor program's artifacts.
3. Never mark a gate/task done without its DoD evidence link.
4. Never write a user-visible string in code (keys only), a color literal in UI (tokens only), or an error ad hoc (taxonomy only).
5. Never let AI output enter a document without preview + explicit insert + pre-edit version (Law L10).
6. Never report a blocked gesture as FAIL — or as PASS (§21.2 cuts both ways).
7. Never claim calculator-grade parity claims about formats: state the fidelity level achieved (§18.2).
8. Never bundle a model into the APK (C5) or fetch anything at runtime (C1).
9. Never ship a rebuild without re-running the gates on that exact artifact (§28.6).
10. Never "temporarily" disable a test to go green — quarantine with a defect id is the only honest path (§24.3).

30.5 A note to the builder

If you are the agent holding this document: everything you need is here on purpose. The laws tell you what must never break; the catalogues tell you what must exist; the gates tell you when to stop and prove; the ledgers tell you how to remember. Build small, prove constantly, write everything down — and the owner will never need to trust you, because they can always verify you. That is the whole method.

30.6 Verifying this document (for any reader)

- Regenerate: `./.venv/bin/python tools/blueprint/build_pdf.py --content docs/onedoc/blueprint/content --out docs/onedoc/blueprint/onedoc-development-blueprint.pdf --doc-title "OneDoc · Application Development Blueprint" --brand-footer "OneDoc · Application Development Blueprint · v1.0 · offline-first / on-device / no sign-in" --cover-art docs/onedoc/blueprint/figures/cover.png`
- Regenerate figures: `./.venv/bin/python tools/onedoc/figures.py`
- Verify catalogues: `./.venv/bin/python tools/onedoc/traceability.py --check --strict` (exit 0 = 90 features, 50 UX criteria, 142 cases, zero open mappings)

APPENDICES

Machine-readable catalogues (features, UX, test cases), gate DoDs, environment and dependency pins, the BLOCKED register, and working templates. The appendices are parsed by tools/onedoc/traceability.py — keep their format exact.

Appendix A — Feature catalogue (machine table)

Format: `| F-id | Name | Priority | Bindings |`. The bindings column lists case ids (comma-separated). This table is normative; Chapter 4 is its prose mirror.

F-id	Name	Priority	Bindings
F-001	One-tap blank document	P0	TC-LIB-001, TC-UX-001
F-002	Inline rename; title mirrors first heading	P0	TC-LIB-002
F-003	Autosave via journal + debounced save	P0	TC-STATE-001, TC-LAW-001
F-004	Star/pin	P1	TC-LIB-003
F-005	Duplicate document	P1	TC-LIB-004
F-006	Trash with undo snackbar	P0	TC-LIB-005
F-007	Trash retention 30 days, restore, permanent delete	P1	TC-LIB-006
F-008	Sorting modes persisted	P2	TC-LIB-007
F-009	Empty states with CTA	P1	TC-LIB-008
F-010	Document info sheet	P2	TC-LIB-009
F-011	Multi-select bulk actions	P1	TC-LIB-010
F-012	Cold start < 1.2 s Tier B / 200 docs	P0	TC-PERF-001
F-013	Block editing with IME composition safety	P0	TC-EDIT-001, TC-EDIT-IME-01
F-014	Enter split / Backspace merge	P0	TC-EDIT-002
F-015	Undo/redo global stack, coalesced, grouped composites	P0	TC-EDIT-003, TC-PROP-P01
F-016	Selection handles + floating toolbar	P0	TC-EDIT-004
F-017	Marks set with non-overlap invariant	P0	TC-EDIT-005, TC-LAW-002
F-018	Headings H1–H3, outline-driven	P0	TC-EDIT-006
F-019	Lists: bullet, numbered, checklist	P0	TC-EDIT-007
F-020	Quote and code blocks	P1	TC-EDIT-008
F-021	Per-block alignment	P1	TC-EDIT-009
F-022	Divider block	P2	TC-EDIT-010
F-023	Paste (text/HTML subset/image), sanitized	P0	TC-EDIT-011
F-024	Cut/copy multi-block; paste round-trip	P0	TC-EDIT-012
F-025	Find & replace with pre-edit version on replace-all	P0	TC-EDIT-013
F-026	Find bar counts and navigates hits	P0	TC-EDIT-014
F-027	Live word/char count + info sheet	P1	TC-EDIT-015
F-028	Hardware keyboard shortcuts	P1	TC-EDIT-016
F-029	Typing P95 < 50 ms; no burst frame drops	P0	TC-PERF-002
F-030	Orientation/process-death state restore	P0	TC-STATE-002
F-031	Tables: structure ops, marks in cells	P0	TC-TBL-001
F-032	Table width presets, wrap, no horizontal scroll	P0	TC-TBL-002
F-033	Image insert (gallery/camera), captions, viewer	P0	TC-IMG-001
F-034	Image resize presets + alt text	P1	TC-IMG-002
F-035	TOC block from headings	P1	TC-STRUCT-001
F-036	Document map sheet	P2	TC-STRUCT-002
F-037	Page setup (paper/margins)	P1	TC-STRUCT-003

F-id	Name	Priority	Bindings
F-038	Page-break guides parity with PDF ±1 line	P1	TC-STRUCT-004
F-039	Footnotes (editor + PDF)	P2	TC-STRUCT-005
F-040	Cover block	P2	TC-STRUCT-006
F-041	Drag-to-reorder blocks + a11y equivalent	P1	TC-STRUCT-007
F-042	Duplicate/delete block	P1	TC-STRUCT-008
F-043	Code block language label	P2	TC-STRUCT-009
F-044	Asset manager, unused cleanup	P2	TC-IMG-003
F-045	12 built-in templates	P0	TC-TPL-001
F-046	Template preview	P1	TC-TPL-002
F-047	Template instantiate-as-copy	P0	TC-TPL-003
F-048	Placeholders incl. auto date	P1	TC-TPL-004
F-049	Save-as-template from documents	P1	TC-TPL-005
F-050	Template gallery states, user section	P2	TC-TPL-006
F-051	PDF export (Print framework, embedded fonts)	P0	TC-IO-001, TC-STRUCT-004
F-052	.odoc export lossless	P0	TC-IO-002, TC-LAW-001
F-053	Markdown/txt/HTML export per fidelity	P0	TC-IO-003
F-054	DOCX export (OOXML subset)	P1	TC-IO-004
F-055	Import odoc/md/txt/html/docx with notes	P0	TC-IO-005
F-056	Refuse unsupported with reason	P0	TC-IO-006
F-057	Share via export + FileProvider	P0	TC-IO-007, TC-SEC-IO-02
F-058	Open-in intent filters	P1	TC-IO-008
F-059	SAF import/export	P0	TC-IO-009
F-060	Encrypted weekly backup to SAF (off default)	P2	TC-IO-010, TC-IO-013
F-061	Versions: auto/manual/pre-edit policy	P0	TC-STATE-003
F-062	Version preview + non-destructive restore	P0	TC-STATE-004
F-063	Journal durability (kill -9 zero loss)	P0	TC-STATE-005, TC-LAW-003
F-064	Storage accounting + asset cleanup	P2	TC-STATE-006
F-065	Schema migration chain tested	P0	TC-STATE-007
F-066	Multi-window save-on-focus-loss	P1	TC-STATE-008
F-067	FTS search < 300 ms / 200 docs	P0	TC-SRCH-001
F-068	Deep-link to hit block with highlight	P0	TC-SRCH-002
F-069	Shared highlight system with find bar	P1	TC-SRCH-003
F-070	Semantic search when pack present, labeled	P2	TC-SRCH-004
F-071	Scan-to-doc with confidence marks	P0	TC-SCAN-001, TC-AI-001
F-072	Multi-page scan queue + reorder	P1	TC-SCAN-002
F-073	OCR on-device only; graceful absence	P0	TC-AI-002
F-074	Dictation with live partials, explicit insert	P0	TC-AI-003
F-075	Assistant verbs with preview cards	P0	TC-AI-004
F-076	gemma-2B Tier A; extractive fallbacks elsewhere	P0	TC-AI-005
F-077	Streamed, cancellable, pre-edit snapshotted	P0	TC-AI-006
F-078	Pack manager: side-load, SHA verify, delete	P0	TC-AI-PACK-01
F-079	Corrupt pack refused; ladder descent	P0	TC-AI-PACK-02
F-080	Read aloud with sentence highlight	P1	TC-AI-007, TC-READ-001

F-id	Name	Priority	Bindings
F-081	Read-aloud ranges + system voice/rate	P2	TC-AI-008
F-082	Unavailable AI hides with reason	P0	TC-AI-009
F-083	Reader mode	P1	TC-READ-001
F-084	TalkBack block traversal + actions	P0	TC-A11Y-001
F-085	Contrast ≥ 4.5:1 all themes	P0	TC-A11Y-002
F-086	200% dynamic type, no clipping	P0	TC-A11Y-003
F-087	8 themes, match-system default	P1	TC-SET-001
F-088	Editor settings (font/size/spell)	P1	TC-SET-002
F-089	About + diagnostics + self-test	P1	TC-SET-003, TC-DIAG-001
F-090	Locales en/hi/mr, RTL mirroring	P2	TC-SET-004

Appendix B — UX acceptance criteria (machine table)

UX-id	Criterion (short)	Bindings
UX-01	FAB → editor directly, doc on first keystroke	TC-UX-001
UX-02	Blank→first char ≤ 2 taps, ≤ 2 s	TC-UX-001, TC-PERF-001
UX-03	Export PDF ≤ 3 taps	TC-UX-002
UX-04	Import ≤ 2 taps	TC-UX-003
UX-05	Find bar remembers term, clean dismiss	TC-UX-004
UX-06	AI/paste inserts undo as one group	TC-UX-005, TC-EDIT-003
UX-07	Restore ≤ 30 s / 50 pages	TC-STATE-004
UX-08	Search results from ≤ 2 chars	TC-SRCH-001
UX-09	Scan page → text ≤ 60 s Tier B	TC-SCAN-001
UX-10	Dictation first partial ≤ 1.5 s	TC-AI-003
UX-11	kill mid-typing: zero loss	TC-STATE-005
UX-12	Rotation/split keeps cursor + composition	TC-STATE-002
UX-13	No save dialogs, autosave silent	TC-STATE-001
UX-14	Trash undo 7 s; 30-day retention	TC-LIB-005, TC-LIB-006
UX-15	Failed import: no empty doc, clipboard intact	TC-IO-006
UX-16	Replace-all: confirm + pre-edit version + undo	TC-EDIT-013, TC-STATE-004
UX-17	Storage-full: in-memory continue, auto retry	TC-STATE-009
UX-18	Export failure: no partial file, specific reason	TC-IO-011
UX-19	Toolbar never covers selection	TC-UX-006
UX-20	Targets ≥ 48 dp (44 toolbar)	TC-UX-006
UX-21	System-standard long-press select; stylus/mouse	TC-EDIT-004
UX-22	IME composition: no flicker/dup/loss	TC-EDIT-IME-01
UX-23	Enter continues list; exits on empty	TC-EDIT-007
UX-24	Style switch preserves marks	TC-EDIT-006
UX-25	Scroll anchored during IME show/hide	TC-UX-007
UX-26	No whole-doc reflow while typing	TC-PERF-002
UX-27	Scroll mid-composition cancels cleanly	TC-EDIT-IME-02
UX-28	Hardware shortcuts match platform; undo parity	TC-EDIT-016
UX-29	AI only via preview + explicit insert	TC-AI-006
UX-30	Unavailable AI says why in one line	TC-AI-009
UX-31	No notifications beyond user-created/progress	TC-SEC-P-02
UX-32	One-card first launch; permissions at use	TC-UX-008
UX-33	Delete warns version count; trash double-confirm	TC-LIB-006
UX-34	Diagnostics on-device only, self-test ≤ 20 s	TC-SET-003, TC-DIAG-001
UX-35	Destructive: haptic + undoable/double-confirm	TC-UX-009
UX-36	Empty states one line one action	TC-LIB-008
UX-37	TalkBack-complete letter flow	TC-A11Y-001
UX-38	Alt-text prompted at insert	TC-IMG-001
UX-39	200% scale: no clipping	TC-A11Y-003
UX-40	Dark + high-contrast themes pass contrast	TC-SET-001

UX-id	Criterion (short)	Bindings
UX-41	Color never sole carrier	TC-AI-001
UX-42	Primary actions bottom 40%	TC-UX-010
UX-43	RTL mirroring correct	TC-SET-004
UX-44	Reduced-motion honored	TC-UX-011
UX-45	Import states achieved fidelity	TC-IO-005
UX-46	OCR low-confidence marked/fixable; export honest	TC-AI-001
UX-47	AI results label method	TC-AI-004, TC-AI-005
UX-48	Pack install shows SHA; failure shows both	TC-AI-PACK-01, TC-AI-PACK-02
UX-49	Word count matches export ±1%	TC-IO-001
UX-50	No marketing/network content ever	TC-SEC-P-03

Appendix AN — document.json schema reference (field tables)

Root: `schema:int` (3) · `docId:string` (`d_`+8 hex) · `meta{title, created, modified?, wordCount, pageSetup{paper:A4|Letter, margin:normal|narrow|wide}}` · `blocks:Block[]`.

Block (all kinds): `id:string` (`b`+n, unique) · `type:Kind` · optional `align:left|center|right|justify` (non-heading). Per kind:

Kind	Extra fields
para	`text:string`, `marks:Mark[]`
h1/h2/h3	`text`, `marks`
list	`listType:bullet|number|check`, `items:[{id, text, marks, checked:bool|null}]`, `level:int (0–4)`
quote	`text`, `marks`
code	`text` (marks empty, enforced), `language:string?`
table	`header:bool`, `cols:[{width:equal|content}]`, `rows:[{id, cells:[{id, blocks:Block[] (depth 1: para/list only)}]}]`
image	`assetId`, `size:S|M|L|full`, `alt:string?`, `caption:blockId?`
divider	—
toc	`depth:1|2|3` (rendered, not stored per entry)
footnote	`text`, `marks`
cover	`title`, `subtitle?`, `date?`

Mark: `type:strong|em|u|strike|code|highlight|link` · `start:int` · `end:int` (half-open, UTF-16 units) · `href:string?` (link only).

Validation (I1–I4, §14.2) runs on open, on import, and after every op batch in debug. The schema is versioned by `schema`; migrations map N→N+1 only (§11.4); unknown `type` values are preserved verbatim (L8) with round-trip tests (TC-LAW-004).

Appendix C — Test catalogue (case tables, part 1)

Format per area: `| Case | Binding | Proves | Level |`. Levels: L1 JVM, L2 Robolectric, L4 instrumentation/emulator, L5 physical-only (pre-registered BLOCKED otherwise), G = CI gate. The catalogue is parsed by `tools/onedoc/traceability.py`; `--check` enforces that every F/UX binding above resolves to a case here.

Laws (L)

Case	Binding	Proves	Level
TC-LAW-001	F-003, F-052	persisted doc equals on-screen text+marks after autosave	L2/L4
TC-LAW-002	F-017	normalize restores non-overlap; idempotent	L1
TC-LAW-003	F-063	deterministic engine: same ops → identical results + export bytes	L1
TC-LAW-004	F-052, F-055	unknown block kinds round-trip verbatim	L1/L4

Library (LIB)

Case	Binding	Proves	Level
TC-LIB-001	F-001	FAB opens editor; doc created on first keystroke	L4
TC-LIB-002	F-002	rename inline; untitled mirrors first heading	L2/L4
TC-LIB-003	F-004	star pins to top on All	L2/L4
TC-LIB-004	F-005	duplicate copies content+assets, new id	L2
TC-LIB-005	F-006	trash + 7 s undo snackbar restores	L4
TC-LIB-006	F-007	retention timer, restore, permanent delete confirm	L2/L4
TC-LIB-007	F-008	sort modes persist across launch	L2/L4
TC-LIB-008	F-009, UX-36	empty states render with CTA	L4
TC-LIB-009	F-010	info sheet numbers correct (words/chars/size)	L2/L4
TC-LIB-010	F-011	multi-select actions on 3 docs	L4

Editor core (EDIT)

Case	Binding	Proves	Level
TC-EDIT-001	F-013	per-block typing; caret behavior at block edges	L4
TC-EDIT-002	F-014	split/merge preserves marks per rules	L2/L4
TC-EDIT-003	F-015, UX-06	undo/redo semantics incl. grouped composites	L2
TC-EDIT-004	F-016, UX-21	selection handles, toolbar actions, stylus/mouse	L4
TC-EDIT-005	F-017	each mark type on/off/overlap normalization	L2
TC-EDIT-006	F-018, UX-24	heading switches keep marks; outline updates	L2/L4
TC-EDIT-007	F-019, UX-23	list continue/exit, renumber, checkbox toggle	L2/L4
TC-EDIT-008	F-020	quote/code blocks; code disables marks	L2/L4
TC-EDIT-009	F-021	four alignments render and persist	L2/L4
TC-EDIT-010	F-022	divider insert/delete semantics	L2
TC-EDIT-011	F-023	paste text/HTML subset/image, sanitized	L2/L4
TC-EDIT-012	F-024	multi-block cut/copy/paste exact round-trip	L2/L4
TC-EDIT-013	F-025, UX-16	find/replace options; replace-all versions first	L2/L4
TC-EDIT-014	F-026	hit count, navigation, non-destructive highlight	L2/L4
TC-EDIT-015	F-027	live counts correct on edit/undo	L2
TC-EDIT-016	F-028, UX-28	hardware shortcuts + undo parity	L4
TC-EDIT-017	F-013	emoji/surrogate boundaries edit atomically	L2

IME (EDIT-IME)

Case	Binding	Proves	Level
TC-EDIT-IME-01	F-013, UX-22	glide/autocorrect full-replace lands as one op	L4
TC-EDIT-IME-02	UX-27	scroll mid-composition cancels cleanly	L4
TC-EDIT-IME-03	F-013	Devanagari clusters never split	L2/L4
TC-EDIT-IME-04	F-013	IME fuzzer: laws hold under random valid sequences	L2/L4

Property suite (PROP) — `core-model`/`core-editor`

Case	Binding	Proves	Level
TC-PROP-P01	F-015	undoAll(apply(D,O)) == D	L1
TC-PROP-P02	F-017	normalize idempotent	L1
TC-PROP-P03	F-003	plainText integrity, length sums	L1
TC-PROP-P04	F-014	split/merge inverses	L1
TC-PROP-P05	F-041	move/remove preserve ids, no holes	L1
TC-PROP-P06	F-031	table ops never orphan cells; depth-1	L1
TC-PROP-P07	F-052, F-055	export/import round-trip per fidelity	L1
TC-PROP-P08	F-063	journal replay == in-memory application	L1
TC-PROP-P09	F-062	restore leaves valid cursor	L1
TC-PROP-P10	F-013	random Unicode planes never corrupt offsets	L1

State & durability (STATE)

Case	Binding	Proves	Level
TC-STATE-001	F-003, UX-13	silent autosave; exit never asks	L2/L4
TC-STATE-002	F-030, UX-12, L9	rotation/split restore cursor+selection	L4
TC-STATE-003	F-061	snapshot policy fires at 5 min/50 ops/pre-edit	L2/L4
TC-STATE-004	F-062, UX-07	preview + restore non-destructive, ≤ 30 s	L4
TC-STATE-005	F-063, UX-11	100 kill trials: ≤ 2 s loss bound (in practice 0)	L4
TC-STATE-006	F-064	storage accounting matches files; cleanup safe	L2
TC-STATE-007	F-065	migration chain v1→vN on seeded corpora	L4
TC-STATE-008	F-066	split-screen focus-loss saves	L4
TC-STATE-009	UX-17	storage-full: memory continue, auto retry	L2/L4

Tables / Images / Structure

Case	Binding	Proves	Level
TC-TBL-001	F-031	table ops + marks in cells + editing	L2/L4
TC-TBL-002	F-032	wrap, presets, no horizontal doc scroll	L4
TC-IMG-001	F-033, UX-38	insert flows, EXIF strip, alt-text prompt, viewer	L4
TC-IMG-002	F-034	resize presets + alt-text editor	L2/L4
TC-IMG-003	F-044	asset manager, unused cleanup with confirm	L2/L4
TC-STRUCT-001	F-035	TOC generation, scroll, refresh	L2/L4
TC-STRUCT-002	F-036	map sheet live position	L4
TC-STRUCT-003	F-037	page setup changes measure + export	L2/L4
TC-STRUCT-004	F-038, F-051	guide/PDF pagination parity ±1 line	L2/L4
TC-STRUCT-005	F-039	footnotes editor + PDF	L2/L4
TC-STRUCT-006	F-040	cover block renders + exports	L2/L4
TC-STRUCT-007	F-041	drag reorder + a11y move actions	L4
TC-STRUCT-008	F-042	duplicate/delete block	L2/L4
TC-STRUCT-009	F-043	code language label persists	L2

Appendix C — Test catalogue (case tables, part 2)

Templates (TPL)

Case	Binding	Proves	Level
TC-TPL-001	F-045	12 built-ins listed, each instantiates valid doc	L2/L4
TC-TPL-002	F-046	preview renders read-only, placeholders marked	L4
TC-TPL-003	F-047	instantiate copies; template file unmutated	L2
TC-TPL-004	F-048	placeholder fill incl. locale-aware date	L2/L4
TC-TPL-005	F-049	save-as-template round-trip	L2/L4
TC-TPL-006	F-050	user section delete; gallery states	L4

Files & formats (IO)

Case	Binding	Proves	Level
TC-IO-001	F-051, UX-49	PDF export: fonts, selection text, pagination, count ±1%, < 10 s	L4
TC-IO-002	F-052	odoc lossless round-trip incl. assets	L1/L4
TC-IO-003	F-053	md/txt/html exports at declared fidelity	L1/L4
TC-IO-004	F-054	docx export valid per OOXML subset; re-import parity	L1/L4
TC-IO-005	F-055, UX-45	imports with correct fidelity + conversion notes	L2/L4
TC-IO-006	F-056, UX-15	unsupported refused with reason; no empty doc	L2/L4
TC-IO-007	F-057	share sheet hands FileProvider URI, cache cleaned	L4
TC-IO-008	F-058	open-in for odoc/md/txt/html from file manager	L4
TC-IO-009	F-059	SAF create/open flows	L4
TC-IO-010	F-060	backup zip encrypted, manifest SHAs correct	L2/L4
TC-IO-011	UX-18	SAF denial mid-export: no partial file + reason	L4
TC-IO-012	F-057, F-058	share→receive→import→export parity loop	L4
TC-IO-013	F-060	backup→wipe→restore byte-identical at F3	L4

Search (SRCH)

Case	Binding	Proves	Level
TC-SRCH-001	F-067, UX-08	FTS speed + snippets + counts on corpus	L2/L4
TC-SRCH-002	F-068	deep-link opens hit block highlighted	L4
TC-SRCH-003	F-069	shared highlight with find bar	L2
TC-SRCH-004	F-070	semantic "related" merge labeled; absent without pack	L4
TC-SRCH-005	F-067	indexing 10k words < 150 ms background	L2

Scan & OCR (SCAN / AI)

Case	Binding	Proves	Level
TC-SCAN-001	F-071, UX-09	capture→crop→OCR→review→insert ≤ 60 s/page	L4
TC-SCAN-002	F-072	multi-page queue, reorder, batch OCR	L4
TC-AI-001	F-071, UX-41, UX-46	fixture accuracy ≥ 95%; confidence marks color+style	L4
TC-AI-002	F-073	no pack: flow ends at save-image with reason	L4
TC-AI-003	F-074, UX-10	dictation partials ≤ 1.5 s; explicit insert	L4
TC-AI-004	F-075, UX-47	four verbs produce labeled preview cards	L4
TC-AI-005	F-076	gemma on Tier A; extractive rung on Tier B labeled	L4
TC-AI-006	F-077, UX-29	streaming, cancel, pre-edit snapshot, single undo	L2/L4
TC-AI-007	F-080	TTS play/pause/stop + sentence highlight	L4
TC-AI-008	F-081	range start at cursor; system rate honored	L4
TC-AI-009	F-082, UX-30	controls hidden with reason per tier/pack/ram	L4
TC-AI-010	§19.5	prompt fixtures: runtime invariance + faithfulness rubric	L1

Packs (AI-PACK / SEC-PK)

Case	Binding	Proves	Level
TC-AI-PACK-01	F-078, UX-48	install flow: SHA shown, verify, capability appears	L2/L4
TC-AI-PACK-02	F-079, UX-48	corrupt/wrong-SHA refused, both hashes shown, ladder descent	L2/L4

Reading & a11y (READ / A11Y)

Case	Binding	Proves	Level
TC-READ-001	F-083, F-080	reader mode + TTS entry from reader	L4
TC-A11Y-001	F-084, UX-37	TalkBack-complete letter: create/edit/format/export	L4
TC-A11Y-002	F-085, UX-40	contrast across 8 themes	G (snapshot)
TC-A11Y-003	F-086, UX-39	200% scale, no clipping, all screens	L4

Settings & diagnostics (SET / DIAG)

Case	Binding	Proves	Level
TC-SET-001	F-087, UX-40	8 themes + match-system; accent choices	L4
TC-SET-002	F-088	editor settings apply + persist	L2/L4
TC-SET-003	F-089, UX-34	about/diagnostics facts; self-test ≤ 20 s offline	L4
TC-SET-004	F-090, UX-43	locale strings + RTL mirroring	L4
TC-DIAG-001	F-089	diagnostics ring buffer, no PII, never written out	L2

Security (SEC)

Case	Binding	Proves	Level
TC-SEC-P-01	C1	merged manifest permission set == §20.5 exactly	G
TC-SEC-P-02	UX-31	no notifications except user-created/progress	L4
TC-SEC-P-03	UX-50, C5	network silence: zero sockets across all flows on release APK	L4
TC-SEC-P-04	§20.4	clipboard read only on paste; no background reads	L4
TC-SEC-IO-01	§20.4	hostile HTML/scripts stripped structurally	L1/L4
TC-SEC-IO-02	F-057	FileProvider grants scoped; cache cleaned	L4
TC-SEC-ST-01	§20.3	encryption-at-rest smoke with keystore	L4
TC-SEC-ST-02	§20.3	export temp lifecycle (created granted, deleted)	L4
TC-SEC-PK-01	§20.4	tampered pack zip refused (hash mismatch)	L1/L4
TC-SEC-PK-02	§20.4	path-traversal pack entry names refused	L1/L4

The SEC suite is deliberately small and absolute: each case is a structural proof, not a scan. P-cases run on the release APK every RC; IO/PK cases run in JVM plus on-APK variants; a security suite red blocks the gate regardless of every other green (§20.6).

UX platform cases (UX-TC)

Case	Binding	Proves	Level
TC-UX-001	UX-01, UX-02	FAB→editor speed and directness	L4
TC-UX-002	UX-03	export ≤ 3 taps	L4
TC-UX-003	UX-04	import ≤ 2 taps via share/open-in	L4
TC-UX-004	UX-05	find bar memory + clean dismiss	L4
TC-UX-005	UX-06	composite insert single undo	L2
TC-UX-006	UX-19, UX-20	toolbar placement + target sizes	L4
TC-UX-007	UX-25	IME anchor scrolling	L4
TC-UX-008	UX-32	first-launch card; permission timing	L4
TC-UX-009	UX-35	destructive-action haptic + confirm/undo	L4
TC-UX-010	UX-42	reachability: primary actions bottom 40%	G (layout audit)
TC-UX-011	UX-44	reduced-motion honored	L4

Performance & environments (PERF / ENV)

Case	Binding	Proves	Level
TC-PERF-001	F-012, UX-02	cold start < 1.2 s, 200 docs, API 33	L4
TC-PERF-002	F-029, UX-26	typing P95 < 50 ms; no burst jank	L4
TC-ENV-001	C6	API 26 smoke: full P0 subset	L4
TC-ENV-002	C6	API 36 smoke incl. Tier A AI	L4
TC-ENV-003	C6	tablet/landscape layout sweep	L4
TC-ENV-004	C1	airplane-mode full journey pass	L4

Journeys (J)

Case	Binding	Proves	Level
TC-J1	§27.1	letter journey end-to-end on APK	L4
TC-J2	§27.2	scan journey (+ -degraded variant, no pack)	L4
TC-J3	§27.3	dictate→tidy→docx journey (+ -llm variant Tier A)	L4
TC-J4	§27.4	search journey (+ -semantic variant with pack)	L4
TC-J5	§27.5	deletion-recovery journey incl. version restore	L4

Journey variant matrix

Variants are procedure alternates of their base case (§24.5) — one manifest row, several runs:

Variant	Base	Difference	Runs when
TC-J2-degraded	TC-J2	no OCR pack; flow ends at save-image with reason (F-073)	G6 matrix on API 33 without packs
TC-J3-llm	TC-J3	gemma rung on Tier A; method label asserted	API 36 with llm pack
TC-J4-semantic	TC-J4	"related" section asserted with embed pack	API 33/36 with embed pack
TC-EDIT-004-draglong	TC-EDIT-004	hold-then-drag via UiAutomation injection (O.4)	nightly API 33

Appendix D — Gate definitions of done (full)

G0 — Charter accepted

- [ ] Owner accepts this blueprint v1.0 as the build contract (the acceptance act is recorded in `reports/onedoc/gates/G0.md`).
- [ ] `docs/onedoc/plan/` seeded: kickoff, task graph (G1+G2 expanded), decisions (D-000 example), sessions (S-000).
- [ ] Traceability green: `tools/onedoc/traceability.py --check --strict` exit 0.

G1 — Toolchain & shell

- [ ] JDK 17, SDK platform 36, build-tools 36.0.0, platform-tools installed; licenses accepted.
- [ ] AVDs per §23.1 created and boot-verified (API 26/33/35/36).
- [ ] Gradle multi-module skeleton per §10.1; `full` and `writer` flavors assemble; debug APK installs on all AVDs.
- [ ] CI pipelines per §22.2 registered; `pr.yml` green with ≤ 8 jobs; budgets enforced (≤ 3 runners).
- [ ] App icon, naming, `app.onedoc` applicationId, versioning scheme (§26.3) in place.

G2 — Model & storage

- [ ] `core-model` complete: block kinds, marks, ops, normalize; invariants I1–I4 enforced.
- [ ] Property suite P1–P10 green in JVM.
- [ ] `core-storage`: Room schema v3 exported; journal write/replay/compact; kill-trial harness passes locally (API 33, 20 trials).
- [ ] Golden files for odoc container; .odoc save/open round-trip.

G3 — Editor core

- [ ] Editor screen: block list, caret, selection, toolbar, tab row skeleton.
- [ ] IME contract implemented (§15); fuzzer harness running on all AVDs green.
- [ ] Undo/redo with coalescing; laws L1–L10 suites green.
- [ ] Typing budget met: P95 < 50 ms at 10k words on API 33 (TC-PERF-002 evidence).

G4 — Rich features

- [ ] Tables, images, TOC/map, page setup + guides, templates (12 built-ins), find/replace, versions UI, reader mode.
- [ ] All EDIT/TBL/IMG/STRUCT/TPL/STATE cases green.
- [ ] DQ-1…DQ-8 checklist passes for every new screen.

G5 — Formats & IO

- [ ] Exporters: pdf/odoc/md/txt/html/docx with goldens; importers: odoc/md/txt/html/docx with notes; refusal paths with reasons.
- [ ] Round-trip property P7 green; conversion-note UX verified.
- [ ] Share/open-in/SAF cases green (TC-IO-*).

G6 — On-device AI (feature-flagged until G7)

- [ ] Pack manager + 4 pack types installable/verifiable/removable; ladder implemented with hidden-with-reason UX.
- [ ] Scan, dictation, assistant, semantic search, read-aloud flows green on their tiers (API 36 full; API 33 extractive/system rungs; API 26 degradation).
- [ ] Network-silence probe green across all AI flows (TC-SEC-P-03).

G7 — Quality pass

- [ ] Full catalogue on release APK across §23.1 matrix; journeys J1–J5 green; evidence PDF assembled.
- [ ] Performance budgets green (§21.4); size ≤ 25 MB.
- [ ] Security suite green; zero open S1/S2; BLOCKED register reviewed with owner-visible reasons.

G8 — Release

- [ ] Release checklist (§28.1) complete; provenance bundle committed; artifacts attached to GitHub Release; defect tracker final state; owner handoff note.

Appendix E — Environment pins

Tool	Pin
JDK	17 (Temurin)
Android SDK	platforms;android-36, build-tools;36.0.0, platform-tools latest stable, cmdline-tools latest stable
Emulator images	system-images;android-26;google_apis;x86_64 · android-33 · android-35 · android-36 (google_apis, no playstore)
AVDs	`od-api26` Pixel 2 2 GB · `od-api33` Pixel 4 3 GB · `od-api35` Pixel 6 4 GB · `od-api36` Pixel 8 8 GB
Emulator boot	`-no-window -no-audio -gpu swiftshader_indirect -no-snapshot-save` (snapshots only for boot speed with `-snapshot` goldens)
CI runners	ubuntu-latest ×3 max parallel; concurrency cancel-in-progress per branch

Appendix F — Dependency and runtime pins (verify at adoption; record exacts in D-nnn)

Concern	Choice	Notes
Language/build	Kotlin (latest stable at G1), AGP pinned, JDK 17	version catalog; no dynamic versions
UI	Android Views + Material Components (m3)	no experimental UI toolkits (API 26 safety)
DI	Hilt	standard
DB	Room (schema export on) + FTS5	external-content FTS
Editor rendering	RecyclerView + custom block views + `EditText`-free text stack (custom `BlockTextView` over `Layout`)	the IME bridge is ours, §15
PDF	android.print framework + PdfDocument	fonts subset via bundled Inter
OCR	ONNX Runtime Mobile (or TFLite) hosting DBNet-lite + CRNN int8	pack: `ocr-en`
ASR	whisper.cpp (pinned release), ggml int8 model	pack: `asr-en`
LLM	llama.cpp-compatible pinned runtime, gemma-2B-it 4-bit (q4_0)	pack: `llm-gemma2b`; Tier A only
Embeddings	ONNX MiniLM-L6 int8, tokenizers bundled in pack	pack: `embed-minilm`
TTS	android.speech.tts (system)	no pack
Crypto	Android Keystore + Tink-free standard JCE (AES/GCM, PBKDF2)	no crypto SDKs
Tests	JUnit4/5, truth, MockK, Robolectric, espresso/uiAutomator, macrobenchmark	property harness custom (§16)

Model pack contents record exact file names + SHA-256 in their `manifest.json`; runtimes pin exact upstream versions in the version catalog and in the first D-nnn that adopts them.

Appendix G — Pre-registered BLOCKED register

Empty at v1.0 by design. Doctrine (§21.2): emulator/input/accessibility limitations pre-register here with environment fingerprint the first time they appear; each entry is re-tried every cycle with current tooling; removal requires either a PASS or a defect id. Inherited examples that will likely apply: physical-device-only behaviors (stylus pressure, multi-touch edge cases) are L5 and pre-BLOCKED on emulators by definition.

Appendix H — Templates and index

H.1 Decision entry (decisions.md)

```
### D-003 · 2026-XX-XX · Export: DOCX subset boundary
Context: DOCX has constructs OneDoc cannot represent (tracked changes, SmartArt).
Decision: writer covers §18.2 subset; beyond it, per-paragraph conversion notes; never guess.
Consequences: fidelity promise is testable (TC-IO-004/005); risk of user surprise mitigated by
     UX-45 note.
Reversal condition: owner asks for tracked-changes preservation → new decision supersedes.
```

H.2 Session entry (sessions.md)

```
## S-002 · 2026-XX-XX · G2 model+storage
Goal: land core-model + property suite.
Done: blocks/marks/ops; P1–P7 green; odoc goldens.
Failed/Blocked: P8 flaky on replay timing — root-caused to unsynced fsync; fix in S-003.
Next: kill-trial harness; schema v3 export.
```

H.3 Index of normative tables

Topic	Where
Constraints C1–C10	§1.3
Journeys J1–J5	§2.4, §27
Features F-001…F-090	§4, App. A
UX-01…UX-50	§5, App. B
Tokens/themes	§6
Figma contract + registry	§7
Design-QA DQ-1…DQ-8	§9.2
Architecture/threads	§10
Schema/journal/migrations	§11
Snapshot policy	§12.2
Laws L1–L10	§13
Document model + invariants	§14
IME contract	§15
Property suite P1–P10	§16
Feature engineering contracts	§17
Fidelity matrix F0–F3	§18.2
AI stack/ladder/packs	§19
Security suite	§20
Budgets + gates	§21
CI pipelines + budgets	§22
Emulator matrix	§23
Direct-APK QA + evidence PDF	§24
Defects	§25
Build/flavors/versioning	§26
Release checklist/provenance	§28
Way of working	§29
Env/dependency pins	App. E/F

Appendix I — Case procedures (executor-ready one-liners)

Format per case: **Case** — *Given* state → *When* action(s) → *Then* assertion(s). The QA harness expands these into `qa/onedoc/procedures/<case>.json` (§24.5). One line each; the catalogue (Appendix C) remains the binding index.

I.1 Laws

- **TC-LAW-001** — Given a doc with mixed marks typed 2 s ago → When autosave completes → Then `document.json` on disk parses to text+marks identical to on-screen.
- **TC-LAW-002** — Given any reachable model state → When `normalize()` runs twice → Then second run is a no-op and same-type marks never improperly overlap.
- **TC-LAW-003** — Given a fixed op list and seed → When applied on two clean instances → Then resulting documents and export bytes (minus timestamps) are identical.
- **TC-LAW-004** — Given an odoc containing a future-kind block → When opened → Then block preserved verbatim; conversion note lists it; saving keeps bytes.

I.2 Library

- **TC-LIB-001** — Given home → When FAB tapped then a key typed → Then editor visible ≤ 2 s, doc exists with that char, no dialogs seen.
- **TC-LIB-002** — Given a doc titled only by first H1 → When renamed via long-press sheet → Then card title updates; untitled docs mirror first heading on edit.
- **TC-LIB-003** — Given 3 docs → When one starred → Then it pins to top of All; state survives relaunch.
- **TC-LIB-004** — Given a doc with an image asset → When duplicated → Then copy has new id, same content, independent asset refs.
- **TC-LIB-005** — Given a doc → When deleted → Then snackbar with undo 7 s; undo restores at same position in list.
- **TC-LIB-006** — Given a trashed doc aged 29 days → When library opened → Then remaining time shown; restore works; permanent delete asks confirm twice.
- **TC-LIB-007** — Given sort switched to title → When app relaunched → Then ordering persists.
- **TC-LIB-008** — Given empty library / empty trash / empty template-user-section → When each opened → Then single-line empty state with one CTA renders (no blank screens).
- **TC-LIB-009** — Given a 1,234-word doc with 2 images → When info sheet opened → Then words/chars/reading time/size/version count match computed truth.
- **TC-LIB-010** — Given 3 docs multi-selected → When export+delete applied → Then zip contains 3 exports; 3 cards vanish; undo restores them.

I.3 Editor core

- **TC-EDIT-001** — Given cursor mid-paragraph → When typing fast (scripted 300 cps) → Then text lands at caret; neighbors unaffected; caret stays visual position.
- **TC-EDIT-002** — Given "aaa|bbb" in one block and marks spanning the middle → When Enter → Then two blocks with marks redistributed; Backspace merges back to the

original spanned marks.

- **TC-EDIT-003** — Given type(10 chars)+paste+bold+type(5) → When undo×4/redo×4 → Then each step matches a previously rendered state; final = original start.
- **TC-EDIT-004** — Given a selection → When handles dragged, toolbar tapped, stylus/mouse used → Then actions apply to selection; toolbar never covers it; active states correct.
- **TC-EDIT-005** — Given text with nested same-type marks → When toggle bold on overlapping range → Then normalize yields single merged range; rendering matches.
- **TC-EDIT-006** — Given paragraph with bold middle → When switched H1→H2→para→list→para → Then bold survives every transition; outline updates.
- **TC-EDIT-007** — Given list item 3 of 5 → When Enter at end / at empty item / Tab / Shift+Tab → Then continue+renumber / exit-list / indent / outdent exactly as Docs.
- **TC-EDIT-008** — Given code block → When typing and toggling marks → Then code text takes no marks; quote allows them.
- **TC-EDIT-009** — Given paragraph → When each alignment set → Then rendering + persistence + export carry align.
- **TC-EDIT-010** — Given divider between paragraphs → When deleting adjacent empty para → Then divider remains single removable block.
- **TC-EDIT-011** — Given clipboard from Docs HTML (marks+list+table) → When pasted → Then subset converts per F2; scripts/styles dropped; clipboard unchanged on failure.
- **TC-EDIT-012** — Given 3-block selection → When cut then pasted elsewhere → Then target content byte-equal; source collapsed correctly; one undo reverts both.
- **TC-EDIT-013** — Given 7 matches of "q3" → When replace-all with "Q3" (case on) → Then confirm + pre-edit version created; one undo restores all 7.
- **TC-EDIT-014** — Given matches across pages → When next/prev in find bar → Then count updates, view scrolls, highlights non-destructive after close.
- **TC-EDIT-015** — Given live typing incl. emoji → When count chip read → Then words/chars match ICU grapheme truth.
- **TC-EDIT-016** — Given hardware keyboard → When Ctrl+B/Z/Y/F, arrows, Tab in list/table → Then behaviors per §15.4; undo parity with soft path.
- **TC-EDIT-017** — Given "a□b" → When deleting between glyphs / splitting → Then surrogate pair moves/deletes atomically; offsets stay valid.

I.4 IME

- **TC-EDIT-IME-01** — Given glide-typed word with autocorrect full-replace → When composition finishes → Then single op; prior marks intact; no ghost text.
- **TC-EDIT-IME-02** — Given active composition → When two-finger scroll → Then composition cancels cleanly, text = last committed, no corruption.
- **TC-EDIT-IME-03** — Given Hindi transliteration of "namaste" → When composed → Then combining clusters never split across ops; rendering stable.
- **TC-EDIT-IME-04** — Given IME fuzzer with random valid sequences + ops → When run 10k steps → Then L2/L4/L5 hold after every step (seeds on failure).

I.5 Properties, State, Tables/Images/Structure

- **TC-PROP-P01…P10** — Given generators per §16.3 → When 5k cases per property (PR) → Then all hold; failures shrink to minimal counterexamples with seeds.
- **TC-STATE-001** — Given edits 0.8 s old → When back/home → Then no dialog, state present on reopen (silent autosave).
- **TC-STATE-002** — Given mid-sentence caret + selection + IME composing → When rotate/split resize → Then caret, selection, composition state survive; scroll anchored.
- **TC-STATE-003** — Given 50 op batches in 2 min → When thresholds cross → Then auto snapshot lands exactly per policy table.
- **TC-STATE-004** — Given 5 versions on 50-page doc → When preview then restore → Then ≤ 30 s, pre_edit snapshot of current first, redo cleared, content equals chosen version.
- **TC-STATE-005** — Given typing loop → When `am kill` at 100 randomized instants → Then relaunch content always within 2 s of typing (observed: 0 loss).
- **TC-STATE-006** — Given doc with unused assets → When info/asset manager opened → Then accounting matches `du` truth; cleanup requires confirm and re-checks refs.
- **TC-STATE-007** — Given synthetic v1…vN-1 databases → When sequential migrations run → Then each up-test passes; chain lands on vN; dry-run corpus valid.
- **TC-STATE-008** — Given editor in split-screen → When focus lost to companion app → Then autosave fires; nothing lost on kill right after.
- **TC-STATE-009** — Given storage 99% full → When typing continues → Then editor keeps working in memory; warning chip; autosaves succeed after space freed (auto retry).
- **TC-TBL-001** — Given 3×3 table → When add/remove row/col, header toggle, bold inside cell → Then ops atomic in undo; marks render in cells; export keeps them.
- **TC-TBL-002** — Given table with long cell text → When rendered → Then columns wrap to presets; document never scrolls horizontally.
- **TC-IMG-001** — Given gallery/camera/scan inserts → When image added → Then EXIF GPS stripped, alt-text prompt focused, caption editable, viewer shows alt for TalkBack.
- **TC-IMG-002** — Given image → When S/M/L/full applied → Then measure-relative size persists; alt editor edits text read by reader.
- **TC-IMG-003** — Given doc with orphan asset after image delete → When asset manager opened → Then orphan listed; remove requires confirm; references double-checked.
- **TC-STRUCT-001** — Given doc with H1–H3 → When TOC block present and headings edited → Then entries update; tap scrolls to target with animation.
- **TC-STRUCT-002** — Given long doc scrolled → When map sheet opened → Then live position dot matches; tap navigates.
- **TC-STRUCT-003** — Given A4 normal vs Letter narrow → When switched → Then measure, guides, PDF pagination all change consistently.
- **TC-STRUCT-004** — Given 40-page doc → When guides compared to exported PDF breaks → Then within ±1 line on every page.
- **TC-STRUCT-005** — Given footnote → When exported → Then sup in body, note at page bottom in PDF; editor tap opens bottom sheet.
- **TC-STRUCT-006** — Given cover block filled → When exported md/html/pdf → Then styled per format capabilities; nothing lost silently.

- **TC-STRUCT-007** — Given 12 blocks → When drag-reordered (and via TalkBack actions) → Then order changes as one moveBlock op each; ids stable.
- **TC-STRUCT-008** — Given block menu → When duplicate/delete → Then new block copy with fresh ids / removal with single undo.
- **TC-STRUCT-009** — Given code block → When language label set to "kotlin" → Then label persists through all exports that support it, noted otherwise.

I.6 Templates, IO, Search

- **TC-TPL-001…006** — Given gallery → When each of 12 templates instantiated, previewed, placeholder-filled, saved-as, deleted (user section) → Then behaviors per §17.5 with template files never mutating.
- **TC-IO-001** — Given 50-page doc → When PDF exported → Then fonts embedded (no fallback boxes), text selectable, breaks match §17.4, word count ±1%, < 10 s.
- **TC-IO-002** — Given doc with all 13 block kinds + image → When odoc export→import → Then F3-identical (bytes of document.json; asset SHAs equal).
- **TC-IO-003** — Given marks/tables/code docs → When md/txt/html exported → Then per §18.2 fidelity; unsupported constructs degrade with notes (never silent).
- **TC-IO-004** — Given styled doc → When docx exported then re-imported → Then headings/marks/lists/tables/images parity within F2; OOXML valid.
- **TC-IO-005** — Given md/html/docx/odoc fixtures → When imported → Then conversion notes state kept/dropped counts truthfully (UX-45).
- **TC-IO-006** — Given .exe/.pdf/random binary named .odoc → When import attempted → Then refusal with specific reason; no empty doc; clipboard intact.
- **TC-IO-007** — Given export → When share sheet invoked then cancelled → Then FileProvider URI grants scoped; temp cleaned in both paths.
- **TC-IO-008** — Given file-manager "open with" for each supported type → When tapped → Then import preview opens with correct sniffing.
- **TC-IO-009** — Given SAF locations (Downloads, custom folder) → When create/open through SAF → Then streams complete; sizes honored; caps warn at 50 MB.
- **TC-IO-010** — Given backup enabled + folder picked → When weekly trigger fires → Then encrypted zip lands with manifest SHAs; passphrase prompt is the only security dialog.
- **TC-IO-011** — Given export to SAF → When provider revoked mid-write → Then no partial file remains; reason shown; retry clean.
- **TC-IO-012** — Given share→receive→import→export loop per format → When completed → Then parity within declared fidelity each hop.
- **TC-IO-013** — Given backup → wipe app data → restore → Then library byte-identical at F3; report lists counts; conflicts import as versions.
- **TC-SRCH-001** — Given 200-doc corpus → When "warranty" typed (2 chars on) → Then results stream; snippet+count correct; < 300 ms end-to-end on API 33.
- **TC-SRCH-002** — Given a result → When tapped → Then editor opens at hit block, term highlighted, back returns to results.
- **TC-SRCH-003** — Given highlight in find bar → When search deep-link arrives → Then same highlight system serves both (no double-style).
- **TC-SRCH-004** — Given embed pack + query "what covers repairs" → When results merged → Then "related" section labeled; absent pack → section absent, no error.

- **TC-SRCH-005** — Given 10k-word doc saved → When index updated → Then < 150 ms on background executor; main thread untouched.

I.7 Scan, AI, Packs, Read/A11y

- **TC-SCAN-001** — Given fixture page → When capture→crop→OCR→review→fix→insert → Then ≤ 60 s/page on API 33; confidence marks on < 0.82 words; fixed word lands.
- **TC-SCAN-002** — Given 3 pages queued → When reordered then OCR batch → Then paragraphs in chosen order; page separators correct.
- **TC-AI-001** — Given 10-page OCR fixture set → When processed → Then accuracy ≥ 95%; marks use color+underline; export of scan-sourced doc flags unreviewed low-confidence words (UX-46).
- **TC-AI-002** — Given no OCR pack → When scan flow run → Then ends at save-image with one-line reason; no crash; retry offers pack manager.
- **TC-AI-003** — Given fixture audio injected → When dictation runs → Then first partial ≤ 1.5 s; stop → editable preview; insert = one grouped op.
- **TC-AI-004** — Given selections of varied size → When each verb run → Then labeled preview cards; Insert/Replace/Discard behave; nothing auto-applies.
- **TC-AI-005** — Given API 36 (Tier A) and API 33 (Tier B) → When Summarize → Then gemma rung on A; extractive rung on B; card labels match (UX-47).
- **TC-AI-006** — Given generation streaming → When cancel mid-stream / insert after completion → Then no residue on cancel; pre_edit snapshot before insert; single undo reverts.
- **TC-AI-007** — Given reader mode → When TTS play/pause/stop → Then sentence highlight follows; controls respond < 200 ms.
- **TC-AI-008** — Given cursor mid-doc → When read-from-cursor → Then range honored; system rate changes apply next utterance.
- **TC-AI-009** — Given tiers C/B/A and pack combos → When surfaces render → Then unavailable verbs hidden with long-press reason naming tier/pack/ram (never disabled buttons).
- **TC-AI-010** — Given prompt fixtures → When assistant runs twice per fixture → Then identical outputs (pinned seed); faithfulness rubric passes (no invented facts in 20 fixtures).
- **TC-AI-PACK-01** — Given valid .odpack via SAF → When install confirmed → Then SHA shown pre-confirm; verify passes; capability appears; delete frees space and capability disappears.
- **TC-AI-PACK-02** — Given pack with flipped byte in model file → When installed/loaded → Then refused with both SHAs; capability descends ladder; no crash/loop.
- **TC-READ-001** — Given reader mode with TTS → When font size changed and play pressed → Then render + speech continue from visible block.
- **TC-A11Y-001** — Given TalkBack on → When full letter flow (create→format→export) → Then completable sightless; traversal matches visual order; actions present in menu.
- **TC-A11Y-002** — Given all 8 themes → When contrast audited → Then body ≥ 4.5:1, large ≥ 3:1, HC themes ≥ 7:1 (snapshot gate fails otherwise).
- **TC-A11Y-003** — Given font scale 2.0 → When all screens walked → Then no clipped/truncated text; containers grow or scroll.

I.8 Settings, Security, UX-platform, Perf/Env, Journeys

- **TC-SET-001…004, TC-DIAG-001** — Given settings surfaces → When themes/editor/files/about exercised per §8.6 → Then instant persistence; self-test ≤ 20 s offline; locales+RTL correct; diagnostics never leaves device.
- **TC-SEC-P-01** — Given release APK → When `apkanalyzer manifest` → Then permission set equals §20.5 exactly (CI gate fails on any delta).
- **TC-SEC-P-02** — Given full journey run → When notification shade inspected → Then only user-created/export-progress notifications ever appeared.
- **TC-SEC-P-03** — Given airplane-mode emulator + strace/logcat probes → When ALL flows incl. AI run on release APK → Then zero socket syscalls; zero resolver attempts; app fully functional.
- **TC-SEC-P-04** — Given clipboard seeded → When app foregrounded/idled 5 min → Then no clipboard reads outside explicit paste (Android 13 toast never appears).
- **TC-SEC-IO-01/02, TC-SEC-ST-01/02, TC-SEC-PK-01/02** — Given hostile fixtures (scripted HTML, zip-bomb, traversal pack, oversize DOCX) and storage probes → When processed → Then refusals with reasons; grants scoped; cache cleaned; encryption smoke passes; no path escapes.
- **TC-UX-001…011** — Given each criterion context (§5) → When measured on API 33 → Then all pass with numbers (taps, seconds, dp, %).
- **TC-PERF-001/002** — Given cold start / typing burst harness on API 33 → When run → Then < 1.2 s / P95 < 50 ms with frame stats attached.
- **TC-ENV-001…004** — Given each AVD and airplane mode → When P0 subset / Tier A / tablet-landscape / offline journeys run → Then green with evidence per environment.
- **TC-J1…TC-J5 (+variants)** — Given journey scripts §27.1–27.5 → When executed on release APK per matrix → Then all steps assert; screenshots + artifacts attached; variants (degraded/llm/semantic) follow their ladder rungs.

Appendix J — Fixture corpus

J.1 Documents (seed + test fixtures, `tools/onedoc/testdata/corpus/`)

Fixture	Purpose	Used by
`empty.odoc`	zero blocks; first-run edge	LIB, EDIT
`letter-1p.odoc`	180-word letter, marks + H1	J1, IO goldens
`report-50p.odoc`	50 pages, H1–H3, 3 tables, 4 images, TOC, footnotes	STRUCT-004, IO-001, STATE-004
`walls-20k.odoc`	single 20k-char paragraph block	renderer guard §14.7, PERF
`unicode-planfeast.odoc`	emoji planes 0/1/2/14, Devanagari, CJK, RTL runs	EDIT-017, IME-03, P10
`tables-heavy.odoc`	12 tables incl. 1×1 and 20-col	TBL, P6
`marks-kitchen.odoc`	every mark type, overlapping boundary cases	EDIT-005, P2
`lists-deep.odoc`	bullet/number/check nests, 1–5 levels	EDIT-007
`seed-library-200/`	200 synthetic docs (titles+1–5k words each)	PERF-001, SRCH-001
`manual-appliances.odoc`	6-page manual; "warranty" once oblique ("the manufacturer undertakes repairs for 24 months")	J4
`future-kind.odoc`	hand-crafted block kind `widget_v9`	LAW-004 round-trip
`legacy-v1.odoc`	schema v1 synthetic	STATE-007 migrations

J.2 Images and audio

- `scan/page-0[1-3].jpg` — 300 dpi printed-letter photos with perspective skew (OCR fixtures, accuracy bar TC-AI-001).
- `scan/whiteboard-01.jpg` — low-contrast handwriting (confidence-marking demo, UX-46).
- `images/exif-gps.jpg` — GPS-tagged test image (proves EXIF strip, TC-IMG-001).
- `audio/minutes-en-60s.wav` — 60 s meeting speech, 2 speakers (dictation fixture, TC-AI-003).
- `audio/hi-sample-30s.wav` — Hindi sample (locale path, TC-SET-004 co-verification).

Fixtures are committed to the repo (they are small, synthetic, license-clean) so every run of every cycle is reproducible byte-for-byte — the same discipline Law L6 demands of exports.

Appendix K — Glossary (owner-friendly)

Term	Meaning
Block	One structural unit of a document: paragraph, heading, list, table, image…
Mark	Inline formatting on a text range: bold, italic, link…
Op	The smallest change the engine can apply/undo (typed char, split, mark toggle)
Journal	Append-only log that makes zero-keystroke-loss possible (L3)
Snapshot / Version	A saved copy of a document at a point in time (auto/manual/pre-edit)
Fidelity (F0–F3)	Honesty level of a file conversion: refused / best-effort / subset-lossless / lossless
Pack (.odpack)	A side-loaded, SHA-verified zip containing an AI model — never downloaded by the app
Ladder	The ordered fallback: best model → lighter model → system service → hidden with reason
Tier A/B/C	Device classes by RAM: 8+ / 4–8 / <4 GB — decide which AI runs
Law (L1–L10)	A release-blocking editor behavior with a dedicated proof
Gate (G0–G8)	A phase boundary that closes only with checked DoD evidence
Direct-APK QA	Tests that install the real release APK on an emulator and drive it — the only proof that counts for shipping
BLOCKED	"Could not test fairly" — never counted as pass or failure; logged with reason and retried
Evidence PDF	The owner's report: every claim traced to a case id with screenshots and provenance
D-nnn / S-nnn	Decision ledger / session log entries (append-only)

Appendix L — Risk register (full)

#	Risk	P	Impact	Mitigation	Trigger → response
R1	IME diversity breaks composition invariants	M	S1-class (text loss)	fuzzer + scripts per keyboard (§15)	any fuzzer red → feature freeze, IME-only gate
R2	LLM too heavy for claimed Tier A floor	M	capability demotion	tier gate + budget timings (§19.8)	2× latency miss → raise tier floor, D-nnn
R3	Pagination guide/PDF drift	M	trust (WYSIWYG claim)	single metrics engine (§17.4)	±1 line breach → font-fallback freeze + fix
R4	APK size creep	H	budget breach, cheap-phone exclusion	per-PR size gate (§22.4)	+0.5 MB PR → shrink pass first
R5	Figma/registry drift	M	UI inconsistency	DQ-3 + nightly drift check	drift found → re-sync task before UI merges
R6	DOCX import surprises (real-world files)	H	fidelity trust	refusal+notes posture (§18.1), corpus tests	new malformed class → parser fix + corpus case
R7	Property suite runtime bloat slows PRs	M	dev velocity	shard nightly; PR = 5k cases	PR pipeline > 15 min → move suites to nightly+merge
R8	Emulator flakiness pollutes verdicts	H	wasted cycles, mistrust	boot snapshots, quarantine rules (§24.3)	flaky-twice → quarantine with defect id
R9	Owner keystore delayed at G8	M	release slip	RC keystore path defined (§26.1)	delay → ship RC-signed artifact + re-sign runbook
R10	Scope pressure ("just one network feature")	L	promise break	§1.4 list + C1 gate	any proposal → invalid by charter; D-nnn refuses

Appendix M — Release-notes template (voice contract)

```
OneDoc v<X.Y.Z> (<versionCode>) — build <sha7>

New
- <feature>: <one line, user language>

Improved
- <improvement>: <one line>

Fixed
- <OD-xxx> <one line>

Honest notes
- <fidelity/BLOCKED/limit lines — no "known issues" framing of anything S1/S2 (they cannot exi
    st at G8)>

Verify
- APK SHA-256: <hash>
- Evidence: <reports link> · Provenance: releases/<ver>/provenance.json
```

Rules: ≤ 200 words; no marketing adjectives; every Fixed line cites a tracker id; Honest notes are mandatory even when empty ("None — see evidence PDF.").

Appendix N — Executor quick-start and runbooks

N.1 Environment bootstrap (run verbatim)

```
# JDK
sudo apt-get update && sudo apt-get install -y openjdk-17-jdk
java -version   # expect 17.x

# Android SDK (cmdline-tools path)
mkdir -p $HOME/android-sdk/cmdline-tools
curl -o cmdtools.zip https://dl.google.com/android/repository/commandlinetools-linux-11076708_
    latest.zip
unzip -q cmdtools.zip -d $HOME/android-sdk/cmdline-tools && mv $HOME/android-sdk/cmdline-tools
    /cmdline-tools $HOME/android-sdk/cmdline-tools/latest
export ANDROID_HOME=$HOME/android-sdk && export PATH=$PATH:$ANDROID_HOME/cmdline-tools/latest/
    bin:$ANDROID_HOME/platform-tools:$ANDROID_HOME/emulator

# Packages (pins per Appendix E)
yes | sdkmanager --licenses
sdkmanager "platform-tools" "platforms;android-36" "build-tools;36.0.0" \
  "emulator" "system-images;android-26;google_apis;x86_64" \
  "system-images;android-33;google_apis;x86_64" \
  "system-images;android-35;google_apis;x86_64" \
  "system-images;android-36;google_apis;x86_64"
```

N.2 AVD creation (§23.1, verbatim)

```
echo no | avdmanager create avd -n od-api26 -k "system-images;android-26;google_apis;x86_64" -
    d pixel_2 --force
echo no | avdmanager create avd -n od-api33 -k "system-images;android-33;google_apis;x86_64" -
    d pixel_4 --force
echo no | avdmanager create avd -n od-api35 -k "system-images;android-35;google_apis;x86_64" -
    d pixel_6 --force
echo no | avdmanager create avd -n od-api36 -k "system-images;android-36;google_apis;x86_64" -
    d pixel_8 --force
# RAM profiles per tier: edit config.ini (hw.ramSize=2048|3072|4096|8192)
```

N.3 Emulator boot + clean state

```
emulator -avd od-api33 -no-window -no-audio -gpu swiftshader_indirect -no-snapshot-load -port
    5554 &
adb wait-for-device
adb shell 'while [[ -z $(getprop sys.boot_completed) ]]; do sleep 1; done'
# golden boot snapshot (boot speed only; app data ALWAYS cleared per run):
adb -s od-api33 emu snapshot save golden-boot 2>/dev/null || true
```

Rule: every QA run starts `adb uninstall app.onedoc || true` — no run ever sees a previous run's data (§23.3).

N.4 Build + install + drive (the direct-APK loop)

```
./gradlew :app:assembleFullRelease            # release APK (RC keystore per §26.1)
APK=app/build/outputs/apk/full/release/app-full-release.apk
sha256sum $APK | tee artifacts/apk-sha.txt
apkanalyzer manifest permissions $APK > artifacts/manifest-permissions.txt   # gate: == §20.5
adb install -r $APK
# drive via instrumentation QA apk (same run id, separate module :qa-driver):
adb shell am instrument -w -e case TC-J1,TC-J5,TC-STATE-005 \
  app.onedoc.test/androidx.test.runner.AndroidJUnitRunner | tee artifacts/instr.log
# artifacts: screenshots per case at /sdcard/onedoc-qa/<case>/ pulled to artifacts/
adb pull /sdcard/onedoc-qa artifacts/screens
```

N.5 Release runbook (G8, ordered)

1. Verify G7 evidence is green for the EXACT release SHA (re-tag build if not).
2. `./gradlew :app:assembleFullRelease :app:bundleFullRelease` with release keystore (owner-provided) or RC keystore per §26.1.
3. Static gates: manifest, deny-list, size ≤ 25 MB, SBOM generation.
4. Direct-APK QA on the release build (full catalogue + journeys) → evidence PDF.
5. Defect tracker final pass; release notes per Appendix M; handoff note per §28.4.
6. `provenance.json` per §28.2; tag `v<X.Y.Z>`; GitHub Release with APK + PDF attached.
7. Re-sign runbook (if owner keystore arrives late): build RC-signed now; when keystore arrives, rebuild same commit with release key, re-run the static gates + smoke (not the full matrix — same bytes logic), update provenance; never patch a signed APK binary.

N.6 The nightly all-green ritual

Nightly pipeline = full matrix + 50k-case properties + IME fuzzer + drift checks (Figma registry if token provided, tokens lint, dependency audit). Morning rule: a red nightly is the first task of the day — before any feature work (§29.4 blockage discipline). The ritual is the reason G7 surprises don't exist.

Appendix O — Worked examples (engine traces for the implementer)

O.1 Trace: six keystrokes from screen to disk

Scenario: new document; user types `Hi!` then Enter then `Q3 report`, with `Q3` pre-typed bold via toolbar. Timeline (T = ms since first key):

T	Event	Op applied	Journal record (after CRC, framing)	Notes
0	'H'	insertText(b01, 0, "H")	rec#1	block b01 created as `para` (new doc auto-creates one empty para; Law L1: the file contains it)
120	'i'	insertText(b01, 1, "i")	rec#2	coalescing window open (600 ms)
300	'!'	insertText(b01, 2, "!")	rec#3	same batch
980	Enter	splitBlock(b01, 3)	rec#4	new batch (idle > 600 ms); b02 created
1100	select "Q3" + Bold	setMarks(b02, strong, 0, 2)	rec#5	toolbar one-tap; snapshot? no — policy §12.2 not met yet
1500	type 'Q','3',' ','r'...	insertText×4 coalesced	rec#6	bold extends per §14.5

At T=2,300 ms (800 ms after last op): debounced save folds rec#1–6 into checkpoint `checkpoints/0.json.zip` and updates Room row (title="Hi", wordCount≈3). Journal truncates to empty after fold (§11.3 compaction bound). Kill at any T: replay produces exactly the table's states — this trace is TC-STATE-005's determinism fixture.

O.2 Trace: mark normalization on merge

Before: block X: `[bold 0,5)`, block Y: `[bold 0,3)`; delete Y's first paragraph mark boundary then merge Y into X (Backspace at Y start) with Y text "abc def" → X "…xyzabc def":

```
X.text = "…xyz", Y.text = "abc def"
merge → text "…xyzabc def"
marks before normalize: [bold 0,5), [bold 5,8)      # adjacent same-type
normalize → [bold 0,8)                               # merged (idempotent check: second pass n
    o-op)
```

Counter-case (different types): `[bold 0,5)`, `[italic 5,8)` → normalize keeps both (different types may abut); `[highlight 2,6)`, `[highlight 5,9)` → impossible post-op state; normalize folds to `[highlight 2,9)` and logs invariant repair (debug: fail-fast). P2 generates thousands of these; the trace above is the human-readable contract.

O.3 Trace: one document through four exporters

Source: H1 "Invoice", para "Total **$42** due Friday", checklist 2 items (1 checked).

- **Markdown (F2):**

```
# Invoice

Total **$42** due Friday

- [x] Send to Ada
- [ ] Archive receipt
```

- **HTML (F2):**

```
<h1>Invoice</h1>
<p>Total <strong>$42</strong> due Friday</p>
<ul><li data-check="done">Send to Ada</li><li>Archive receipt</li></ul>
```

- **DOCX (F2):** Heading1 style "Invoice"; paragraph with bold run "$42"; list paragraphs `w:numPr` with checkbox glyphs for check lists (conversion note: "checklist state stored as glyph + custom prop").
- **PDF (F3):** A4 page 1: H1 20/26 at margins; body 15/22; checklist boxes drawn; fonts embedded (Inter subset); text layer selectable; word count 9 (matches editor chip; UX-49 tolerance ±1%).

Each exporter's full golden files for this exact source live at `core-io/src/test/resources/golden/showcase/` — a failing exporter diff explains itself against these.

O.4 Trace: a blocked gesture correctly reported (doctrine in motion)

Claimed case: "drag handle from word A to selection end across 3 blocks fails". Scripted `adb input swipe` cannot perform a stationary hold-then-drag with intermediate events (predecessor doctrine). Verdict path: run → driver marks `TC-EDIT-004-draglong` **BLOCKED**, reason "input synthesis cannot reproduce hold-then-drag; requires physical device or instrumentation gesture injection", fingerprint `(api33, uiautomator2)` → register Appendix G entry → instrumented variant runs `MotionEvent` injection via `UiAutomation` → if that PASSes, case closes as PASS with both evidences; if it FAILs, defect OD-0nn with repro video frames. At no point does the blocked attempt count as a product failure or a health signal.

Appendix P — Strings dictionary (normative, en; hi/mr mirror keys)

Rules: sentence case; no exclamation marks; no emoji; "document" not "file" for user content; numbers humanized; every string is a key in `core/designsystem/strings.csv` — feature code references keys, never literals (lint-enforced; TC-SET-004).

P.1 Core actions

Key	String
action.new.doc	New document
action.save.version	Save version
action.export	Export
action.share	Share
action.rename	Rename
action.duplicate	Duplicate
action.star / action.unstar	Star / Unstar
action.move.trash	Move to trash
action.restore	Restore
action.delete.forever	Delete forever
action.undo / action.redo	Undo / Redo
action.find.replace	Find and replace
action.save.template	Save as template
action.print	Print

P.2 Editor

Key	String
editor.tab.text / insert / ai / format	Text · Insert · AI · Format
editor.count	{words} words · {chars} characters
editor.find.count	{n} of {total}
editor.replace.all.confirm	Replace {n} matches? A version is saved first.
editor.link.hint	Paste or type a link
editor.image.alt.hint	Describe this image (helps screen readers)
editor.composing.protected	— (no string: composition is silent, L5)

P.3 AI surfaces

Key	String
ai.summarize / rewrite / continue / fix	Summarize · Rewrite · Continue writing · Fix grammar
ai.method.llm	On-device gemma-2B
ai.method.extractive	On-device extractive summary
ai.method.system	Android speech service
ai.insert / discard / replace	Insert · Discard · Replace
ai.cancel	Stop
ai.unavailable.pack	Needs the {pack} pack — install it from Assistant.
ai.unavailable.ram	{model} needs {need} GB RAM — this device has {have}.
ai.unavailable.tier	{feature} needs a stronger phone than this one.
ai.descended	Switched to the lighter on-device method (thermal).
pack.install	Install pack
pack.verify.fail	Verification failed. Expected {expected}, found {found}.
pack.license	License: {spdx}

P.4 Files and errors (canonical §18.8 set + these)

Key	String
export.pdf / odoc / md / txt / html / docx	Export as PDF / OneDoc / Markdown / Text / HTML / Word
fidelity.F0/F1/F2/F3	Not supported / Best effort / Subset lossless / Lossless
import.refused.{reason}	per taxonomy §18.8
trash.retention	Deleted documents are kept 30 days. {days} left.
version.auto/manual/pre_edit	Auto · Manual · Before change
restore.done	Restored. Your previous text is saved as a version.
backup.prompt.pass	Set a passphrase to encrypt your backup

P.5 First run and about

Key	String
welcome.title	Your words, your phone.
welcome.body	OneDoc works fully offline. No account, no ads, no tracking.
welcome.cta	Start writing
about.selftest	Run self-test
about.selftest.done	All checks passed ({n}) in {s} s
about.storage	Documents {docs} · Versions {ver} · Packs {packs}

(Complete key list ships as CSV; this appendix pins voice and the strings that gate-acceptance reads aloud.)

Appendix Q — Error catalogue (complete OdError enumeration)

Every error kind the app can surface, with verb (§10.5), user string key, and log treatment. The compiler-enforced enum is generated from this table (a test asserts table == enum == strings.csv coverage — no orphan error paths).

Kind	Verb	String key	Log/diagnostics
E_STORAGE_DISK_FULL	retryable	err.disk.full	auto-retry ≤ 3, backoff 2/8/30 s
E_STORAGE_IO	retryable	err.storage.retry	same
E_JOURNAL_CRC	reportable	err.doc.repair	quarantine file, keep bytes, self-test flags
E_DOC_SCHEMA_FUTURE	actionable	err.doc.newer	refuse open older app; suggest update path (manual)
E_IMPORT_UNSUPPORTED	actionable	import.refused.format	reason names format
E_IMPORT_CORRUPT	actionable	import.refused.corrupt	why: zip/xml/codec
E_IMPORT_TOO_LARGE	actionable	import.refused.size	50 MB cap
E_IMPORT_BINARY	actionable	import.refused.binary	NUL/decode gate
E_EXPORT_SAF_DENIED	actionable	export.denied.folder	one action: pick again
E_EXPORT_WRITE	retryable	export.retrying	no partial file (UX-18)
E_PACK_VERIFY	actionable	pack.verify.fail	both SHAs shown
E_PACK_RUNTIME_OLD	actionable	pack.runtime.old	runtime pin name
E_PACK_TIER	actionable (hidden control variant)	ai.unavailable.tier	long-press reason
E_AI_THERMAL	info (descent)	ai.descended	rung noted in card + diagnostics
E_AI_TIMEOUT	actionable	ai.timed.out	offer lighter method (ladder)
E_TTS_MISSING	info (hidden)	ai.unavailable.tts	reader control hides
E_MIC_PERMISSION	actionable	perm.mic.why	settings deep-link
E_CAMERA_PERMISSION	actionable	perm.camera.why	settings deep-link
E_IME_VIOLATION	reportable (debug fail-fast)	— (never user-visible)	fuzzer alert, defect
E_INVARIANT	reportable (debug fail-fast)	err.something.broke	quarantine + diagnostics ring
E_DB_MIGRATION	reportable	err.upgrade.repair	backup plan §11.4; never silent drop

Reportable errors are the only ones that add a diagnostics entry beyond counts; they always preserve user data first (quarantine, not delete) — the tracker's S1 class is defined by exactly these plus Law violations.

Appendix R — Gesture map (every gesture, its action, and the accessible path)

Gesture	Context	Action	Accessible equivalent (TalkBack menu / a11y action)
Tap	block	place caret	same (default focus)
Double-tap	word	select word	selection action "select word"
Triple-tap	paragraph	select paragraph	"select paragraph"
Long-press	text	word select + drag handles	same via keyboard selection
Long-press	doc card	multi-select enter	custom action "select"
Long-press	block handle	drag-reorder	"Move block up/down"
Long-press	AI verb (hidden context)	show availability reason	reason is the content description
Two-finger scroll	during composition	cancel composition, scroll	same
Pinch	editor	zoom page scale (compose) 80–130%	font-size stepper in settings
Swipe from left edge	editor	back (system)	back action
Swipe down on handle	selected block	dismiss selection	Esc action
Drag corner handles	scan crop	perspective crop	numeric adjust fields
Drag within cell edge	table	(nothing — presets only, §17.1)	column preset menu
Swipe on snackbar	any	dismiss (undo remains via versions)	"Undo" action button

Appendix S — Gate demos (what the owner can watch at each gate)

Gate	30-second demo
G1	`adb install` shell app; icon launches; empty home; CI run link green with ≤ 8 jobs
G2	create doc, type, kill app mid-typing, reopen — words survived (journal proof on terminal)
G3	type with IME, select, bold, undo/redo; typing-latency overlay showing P95
G4	build a table, insert an image with alt text, instantiate "Letter" template, replace-all with version created
G5	export the same doc as PDF/md/docx; open the conversion note; import it back
G6	install OCR pack (SHA shown), scan a page to text; hide the pack → capability explains itself
G7	run the evidence PDF walkthrough: journeys strip, budget table, BLOCKED register
G8	release page: signed APK, SHA, provenance.json, notes; verify-by-checksum demo

Appendix T — Coverage matrix (features × gates × levels)

Rule: a feature ships only after its cases pass at their levels on the matrix AVDs (§23.1). The matrix below is the map the G7 audit walks; the machine check is `traceability.py --check --strict` plus the run manifest.

Area	Features	L1	L2	L4	Gates	P0 cases
Laws	L1–L10	✓	✓	✓	G3, G7	all
Library	F-001…012	—	✓	✓	G2–G7	001,002,003,005,012
Editor core	F-013…030	✓	✓	✓	G3, G4, G7	013–018,023,024,025,026,029,030
Structure	F-031…044	✓ (props)	✓	✓	G4, G7	031–033
Templates	F-045…050	—	✓	✓	G4, G7	045,047
Files/IO	F-051…060	✓ (goldens)	✓	✓	G5, G7	051–057,059
State	F-061…066	✓ (P8)	✓	✓	G2, G4, G7	061,062,063,065
Search	F-067…070	✓	✓	✓	G4, G6, G7	067,068
AI	F-071…082	—	✓	✓	G6, G7	071,073,074,075,076,077,078,079,082
Read/A11y	F-083…086	—	—	✓	G4, G7	084,085,086
Settings	F-087…090	—	✓	✓	G4, G7	—
UX	UX-01…50	—	part	✓	G3–G7	01–03,05,06,08,11,19,22,29,30,32,37,39
Security	TC-SEC-*	✓	✓	✓	G6, G7	all
Perf/Env	TC-PERF/ENV	—	—	✓	G1, G3, G7	001,002, ENV all
Journeys	TC-J1…J5	—	—	✓	G7, G8	all

Appendix U — Run manifest (case → gate → AVD → pipeline)

Every catalogue case appears exactly once here; the QA harness consumes this table as the run manifest (§24.7). `— (JVM)` rows run in pr/nightly JVM shards. Variants (TC-AI-004-b …, TC-J2-degraded …) are procedure variants of their base case (§24.5), not separate manifest rows.

Case	First gate	AVDs	Pipeline	Procedure source
TC-A11Y-001	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-A11Y-002	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-A11Y-003	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-AI-001	G6	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-AI-002	G6	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-AI-003	G6	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-AI-004	G6	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-AI-005	G6	36 + 26/33/35	instrumented + apk-qa	Appendix C/I
TC-AI-006	G6	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-AI-007	G6	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-AI-008	G6	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-AI-009	G6	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-AI-010	G6	36 + 26/33/35	instrumented + apk-qa	Appendix C/I
TC-AI-PACK-01	G6	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-AI-PACK-02	G6	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-DIAG-001	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-EDIT-001	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-EDIT-002	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-EDIT-003	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-EDIT-004	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-EDIT-005	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-EDIT-006	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-EDIT-007	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-EDIT-008	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-EDIT-009	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-EDIT-010	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-EDIT-011	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-EDIT-012	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-EDIT-013	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-EDIT-014	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-EDIT-015	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-EDIT-016	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-EDIT-017	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-EDIT-IME-01	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-EDIT-IME-02	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-EDIT-IME-03	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I

Case	First gate	AVDs	Pipeline	Procedure source
TC-EDIT-IME-04	G3	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-ENV-001	G7	26	apk-qa	Appendix C/I
TC-ENV-002	G7	36 (+33 ladder)	apk-qa	Appendix C/I
TC-ENV-003	G7	26/33/35/36 tablet	apk-qa	Appendix C/I
TC-ENV-004	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-IMG-001	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-IMG-002	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-IMG-003	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-IO-001	G5	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-IO-002	G5	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-IO-003	G5	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-IO-004	G5	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-IO-005	G5	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-IO-006	G5	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-IO-007	G5	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-IO-008	G5	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-IO-009	G5	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-IO-010	G5	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-IO-011	G5	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-IO-012	G5	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-IO-013	G5	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-LAW-001	G3	— (JVM)	pr + nightly	Appendix C/I
TC-LAW-002	G3	— (JVM)	pr + nightly	Appendix C/I
TC-LAW-003	G3	— (JVM)	pr + nightly	Appendix C/I
TC-LAW-004	G3	— (JVM)	pr + nightly	Appendix C/I
TC-LIB-001	G2	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-LIB-002	G2	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-LIB-003	G2	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-LIB-004	G2	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-LIB-005	G2	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-LIB-006	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-LIB-007	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-LIB-008	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-LIB-009	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-LIB-010	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-PERF-001	G7	33	apk-qa	Appendix C/I
TC-PERF-002	G7	33	apk-qa	Appendix C/I
TC-PROP-P01	G3	— (JVM)	pr + nightly	Appendix C/I
TC-PROP-P02	G3	— (JVM)	pr + nightly	Appendix C/I
TC-PROP-P03	G3	— (JVM)	pr + nightly	Appendix C/I
TC-PROP-P04	G3	— (JVM)	pr + nightly	Appendix C/I
TC-PROP-P05	G3	— (JVM)	pr + nightly	Appendix C/I
TC-PROP-P06	G3	— (JVM)	pr + nightly	Appendix C/I

Case	First gate	AVDs	Pipeline	Procedure source
TC-PROP-P07	G3	— (JVM)	pr + nightly	Appendix C/I
TC-PROP-P08	G3	— (JVM)	pr + nightly	Appendix C/I
TC-PROP-P09	G3	— (JVM)	pr + nightly	Appendix C/I
TC-PROP-P10	G3	— (JVM)	pr + nightly	Appendix C/I
TC-READ-001	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-SCAN-001	G6	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-SCAN-002	G6	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-SEC-IO-01	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-SEC-IO-02	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-SEC-P-01	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-SEC-P-02	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-SEC-P-03	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-SEC-P-04	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-SEC-PK-01	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-SEC-PK-02	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-SEC-ST-01	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-SEC-ST-02	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-SET-001	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-SET-002	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-SET-003	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-SET-004	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-SRCH-001	G5	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-SRCH-002	G5	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-SRCH-003	G5	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-SRCH-004	G5	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-SRCH-005	G5	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-STATE-001	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-STATE-002	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-STATE-003	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-STATE-004	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-STATE-005	G2	33 (100 trials)	instrumented + apk-qa	Appendix C/I
TC-STATE-006	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-STATE-007	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-STATE-008	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-STATE-009	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-STRUCT-001	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-STRUCT-002	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-STRUCT-003	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-STRUCT-004	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-STRUCT-005	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-STRUCT-006	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-STRUCT-007	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-STRUCT-008	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I

Case	First gate	AVDs	Pipeline	Procedure source
TC-STRUCT-009	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-TBL-001	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-TBL-002	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-TPL-001	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-TPL-002	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-TPL-003	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-TPL-004	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-TPL-005	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-TPL-006	G4	33 (+26/35/36 P0 sweep)	instrumented + apk-qa	Appendix C/I
TC-UX-001	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-UX-002	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-UX-003	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-UX-004	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-UX-005	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-UX-006	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-UX-007	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-UX-008	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-UX-009	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-UX-010	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I
TC-UX-011	G7	33 (+26/35/36 P0 sweep)	apk-qa	Appendix C/I

Count: 142 rows — must equal the catalogue case count (`traceability.py --check` cross-verifies).

Appendix V — Fixture expected outputs (the oracle tables)

V.1 OCR oracle (`scan/page-0[1-3].jpg`)

The fixture pages are synthetic printed letters (font: DejaVu Serif 11 pt on white). Expected transcripts (normalized whitespace; accuracy bar ≥ 95% word-level):

- **page-01** (82 words): business letter — salutation "Dear Ms. Iyer," body about a service visit on "14 March", closing "Sincerely, R. Menon".
- **page-02** (95 words): same letter page 2 — numbered list of 3 items ("inspect the intake valve", "replace the filter cartridge", "schedule a follow-up").
- **page-03** (77 words): terms paragraph containing the test phrase "warranty remains valid for twenty-four months" (the J4 cross-link target).

Assertions read: `ocr_output` vs oracle with Levenshtein word accuracy ≥ 0.95; low-confidence set must include at least the fixture's two deliberately degraded words (printed at 60% contrast on page-02: "cartridge" and "follow-up") — proving confidence marking works (UX-46), not just accuracy.

V.2 Dictation oracle (`audio/minutes-en-60s.wav`)

60-second two-speaker meeting fixture (synthetic voices, room tone −30 dB). Expected content: date line, three action items ("Ada to send the invoice", "Ben to review the lease", "next meeting Thursday"), two filler disfluencies the final text SHOULD still contain (verbatim ASR, no silent cleanup — the tidy pass is a separate, explicit step §19.3). Assertions: all three action items present; WER vs oracle ≤ 15%; first-partial latency measured from first phoneme.

V.3 Search oracle (seed-library-200)

The synthetic library embeds known hits: doc `s_041` contains "warranty" 3× (title match 0×); `s_112` contains "warranties" once; the oblique fixture `manual-appliances.odoc` contains the phrase once. Assertions for TC-SRCH-001: all three found; `s_041` ranks first (density+recency pins); snippet shows hit context; per-doc counts correct. Semantic variant (TC-J4-semantic): query "what covers repairs" must surface `manual-appliances` in related with cosine ≥ 0.55 using the pinned MiniLM model — this pins embedding pack behavior, not just plumbing.

V.4 Corpus truth table (counts used by count-assertions)

Fixture	Words	Chars	Blocks	Reading time
letter-1p.odoc	180	1,042	9	1 min
report-50p.odoc	11,860	71,338	214	47 min
walls-20k.odoc	3,341	20,000	1	14 min
marks-kitchen.odoc	412	2,688	18	2 min

Info sheet (TC-LIB-009), export counts (UX-49), and index stats (TC-SRCH-005) assert against these exact numbers (word counting = ICU grapheme-clusters per §18.7; the table is generated by the corpus build script and regenerated with any corpus change — never hand-edited).

Appendix W — Accessibility annotations per screen (traversal contracts)

Format: traversal order is the focus order; every row = one a11y node with (role, label source, actions). Full tables live in the registry CSV's `a11y` column; this appendix pins the editor and the two riskiest screens.

W.1 Editor (OD-EDIT-001/002)

Order: app-bar(back: "Back", title, undo, redo, overflow) → page content as block list (each block: role "text field"/"heading N"/"list item"; label = text content with mark prefixes "bold, "; actions: format menu, move up, move down, duplicate, delete) → selection toolbar nodes when visible (each with state on/off) → assistant row (each verb + availability reason) → tab row (4 tabs, selected state). Rules: table blocks expand to cell matrix (rows×cols collection); image blocks announce alt text or "image, no description yet"; page-break guides are non-focusable; IME compose region announced as "composing text".

W.2 Pack manager (OD-AI-001)

Order: app bar → education/empty card (if no packs) → pack rows (label = name+capability+language; state = verified/unverified/failed; actions: install, verify details, delete) → install button (SAF flow). Failure state announces both hashes (UX-48) via live-region polite.

W.3 Version restore (OD-FILE-001)

Order: app bar → list (label = kind + relative time + size; state = kind) → preview action → restore action (announces "Your current text will be saved as a version first"). Preview mode announces "previewing version from {time}; back to return".

Appendix X — Compliance map (constraint → verifying evidence)

One line per hard rule — the audit path at G7/G8:

Rule	Verified by	Where evidence lands
C1 no INTERNET	TC-SEC-P-01 (gate) + TC-SEC-P-03 (run)	CI gate log + evidence PDF §3
C2 no sign-in	dependency scan + string scan + journeys offline	§26 CI job + TC-ENV-004
C3 no ads/trackers	deny-list audit + TC-SEC-P-02/03	CI job + evidence PDF
C4 private storage	TC-SEC-ST-01/02 + FileProvider checks	security suite report
C5 packs never download	pack tests + network probe during AI flows	TC-AI-PACK-* + TC-SEC-P-03
C6 API 26+ matrix	TC-ENV-001…003	matrix results §24 report
C7 laws blocking	LAW suite + G3/G7 gate DoD	gate reports
C8 CI budgets	workflow configs + job counts per run	run links in gate reports
C9 writer flavor assembles	pr.yml flavor matrix job	CI history
C10 provenance	provenance.json + §28.5 runbook	release bundle
Laws L1–L10	LAW suite + property P1–P10	G3/G7 evidence
UX-01…50	UX-TC cases + DQ gate	evidence PDF §4

The map is regenerated with the catalogue (`traceability.py --emit-compliance`) so a rule without evidence is visible in CI, not discovered at the gate.

Appendix Y — The v1.0 build plan (seed tasks per gate)

Sequenced seed tasks; the agent expands just-in-time (§29.3) into `docs/onedoc/plan/task-graph.yaml`. Dependencies are within-gate unless noted. IDs are stable (T-Gn-nnn).

Task	Gate	Depends	Deliverable (DoD hint)
T-G0-001	G0	—	owner acceptance recorded; ledgers seeded; traceability green
T-G1-001	G1	T-G0-001	JDK/SDK/AVDs per App. E installed and boot-verified
T-G1-002	G1	T-G1-001	Gradle skeleton: all modules assemble (full+writer flavors)
T-G1-003	G1	T-G1-002	CI pr.yml green ≤ 8 jobs; budgets enforced (§22)
T-G1-004	G1	T-G1-002	shell app: home empty-state + settings stub + About; icon/theme tokens in
T-G1-005	G1	T-G1-001	qa-driver module skeleton; AVD boot snapshots; N.3 ritual scripted
T-G2-001	G2	T-G1-002	core-model: blocks/marks/ops/normalize; invariants enforced
T-G2-002	G2	T-G2-001	property harness + P1–P10 green (§16)
T-G2-003	G2	T-G2-001	core-storage: Room schema v3 exported; ERD as built (§11)
T-G2-004	G2	T-G2-003	journal write/replay/compact; kill-trial harness local pass (§12)
T-G2-005	G2	T-G2-003	odoc container save/open + goldens; LAW-001/004 green
T-G3-001	G3	T-G2-005	editor screen: block list + caret + keyboard bridge (§15.1)
T-G3-002	G3	T-G3-001	IME contract + fuzzer harness green on 4 AVDs
T-G3-003	G3	T-G3-001	selection engine + toolbar state machine (§15.6/15.7)
T-G3-004	G3	T-G3-003	marks engine + §14.5 semantics + EDIT-005/006
T-G3-005	G3	T-G3-001	undo/redo + coalescing + EDIT-003; typing budget met
T-G3-006	G3	T-G3-005	editor core DQ pass; a11y node contract (W.1)
T-G4-001	G4	T-G3-004	tables (§17.1) + TBL cases
T-G4-002	G4	T-G3-004	images/assets/alt (§17.2) + IMG cases
T-G4-003	G4	T-G3-004	lists + quote/code + EDIT-007/008
T-G4-004	G4	T-G3-003	find bar + replace-all + versions-first (F-025/026)
T-G4-005	G4	T-G2-004	versions UI + restore + STATE-003/004
T-G4-006	G4	T-G2-003	templates: 12 built-ins + save-as (§17.5) + TPL cases
T-G4-007	G4	T-G4-001	TOC/map + page setup/guides (§17.3/17.4)
T-G4-008	G4	T-G4-007	reader mode (§17.11) + READ-001
T-G4-009	G4	T-G1-004	library complete: filters, trash, multi-select, info sheet
T-G4-010	G4	all G4	DQ-1…8 pass on every G4 screen
T-G5-001	G5	T-G2-005	md/txt writers + goldens
T-G5-002	G5	T-G5-001	html writer/reader (tokenizer) + SEC-IO-01
T-G5-003	G5	T-G5-001	docx writer/reader + round-trip gate
T-G5-004	G5	T-G5-002	PDF export via Print framework + STRUCT-004 parity
T-G5-005	G5	T-G5-003	share/open-in/SAF + IO-007/008/009/012
T-G5-006	G5	T-G5-001	conversion-note UX + refusal paths + IO-005/006/011
T-G5-007	G5	T-G4-005	backup/restore encrypted + IO-010/013
T-G6-001	G6	T-G5-003	pack manager + .odpack verify + AI-PACK cases

Task	Gate	Depends	Deliverable (DoD hint)
T-G6-002	G6	T-G6-001	OCR pipeline (§19.12) + scan flows + SCAN/AI-001/002
T-G6-003	G6	T-G6-001	ASR streaming + dictation UX + AI-003
T-G6-004	G6	T-G6-001	LLM runtime (Tier A) + 4 verbs + preview cards + AI-004/005/006/010
T-G6-005	G6	T-G6-001	embeddings + semantic search + SRCH-004/005
T-G6-006	G6	T-G4-008	read aloud + AI-007/008
T-G6-007	G6	T-G6-004	ladder + hidden-with-reason everywhere + AI-009
T-G6-008	G6	T-G6-002	network-silence probe across AI flows (SEC-P-03)
T-G7-001	G7	all	full catalogue on matrix; journeys J1–J5
T-G7-002	G7	T-G7-001	perf budgets + size gate evidence
T-G7-003	G7	T-G7-001	security suite + compliance map (X) current
T-G7-004	G7	T-G7-001	evidence PDF assembled; BLOCKED register reviewed
T-G8-001	G8	T-G7-004	release checklist (§28.1); provenance; GitHub Release; handoff note

Rule: a task moves to `done` only with its evidence link in the graph; `blocked` requires reason + unblock plan (§29.4).

Appendix Z — Redline tables for the remaining surfaces

Z.1 Find & replace bar (OD-EDIT-004)

Bar height 48 dp under app bar; fields: find (auto-focused, last term preselected — UX-05), match count "n of total", prev/next (44 dp), toggle-case (Aa), whole-word (ab|), replace field + Replace / Replace all. Dismiss (×) clears highlights, leaves document untouched. Geometry: bar never pushes content down — it overlays with canvas inset animation 200 ms.

Z.2 Replace-all confirm (dialog)

Title "Replace {n} matches?"; body one line: "A version of this document is saved first."; buttons Cancel / Replace (filled, amber-on-dark in HC themes). Second confirm never appears — the version IS the safety (UX-16).

Z.3 Versions list (OD-FILE-001) redlines

Row: kind chip (Auto gray / Manual teal / Before change amber) + relative time + size; tap row = preview (read-only banner "Preview — Restore to edit"); Restore = primary button on preview, confirm only when current doc has unsaved-seeming deltas > 0 (it never does — autosave) — so no confirm: the pre-edit snapshot is the safety. Empty state: "Versions appear as you work. Nothing to manage yet."

Z.4 Template gallery (OD-TPL-001/002)

Grid 2-col cards: name + 3-line ghost preview (no real text — privacy in screenshots); preview screen = reader-mode render with placeholders as amber pills; create = primary button "Use template" → new doc with placeholders filled (date auto). User section header "Your templates" with delete via long-press → confirm dialog (destructive).

Z.5 Image alt-text editor (modal over editor)

Field focused on open (UX-38); helper "Describe this image (helps screen readers)"; Skip = secondary text button (never silent — dotted outline persists per §17.2); Save = primary. The editor also lists alt text from TalkBack (W.1 contract).

Z.6 TOC insert sheet

Options: depth (H1 / H1–2 / H1–3 default), style (links on tap → scroll). Insert places TOC block at caret; empty-heading edge: TOC renders "Add headings to fill this table." state (DQ-8).

Appendix AB — Journey step tables (driver-ready)

Each step: action (driver command class), assertion, evidence. Verbatim source for TC-J* procedures (§24.5).

AB.1 J1 — letter to PDF

#	Action	Assertion	Evidence
1	cold launch (data cleared)	home empty-state visible ≤ 1.2 s (frame log)	screenshot + startup log
2	tap FAB	editor visible, caret in body, no dialogs	screenshot
3	type 40-word salutation+body (scripted)	text present; count chip = truth	a11y tree dump
4	select first sentence	toolbar visible, not covering selection	screenshot
5	tap Bold	marks applied (tree: bold prefix)	tree
6	cursor to title line, apply H1	heading role on block	tree
7	menu → Export PDF	export sheet, fidelity badge F3	screenshot
8	save via SAF (Downloads)	done state ≤ 10 s (50th-pct of 3 runs)	log + timing
9	open exported PDF (system viewer)	pages render, text selectable (pdftotext on pull)	artifact PDF
10	assertions rollup	≤ 2 taps to editor; word count ±1%; total ≤ 3 min median	steps.json

AB.2 J2 — scan three pages

#	Action	Assertion
1	FAB → Insert → Scan	camera preview (virtual scene fixture) ≤ 2 s
2	capture page-01/02/03	3 thumbnails queued
3	drag page-02 before page-01	order persists in queue
4	run OCR	processing state ≤ 12 s/page; cancel button live
5	review screen	low-confidence words marked (incl. fixture's 2 degraded words)
6	fix "cartridge" via fix field	corrected text replaces; mark cleared
7	Insert	paragraphs in chosen order; page separators; images as assets
8	conversion note	"OCR from 3 pages; 2 low-confidence words flagged; source images attached."
variant	no pack installed	flow ends after crop at save-image with reason (F-073)

AB.3 J3 — dictate, tidy, share DOCX

#	Action	Assertion
1	new doc → dictate	mic permission at first use (UX-32); strip appears
2	play fixture audio	first partial ≤ 1.5 s; partials stream
3	stop	preview card with method label
4	Insert	one grouped op (single undo reverts)
5	H1 the date line; select paragraph	toolbar states correct
6	Summarize	rung-labeled card (extractive on 33; gemma on 36)
7	Insert below	pre_edit version created (versions screen count +1)
8	Export DOCX → share sheet	docx received by test receiver app
9	re-import the docx	headings/list parity per F2; note present

AB.4 J4 — find the warranty clause

#	Action	Assertion
1	seed 200-doc corpus + manual	library shows 201 docs
2	search "warr" (2+ chars)	results streaming ≤ 300 ms; 3 docs hit
3	tap manual result	editor at hit block; "warranty" highlighted
4	back → search "what covers repairs" (embed pack)	related section labels manual; FTS section still primary
variant	no pack	related section absent; no error

AB.5 J5 — recover deleted half

#	Action	Assertion
1	open report-50p.odoc	renders, position top
2	select blocks 100–214, delete	content gone; count drops
3	type replacement sentence	present
4	wait 1 s → `am kill` → relaunch → open doc	ALL content intact (zero loss) — journal replay
5	Versions → preview pre-delete auto snapshot	preview renders read-only
6	Restore	pre_edit snapshot created; deletion undone; redo cleared
7	timing	restore ≤ 30 s (50-page doc)

Appendix AC — Screen-to-feature trace

Screen	Implements features	Cases exercising it
OD-HOME-001/002/003	F-001…012, F-067 (entry)	LIB-001…010, SRCH-001/002, PERF-001
OD-EDIT-001 compose	F-013…030, F-043, F-088	EDIT-*, IME-*, PROP (indirect)
OD-EDIT-002 selection	F-016/017, F-041/042, UX-19/20/21	EDIT-004/005, UX-006, A11Y-001
OD-EDIT-003 insert sheet	F-031…040, F-071 (entry)	TBL/IMG/STRUCT cases, SCAN-001
OD-EDIT-004 find bar	F-025/026, F-069	EDIT-013/014, SRCH-003
OD-TPL-001/002	F-045…050	TPL-001…006
OD-SCAN-001…003	F-071…073	SCAN-001/002, AI-001/002, J2
OD-AI-001…003	F-074…082, F-078/079	AI-*, AI-PACK-*, J3
OD-FILE-001 versions	F-061…064	STATE-003…006, J5
OD-FILE-002/003	F-051…060	IO-*
OD-SET-001…004	F-087…090, F-089 self-test	SET-001…004, DIAG-001

Appendix AD — Locale deck (hi + mr for the P.1–P.5 core keys)

Translation register: formal-polite (□□), same sentence-case discipline (no exclamation marks), numbers in Devanagari-friendly humanized form. Keys mirror Appendix P exactly; the strings lint fails a release if any key is missing in any shipped locale (F-090).

hi (□□□□□□) — core keys

Key	String
action.new.doc	□□□ □□□□□□□□□
action.export	□□□□□□□
action.rename	□□□ □□□□□
action.move.trash	□□□□□ □□□ □□ □□□□
action.restore	□□□□□□□□□□□ □□□□
editor.count	{words} □□□□ · {chars} □□□□□
editor.find.count	{total} □□□ □□ {n}
ai.summarize / rewrite / continue / fix	□□□□□□ · □□□ □□ □□□□□ · □□□ □□□□□ · □□□□□□□ □□□ □□□□
ai.unavailable.ram	{model} □□ {need} GB RAM □□□□□ — □□ □□□□ □□□ {have} GB □□□
pack.verify.fail	□□□□ □□□□□ □□□□□□□□ {expected}, □□□□ {found}।
trash.retention	□□□□□ □□□□□□□□□ 30 □□□ □□□□ □□□□ {days} □□□ □□□□□
welcome.title	□□□□ □□□□, □□□□ □□□□□
welcome.body	OneDoc □□□□ □□□ □□□□□□□ □□□□ □□□ । □□□□□□, । □□□□□□□□, । □□□□□□□□□
welcome.cta	□□□□□ □□□□ □□□□

mr (□□□□□) — core keys

Key	String
action.new.doc	□□□□ □□□□□□□
action.export	□□□□□□□
action.rename	□□□ □□□□
action.move.trash	□□□□□ □□□□
action.restore	□□□□□□ □□□
editor.count	{words} □□□□ · {chars} □□□□□
editor.find.count	{total} □□□□ {n}
ai.summarize / rewrite / continue / fix	□□□□□□ · □□□□□□ □□□□ · □□□□ □□□□ · □□□□□□□ □□□□□□□
ai.unavailable.ram	{model} □□ {need} GB RAM □□□□□ — □□ □□□□□□□□ {have} GB □□□.
pack.verify.fail	□□□□□□ □□□□□□□. □□□□□□□□ {expected}, □□□□□ {found}.
trash.retention	□□□□□□□ □□□□□□□ 30 □□□□ □□□□□□. {days} □□□□ □□□□□□.
welcome.title	□□□□□ □□□□, □□□□□ □□□.
welcome.body	OneDoc □□□□□□□□ □□□□□□ □□□□□. □□□□ □□□□, □□□□□□□□ □□□□, □□□□□□□□ □□□□.
welcome.cta	□□□□□□□ □□□□□□□ □□□

(The full 120-key deck lives in `core/designsystem/strings.csv` generated with the build; this appendix pins the register and the highest-traffic strings. Locale runs per §23.5 verify rendering + truncation at 200%.)

Appendix AE — Test-data provenance and privacy statement

All fixtures (Appendix J) are synthetic: generated letters, TTS-spoken audio with no real person, no real names, no GPS. No fixture encodes personal data; the EXIF fixture's GPS point is a null-island coordinate. This matters for two reasons: runs are reproducible byte-for-byte anywhere (Law L6 spirit), and the evidence PDF can be shared publicly without redaction. If the owner later contributes real documents for testing, they enter a separate `contrib/` directory (git-ignored by default, owner-synced), never the committed corpus.

Appendix AO — Owner FAQ (the ten questions, answered once)

1. **Where do my documents live?** Inside the app's private storage on your phone, encrypted where your phone has a lock screen (§11.5). Backups go only where you choose.
2. **Can OneDoc send my text anywhere?** No — there is no internet permission in the app at all (C1); tests on the release APK prove zero network use (TC-SEC-P-03).
3. **What are the packs?** Optional AI models you side-load from files you hold; each is hash-verified; the app never downloads (§19.6).
4. **Why does the assistant disappear on my phone?** Its controls hide when the hardware can't run them — long-press any AI spot for the reason (§19.4). Nothing is broken.
5. **What happens if the app is killed mid-sentence?** Nothing is lost; the journal has it (Law L3, proven by 100 kill tests).
6. **Can I get my work into Word/Google Docs?** Yes — export DOCX/PDF/Markdown/HTML with stated fidelity (§18.2); the conversion note tells you exactly what kept or dropped.
7. **Does it work in airplane mode?** One hundred percent of it — that is the product (P1).
8. **Why no comments/co-editing?** OneDoc is a single-user, single-device studio by charter (§1.4); collaboration needs servers, and servers break every promise above.
9. **How do I know a release is the real one?** Compare the APK's SHA-256 to the provenance file; 5-minute runbook in §28.5.
10. **What if I find a bug?** The evidence PDF and defect tracker (§25) show how every claim is tested; report it and it joins the same system with a test that fails first.

Appendix AF — Complete en strings deck (v1.0, 110 keys)

The canonical table; `core/designsystem/strings.csv` is generated from it (and from the hi/mr deck, Appendix AD). Lint: feature code references keys only.

Key	en
action.new.doc	New document
action.save.version	Save version
action.export	Export
action.share	Share
action.rename	Rename
action.duplicate	Duplicate
action.star	Star
action.unstar	Unstar
action.move.trash	Move to trash
action.restore	Restore
action.delete.forever	Delete forever
action.undo	Undo
action.redo	Redo
action.find.replace	Find and replace
action.insert	Insert
action.print	Print
action.save.template	Save as template
action.select.all	Select all
action.clear.marks	Clear formatting
editor.tab.text	Text
editor.tab.insert	Insert
editor.tab.ai	AI
editor.tab.format	Format
editor.count	{words} words · {chars} characters
editor.find.hint	Find in document
editor.find.count	{n} of {total}
editor.replace.hint	Replace with
editor.replace.all.confirm	Replace {n} matches? A version is saved first.
editor.link.hint	Paste or type a link
editor.image.alt.hint	Describe this image (helps screen readers)
editor.image.alt.skip	Skip for now
editor.composing.protected	
ai.summarize	Summarize
ai.rewrite	Rewrite
ai.continue	Continue writing
ai.fix	Fix grammar
ai.rewrite.polish	Polish

Key	en
ai.rewrite.formal	Formal
ai.rewrite.casual	Casual
ai.method.llm	On-device gemma-2B
ai.method.extractive	On-device extractive summary
ai.method.system	Android speech service
ai.insert	Insert
ai.discard	Discard
ai.replace	Replace
ai.cancel	Stop
ai.unavailable.pack	Needs the {pack} pack — install it from Assistant.
ai.unavailable.ram	{model} needs {need} GB RAM — this device has {have}.
ai.unavailable.tier	{feature} needs a stronger phone than this one.
ai.unavailable.tts	No speech service on this device.
ai.descended	Switched to the lighter on-device method (thermal).
ai.timed.out	That took too long. Try the lighter method?
pack.install	Install pack
pack.delete	Delete pack
pack.verify.fail	Verification failed. Expected {expected}, found {found}.
pack.license	License: {spdx}
pack.tier.req	Needs a Tier {tier} device (this is Tier {have}).
export.pdf	Export as PDF
export.odoc	Export as OneDoc (.odoc)
export.md	Export as Markdown
export.txt	Export as text
export.html	Export as HTML
export.docx	Export as Word (.docx)
fidelity.F0	Not supported
fidelity.F1	Best effort
fidelity.F2	Subset lossless
fidelity.F3	Lossless
import.refused.format	{fmt} isn't supported. Supported: odoc, md, txt, html, docx.
import.refused.corrupt	This file is damaged ({why}). OneDoc won't risk opening it.
import.refused.size	That file is over 50 MB — too big for a phone to convert safely.
import.refused.binary	That looks like a binary file, not a document.
import.note.prefix	Imported as {fmt} — {level}.
export.denied.folder	OneDoc can't write to that folder. Pick a different location.
export.retrying	Writing was interrupted. Retrying…
err.disk.full	Storage is full. Free some space — OneDoc will retry.
err.storage.retry	Storage hiccup. Retrying…
err.doc.repair	This document needed a small repair. Your text is safe.
err.doc.newer	This document was made by a newer OneDoc.
err.something.broke	Something went wrong. Your text is safe.
err.upgrade.repair	Finishing an update to your library…

Key	en
trash.retention	Deleted documents are kept 30 days. {days} left.
version.auto	Auto
version.manual	Manual
version.pre_edit	Before change
restore.done	Restored. Your previous text is saved as a version.
backup.prompt.pass	Set a passphrase to encrypt your backup
backup.enabled	Weekly encrypted backup
storage.usage	Documents {docs} · Versions {ver} · Packs {packs}
perm.mic.why	OneDoc uses the microphone only while you dictate. Nothing is saved or sent.
perm.camera.why	OneDoc uses the camera only to scan pages into text.
welcome.title	Your words, your phone.
welcome.body	OneDoc works fully offline. No account, no ads, no tracking.
welcome.cta	Start writing
about.selftest	Run self-test
about.selftest.done	All checks passed ({n}) in {s} s
about.version	Version {name} ({code}) · build {sha}
about.repo	Source: github.com/tarungittarun/AIA-
announce.export.done	Export finished
announce.restore.done	Restore finished
announce.gen.done	Assistant finished
announce.verify.done	Pack verification finished
hc.storage.full	Storage almost full — OneDoc is keeping your text in memory.
hc.journal.repair	OneDoc repaired this document after an interruption. Details in Diagnostics.
scan.no.pack	OCR needs the ocr-en pack. For now you can save the page image.
scan.low.yield	Not much text found. Retake with better light?
search.related	Related
search.results.hint	Search documents
tpl.use	Use template
tpl.save.as	Saved to your templates
img.deleted.used	This image is used. Delete it from the page first.

Appendix AI — Post-release 30/60/90 sketch (owner-facing, not scheduled work)

Day 30: first maintenance release candidate if tracker holds S2+ fixes; locale deck review with owner; battery/thermal field data from owner's device (diagnostics export, manual share). **Day 60:** v1.1 shelf review — D-R102/103/104 graduate by owner decision or wait; pack ecosystem check (any pack format drift from runtime updates). **Day 90:** dependency refresh audit (pinned bumps, deny-list re-run, SBOM diff), full-matrix re-run on the current release artifact (proof the promise still holds on a stale binary — the strongest offline claim OneDoc can make).

Appendix AG — D-reserved decisions (pre-numbered, awaiting their moment)

Reserved ledger entries the program already knows it will need; each activates with evidence when its trigger fires (§29.2 format):

- **D-R101** — LLM tier floor move (trigger: R2 budget miss on physical Tier A).
- **D-R102** — markdown input shortcuts (trigger: G4 UX debt review; shelf §30.2).
- **D-R103** — neural TTS pack (trigger: owner request; pack format ready).
- **D-R104** — hi/mr ASR packs (trigger: fixture corpus + accuracy bar defined).
- **D-R105** — writer-flavor distribution (trigger: owner opt-in; flavor already green).
- **D-R106** — PDF import via OCR pipeline (trigger: J2 accuracy on real-world PDFs ≥ bar).
- **D-R107** — tracked-changes preservation in DOCX (trigger: owner need; reader work sized first).
- **D-R108** — additional themes (trigger: DQ-4 table extension costed; no token literals ever).

Appendix AH — Owner's guide to the evidence PDF (what "good" looks like)

Cover math must reconcile: case count here = catalogue count; verdict counts sum to it. Journeys: every step green with a screenshot that visibly matches the step (a scan journey whose "review" shot shows a keyboard is a red flag). Budgets: actuals within budgets with headroom — headroom is the safety margin for physical devices. BLOCKED register: entries with reasons you understand in one line; any entry older than two cycles should annoy you — ask for the tooling-change plan. Defects: S1/S2 count zero *at release candidates*; a tracker with fixed-and-verified rows is healthy history, not failure. Provenance: SHA in the PDF = SHA of the APK you downloaded — recompute it once; that single check converts trust into verification, which is the entire point of P5.

Appendix AM — Figure index

File	Where referenced
cover.png	Parts II–X (caption-referenced)
fig-ai-ladder.png	Parts II–X (caption-referenced)
fig-ai-stack.png	Parts II–X (caption-referenced)
fig-architecture.png	Parts II–X (caption-referenced)
fig-defects.png	Parts II–X (caption-referenced)
fig-docmodel.png	Parts II–X (caption-referenced)
fig-editor-wireframe.png	Parts II–X (caption-referenced)
fig-erd.png	Parts II–X (caption-referenced)
fig-figma-workflow.png	Parts II–X (caption-referenced)
fig-formats.png	Parts II–X (caption-referenced)
fig-gates.png	Parts II–X (caption-referenced)
fig-home-wireframe.png	Parts II–X (caption-referenced)
fig-navigation.png	Parts II–X (caption-referenced)
fig-product-map.png	Parts II–X (caption-referenced)
fig-testing.png	Parts II–X (caption-referenced)
fig-tokens.png	Parts II–X (caption-referenced)
fig-versioning.png	Parts II–X (caption-referenced)

Figures regenerate from `tools/onedoc/figures.py`; the PDF embeds them via the content files only — no manual image placement anywhere.

Appendix AK — Feature → bindings quick table (audit view)

F	Name	Pri	Cases
F-001	One-tap blank document	P0	TC-LIB-001, TC-UX-001
F-002	Inline rename; title mirrors first heading	P0	TC-LIB-002
F-003	Autosave via journal + debounced save	P0	TC-STATE-001, TC-LAW-001
F-004	Star/pin	P1	TC-LIB-003
F-005	Duplicate document	P1	TC-LIB-004
F-006	Trash with undo snackbar	P0	TC-LIB-005
F-007	Trash retention 30 days, restore, permanent delete	P1	TC-LIB-006
F-008	Sorting modes persisted	P2	TC-LIB-007
F-009	Empty states with CTA	P1	TC-LIB-008
F-010	Document info sheet	P2	TC-LIB-009
F-011	Multi-select bulk actions	P1	TC-LIB-010
F-012	Cold start < 1.2 s Tier B / 200 docs	P0	TC-PERF-001
F-013	Block editing with IME composition safety	P0	TC-EDIT-001, TC-EDIT-IME-01
F-014	Enter split / Backspace merge	P0	TC-EDIT-002
F-015	Undo/redo global stack, coalesced, grouped composites	P0	TC-EDIT-003, TC-PROP-P01
F-016	Selection handles + floating toolbar	P0	TC-EDIT-004
F-017	Marks set with non-overlap invariant	P0	TC-EDIT-005, TC-LAW-002
F-018	Headings H1–H3, outline-driven	P0	TC-EDIT-006
F-019	Lists: bullet, numbered, checklist	P0	TC-EDIT-007
F-020	Quote and code blocks	P1	TC-EDIT-008
F-021	Per-block alignment	P1	TC-EDIT-009
F-022	Divider block	P2	TC-EDIT-010
F-023	Paste (text/HTML subset/image), sanitized	P0	TC-EDIT-011
F-024	Cut/copy multi-block; paste round-trip	P0	TC-EDIT-012
F-025	Find & replace with pre-edit version on replace-all	P0	TC-EDIT-013
F-026	Find bar counts and navigates hits	P0	TC-EDIT-014
F-027	Live word/char count + info sheet	P1	TC-EDIT-015
F-028	Hardware keyboard shortcuts	P1	TC-EDIT-016
F-029	Typing P95 < 50 ms; no burst frame drops	P0	TC-PERF-002
F-030	Orientation/process-death state restore	P0	TC-STATE-002
F-031	Tables: structure ops, marks in cells	P0	TC-TBL-001
F-032	Table width presets, wrap, no horizontal scroll	P0	TC-TBL-002
F-033	Image insert (gallery/camera), captions, viewer	P0	TC-IMG-001
F-034	Image resize presets + alt text	P1	TC-IMG-002
F-035	TOC block from headings	P1	TC-STRUCT-001
F-036	Document map sheet	P2	TC-STRUCT-002
F-037	Page setup (paper/margins)	P1	TC-STRUCT-003
F-038	Page-break guides parity with PDF ±1 line	P1	TC-STRUCT-004
F-039	Footnotes (editor + PDF)	P2	TC-STRUCT-005

F	Name	Pri	Cases
F-040	Cover block	P2	TC-STRUCT-006
F-041	Drag-to-reorder blocks + a11y equivalent	P1	TC-STRUCT-007
F-042	Duplicate/delete block	P1	TC-STRUCT-008
F-043	Code block language label	P2	TC-STRUCT-009
F-044	Asset manager, unused cleanup	P2	TC-IMG-003
F-045	12 built-in templates	P0	TC-TPL-001
F-046	Template preview	P1	TC-TPL-002
F-047	Template instantiate-as-copy	P0	TC-TPL-003
F-048	Placeholders incl. auto date	P1	TC-TPL-004
F-049	Save-as-template from documents	P1	TC-TPL-005
F-050	Template gallery states, user section	P2	TC-TPL-006
F-051	PDF export (Print framework, embedded fonts)	P0	TC-IO-001, TC-STRUCT-004
F-052	.odoc export lossless	P0	TC-IO-002, TC-LAW-001
F-053	Markdown/txt/HTML export per fidelity	P0	TC-IO-003
F-054	DOCX export (OOXML subset)	P1	TC-IO-004
F-055	Import odoc/md/txt/html/docx with notes	P0	TC-IO-005
F-056	Refuse unsupported with reason	P0	TC-IO-006
F-057	Share via export + FileProvider	P0	TC-IO-007, TC-SEC-IO-02
F-058	Open-in intent filters	P1	TC-IO-008
F-059	SAF import/export	P0	TC-IO-009
F-060	Encrypted weekly backup to SAF (off default)	P2	TC-IO-010, TC-IO-013
F-061	Versions: auto/manual/pre-edit policy	P0	TC-STATE-003
F-062	Version preview + non-destructive restore	P0	TC-STATE-004
F-063	Journal durability (kill -9 zero loss)	P0	TC-STATE-005, TC-LAW-003
F-064	Storage accounting + asset cleanup	P2	TC-STATE-006
F-065	Schema migration chain tested	P0	TC-STATE-007
F-066	Multi-window save-on-focus-loss	P1	TC-STATE-008
F-067	FTS search < 300 ms / 200 docs	P0	TC-SRCH-001
F-068	Deep-link to hit block with highlight	P0	TC-SRCH-002
F-069	Shared highlight system with find bar	P1	TC-SRCH-003
F-070	Semantic search when pack present, labeled	P2	TC-SRCH-004
F-071	Scan-to-doc with confidence marks	P0	TC-SCAN-001, TC-AI-001
F-072	Multi-page scan queue + reorder	P1	TC-SCAN-002
F-073	OCR on-device only; graceful absence	P0	TC-AI-002
F-074	Dictation with live partials, explicit insert	P0	TC-AI-003
F-075	Assistant verbs with preview cards	P0	TC-AI-004
F-076	gemma-2B Tier A; extractive fallbacks elsewhere	P0	TC-AI-005
F-077	Streamed, cancellable, pre-edit snapshotted	P0	TC-AI-006
F-078	Pack manager: side-load, SHA verify, delete	P0	TC-AI-PACK-01
F-079	Corrupt pack refused; ladder descent	P0	TC-AI-PACK-02
F-080	Read aloud with sentence highlight	P1	TC-AI-007, TC-READ-001
F-081	Read-aloud ranges + system voice/rate	P2	TC-AI-008
F-082	Unavailable AI hides with reason	P0	TC-AI-009

F	Name	Pri	Cases
F-083	Reader mode	P1	TC-READ-001
F-084	TalkBack block traversal + actions	P0	TC-A11Y-001
F-085	Contrast ≥ 4.5:1 all themes	P0	TC-A11Y-002
F-086	200% dynamic type, no clipping	P0	TC-A11Y-003
F-087	8 themes, match-system default	P1	TC-SET-001
F-088	Editor settings (font/size/spell)	P1	TC-SET-002
F-089	About + diagnostics + self-test	P1	TC-SET-003, TC-DIAG-001
F-090	Locales en/hi/mr, RTL mirroring	P2	TC-SET-004

Appendix AL — UX criteria → verification method map

UX	Criterion (short)	Automated as
UX-01	FAB → editor directly, doc on first keystroke	driver timing + tap-count
UX-02	Blank→first char ≤ 2 taps, ≤ 2 s	driver timing + tap-count
UX-03	Export PDF ≤ 3 taps	tap-count + state walk
UX-04	Import ≤ 2 taps	tap-count + state walk
UX-05	Find bar remembers term, clean dismiss	tap-count + state walk
UX-06	AI/paste inserts undo as one group	timed procedure (budget)
UX-07	Restore ≤ 30 s / 50 pages	timed procedure (budget)
UX-08	Search results from ≤ 2 chars	timed procedure (budget)
UX-09	Scan page → text ≤ 60 s Tier B	timed procedure (budget)
UX-10	Dictation first partial ≤ 1.5 s	timed procedure (budget)
UX-11	kill mid-typing: zero loss	kill-trial (STATE-005 evidence)
UX-12	Rotation/split keeps cursor + composition	rotation/IME instrumentation
UX-13	No save dialogs, autosave silent	silent-save assert
UX-14	Trash undo 7 s; 30-day retention	snackbar timing probe
UX-15	Failed import: no empty doc, clipboard intact	corruption drill variant
UX-16	Replace-all: confirm + pre-edit version + undo	corruption drill variant
UX-17	Storage-full: in-memory continue, auto retry	corruption drill variant
UX-18	Export failure: no partial file, specific reason	corruption drill variant
UX-19	Toolbar never covers selection	geometry unit test + dp audit
UX-20	Targets ≥ 48 dp (44 toolbar)	geometry unit test + dp audit
UX-21	System-standard long-press select; stylus/mouse	instrumented interaction
UX-22	IME composition: no flicker/dup/loss	IME scripts + recording
UX-23	Enter continues list; exits on empty	instrumented interaction
UX-24	Style switch preserves marks	instrumented interaction
UX-25	Scroll anchored during IME show/hide	instrumented interaction
UX-26	No whole-doc reflow while typing	instrumented interaction
UX-27	Scroll mid-composition cancels cleanly	rotation/IME instrumentation
UX-28	Hardware shortcuts match platform; undo parity	instrumented interaction
UX-29	AI only via preview + explicit insert	UI state + string assertions
UX-30	Unavailable AI says why in one line	UI state + string assertions
UX-31	No notifications beyond user-created/progress	notification listener probe
UX-32	One-card first launch; permissions at use	fresh-install run
UX-33	Delete warns version count; trash double-confirm	flow + confirm-count assert
UX-34	Diagnostics on-device only, self-test ≤ 20 s	self-test timer probe
UX-35	Destructive: haptic + undoable/double-confirm	flow + confirm-count assert
UX-36	Empty states one line one action	state-walk (DQ-8)
UX-37	TalkBack-complete letter flow	TalkBack full-flow instrumentation
UX-38	Alt-text prompted at insert	insert-flow focus assert
UX-39	200% scale: no clipping	emulator walk @200% + screenshot diff
UX-40	Dark + high-contrast themes pass contrast	WCAG compute from tokens (DQ-4)

UX	Criterion (short)	Automated as
UX-41	Color never sole carrier	render + RTL/locale sweep
UX-42	Primary actions bottom 40%	driver timing + tap-count
UX-43	RTL mirroring correct	render + RTL/locale sweep
UX-44	Reduced-motion honored	reduced-motion toggle test
UX-45	Import states achieved fidelity	UI state + string assertions
UX-46	OCR low-confidence marked/fixable; export honest	UI state + string assertions
UX-47	AI results label method	UI state + string assertions
UX-48	Pack install shows SHA; failure shows both	UI state + string assertions
UX-49	Word count matches export ±1%	pdftotext count diff
UX-50	No marketing/network content ever	network probe + string scan

Colophon and revision history

Document provenance

OneDoc Application Development Blueprint v1.0 — generated from the markdown sources in `docs/onedoc/blueprint/content/` by `tools/blueprint/build_pdf.py` (shared engine) with figures from `tools/onedoc/figures.py`. The PDF, the CSV registries, and the traceability tooling are regenerated together; the traceability check (`tools/onedoc/traceability.py --check --strict`) is the machine proof that feature, UX and case catalogues agree with each other.

Revision	Date	Change
1.0	2026-10-31	Initial complete build contract: Parts I–XII, Appendices A–H, 18 figures, 90 features, 50 UX criteria, ~150-case catalogue

The one-paragraph summary (for the owner)

OneDoc is a fully offline document studio for Android 8+: a Docs-class block editor with tables, images, templates, versions and honest file fidelity; on-device AI (scan-to-text, dictation, a writing assistant, semantic search, read-aloud) that installs only as owner-side-loaded, SHA-verified packs and hides itself gracefully where hardware can't run it; no account, no internet permission, no ads, no telemetry — the privacy story is architectural, not promised. The blueprint is written for an autonomous AI developer: every feature maps to test cases, every law has a proof, every gate has a definition of done, and the final deliverable is a signed APK plus an evidence PDF whose every claim traces to a machine-verified run.

Appendix AP — Seed task graph (machine-readable)

This is the exact content of `docs/onedoc/plan/task-graph.yaml` at blueprint v1.0 — the agent copies it, then expands just-in-time (§29.3). 47 tasks; ids unique; dependencies within-gate unless noted in Appendix Y prose.

```
# OneDoc v1.0 task graph — seeded from blueprint Appendix Y
# statuses: todo | doing | blocked(reason) | done(evidence)
tasks:
  - id: T-G0-001
    gate: G0
    depends: []
    status: todo
    dod: >-
      owner acceptance recorded; ledgers seeded; traceability green
  - id: T-G1-001
    gate: G1
    depends: [T-G0-001]
    status: todo
    dod: >-
      JDK/SDK/AVDs installed and boot-verified
  - id: T-G1-002
    gate: G1
    depends: [T-G1-001]
    status: todo
    dod: >-
      Gradle skeleton: all modules assemble (full+writer)
  - id: T-G1-003
    gate: G1
    depends: [T-G1-002]
    status: todo
    dod: >-
      CI pr.yml green <=8 jobs; budgets enforced
  - id: T-G1-004
    gate: G1
    depends: [T-G1-002]
    status: todo
    dod: >-
      shell app: home empty-state + settings stub + About
  - id: T-G1-005
    gate: G1
    depends: [T-G1-001]
    status: todo
    dod: >-
      qa-driver skeleton; AVD snapshots; N.3 ritual scripted
  - id: T-G2-001
    gate: G2
    depends: [T-G1-002]
    status: todo
    dod: >-
      core-model: blocks/marks/ops/normalize + invariants
  - id: T-G2-002
    gate: G2
    depends: [T-G2-001]
    status: todo
    dod: >-
      property harness P1-P10 green
  - id: T-G2-003
    gate: G2
    depends: [T-G2-001]
    status: todo
    dod: >-
      core-storage: Room schema v3 exported
  - id: T-G2-004
    gate: G2
    depends: [T-G2-003]
    status: todo
```

```
    dod: >-
      journal write/replay/compact; kill-trial harness pass
  - id: T-G2-005
    gate: G2
    depends: [T-G2-003]
    status: todo
    dod: >-
      odoc container + goldens; LAW-001/004 green
  - id: T-G3-001
    gate: G3
    depends: [T-G2-005]
    status: todo
    dod: >-
      editor screen: block list + caret + IME bridge
  - id: T-G3-002
    gate: G3
    depends: [T-G3-001]
    status: todo
    dod: >-
      IME contract + fuzzer green on 4 AVDs
  - id: T-G3-003
    gate: G3
    depends: [T-G3-001]
    status: todo
    dod: >-
      selection engine + toolbar state machine
  - id: T-G3-004
    gate: G3
    depends: [T-G3-003]
    status: todo
    dod: >-
      marks engine + 14.5 semantics
  - id: T-G3-005
    gate: G3
    depends: [T-G3-001]
    status: todo
    dod: >-
      undo/redo + coalescing; typing budget met
  - id: T-G3-006
    gate: G3
    depends: [T-G3-005]
    status: todo
    dod: >-
      editor DQ pass; a11y node contract
  - id: T-G4-001
    gate: G4
    depends: [T-G3-004]
    status: todo
    dod: >-
      tables + TBL cases
  - id: T-G4-002
    gate: G4
    depends: [T-G3-004]
    status: todo
    dod: >-
      images/assets/alt + IMG cases
  - id: T-G4-003
    gate: G4
    depends: [T-G3-004]
    status: todo
    dod: >-
      lists + quote/code
  - id: T-G4-004
    gate: G4
    depends: [T-G3-003]
    status: todo
    dod: >-
      find/replace + versions-first
  - id: T-G4-005
    gate: G4
    depends: [T-G2-004]
```

```
    status: todo
    dod: >-
      versions UI + restore
  - id: T-G4-006
    gate: G4
    depends: [T-G2-003]
    status: todo
    dod: >-
      templates: 12 built-ins + save-as
  - id: T-G4-007
    gate: G4
    depends: [T-G4-001]
    status: todo
    dod: >-
      TOC/map + page setup/guides
  - id: T-G4-008
    gate: G4
    depends: [T-G4-007]
    status: todo
    dod: >-
      reader mode
  - id: T-G4-009
    gate: G4
    depends: [T-G1-004]
    status: todo
    dod: >-
      library complete: filters, trash, multi-select, info
  - id: T-G4-010
    gate: G4
    depends: []
    status: todo
    dod: >-
      DQ-1..8 pass on every G4 screen
  - id: T-G5-001
    gate: G5
    depends: [T-G2-005]
    status: todo
    dod: >-
      md/txt writers + goldens
  - id: T-G5-002
    gate: G5
    depends: [T-G5-001]
    status: todo
    dod: >-
      html writer/reader + SEC-IO-01
  - id: T-G5-003
    gate: G5
    depends: [T-G5-001]
    status: todo
    dod: >-
      docx writer/reader round-trip
  - id: T-G5-004
    gate: G5
    depends: [T-G5-002]
    status: todo
    dod: >-
      PDF export + STRUCT-004 parity
  - id: T-G5-005
    gate: G5
    depends: [T-G5-003]
    status: todo
    dod: >-
      share/open-in/SAF
  - id: T-G5-006
    gate: G5
    depends: [T-G5-001]
    status: todo
    dod: >-
      conversion notes + refusal paths
  - id: T-G5-007
    gate: G5
```

```
    depends: [T-G4-005]
    status: todo
    dod: >-
      backup/restore encrypted
  - id: T-G6-001
    gate: G6
    depends: [T-G5-003]
    status: todo
    dod: >-
      pack manager + odpack verify
  - id: T-G6-002
    gate: G6
    depends: [T-G6-001]
    status: todo
    dod: >-
      OCR pipeline + scan flows
  - id: T-G6-003
    gate: G6
    depends: [T-G6-001]
    status: todo
    dod: >-
      ASR streaming + dictation UX
  - id: T-G6-004
    gate: G6
    depends: [T-G6-001]
    status: todo
    dod: >-
      LLM Tier A + 4 verbs + preview cards
  - id: T-G6-005
    gate: G6
    depends: [T-G6-001]
    status: todo
    dod: >-
      embeddings + semantic search
  - id: T-G6-006
    gate: G6
    depends: [T-G4-008]
    status: todo
    dod: >-
      read aloud
  - id: T-G6-007
    gate: G6
    depends: [T-G6-004]
    status: todo
    dod: >-
      ladder + hidden-with-reason
  - id: T-G6-008
    gate: G6
    depends: [T-G6-002]
    status: todo
    dod: >-
      network-silence probe across AI
  - id: T-G7-001
    gate: G7
    depends: []
    status: todo
    dod: >-
      full catalogue on matrix; journeys J1-J5
  - id: T-G7-002
    gate: G7
    depends: [T-G7-001]
    status: todo
    dod: >-
      perf budgets + size gate evidence
  - id: T-G7-003
    gate: G7
    depends: [T-G7-001]
    status: todo
    dod: >-
      security suite + compliance map
  - id: T-G7-004
```

```
    gate: G7
    depends: [T-G7-001]
    status: todo
    dod: >-
      evidence PDF; BLOCKED register review
  - id: T-G8-001
    gate: G8
    depends: [T-G7-004]
    status: todo
    dod: >-
      release checklist; provenance; GitHub Release
```

Appendix AQ — The work order (kickoff, inline)

Canonical source: `docs/onedoc/plan/kickoff.md` (the file wins on divergence); this inline copy makes the PDF self-sufficient for an agent that only has the document.

Status: DRAFT — nothing in this package is approved until the owner accepts the OneDoc blueprint v1.0 (G0). On acceptance: close `reports/onedoc/gates/G0.md`, record D-001, then execute T-G1-001.

0. The go condition

G0 closes when the owner accepts `docs/onedoc/blueprint/onedoc-development-blueprint.pdf` as the build contract. Until then this package is a draft: read it, verify it, do not build from it.

1. Rules of engagement (absolute)

1. Fully local/on-device: no sign-in, no internet permission, no ads, no trackers, no exceptions (blueprint C1–C5).
2. Android 8.0+ (API 26); phones and tablets; portrait and landscape (C6).
3. Editor laws L1–L10 and journeys J1–J5 are release blockers (§13, §27).
4. Prove on the artifact: source tests help, only direct-APK QA passes ship (§21.1).
5. CI budgets: ≤ 3 parallel runners, ≤ 30 jobs/run (C8); publisher green ≠ QA pass (§22.3).
6. BLOCKED discipline: never a product verdict; log, retry, never fake (§21.2).
7. Preserve the predecessor program's artifacts under `reports/` and `qa/` — read-only history (defect tracker, v3.12.x regressions).
8. Ledgers always: D-nnn decisions, S-nnn sessions, gate reports (§29.2).
9. Traceability green after any catalogue change: `tools/onedoc/traceability.py --check --strict`.
10. Work only on the session branch; incremental commits; no history rewrites.

2. Reading order (first 30 minutes)

Blueprint Part XII §29 (loop) → Part I (charter) → Part X §21–22 (quality/CI) → Part V §13 (laws) → Appendix D (gate DoDs). Then Appendix N (environment) before T-G1-001.

3. Environment bootstrap (Day 0)

Follow Appendix N.1–N.3 verbatim: JDK 17, SDK 36, build-tools 36.0.0, AVDs `od-api26/33/35/36`, golden boot snapshots. Recreate the tool venv after any sandbox restore: `python3 -m venv .venv && .venv/bin/pip -q install reportlab pypdf pillow matplotlib pymupdf pyyaml`.

4. First sprint (G1 = T-G1-001…005)

Day	Tasks
1	T-G1-001 environment; T-G1-002 Gradle skeleton assembling both flavors
2	T-G1-003 CI pr.yml green within budgets; T-G1-004 shell app (home empty-state, settings stub, About)
3	T-G1-005 qa-driver skeleton + AVD snapshots; close G1 with DoD evidence

5. Working files

File	Role
`docs/onedoc/plan/task-graph.yaml`	seeded graph (47 tasks) — statuses move, ids never change
`docs/onedoc/plan/decisions.md`	append-only decision ledger
`docs/onedoc/plan/sessions.md`	append-only session log
`reports/onedoc/gates/G0.md`	gate report (draft; closes on acceptance)
`tools/onedoc/traceability.py`	catalogue generator + invariant checks
`qa/onedoc/testcases.csv`	generated catalogue (regenerate, never hand-edit)

6. Not-to-do while in draft

No toolchain installs, no repository restructuring, no CI edits, no app code — G0 pending means reading and verifying only.

7. G0 exit checklist

- [ ] Owner acceptance recorded in `reports/onedoc/gates/G0.md` (verdict CLOSED, date, evidence)
- [ ] D-001 appended (decision: charter accepted)
- [ ] S-001 session block written
- [ ] `traceability.py --check --strict` exit 0

8. Owner actions

1. Accept the blueprint (reply "go" — that acceptance is the G0 evidence).
2. Optional: a Figma personal-access token for the nightly registry-drift check (§7.4) — the agent proceeds without it.
3. Day-6 (G8 prep): the release keystore — an RC keystore covers everything until then (§26.1).

Appendix AR — Package file map (what this program adds to the repo)

Path	Role
docs/onedoc/blueprint/onedoc-development-blueprint.pdf	this document (v1.0)
docs/onedoc/blueprint/content/*.md	its sources (34 files)
docs/onedoc/blueprint/figures/*.png	17 figures (regenerated by tools/onedoc/figures.py)
docs/onedoc/blueprint/ui-screens.csv	screen/frame registry (§7.5)
docs/onedoc/plan/kickoff.md	the work order (Appendix AQ)
docs/onedoc/plan/task-graph.yaml	seeded task graph, 47 tasks (Appendix AP)
docs/onedoc/plan/decisions.md · sessions.md	append-only ledgers (§29.2)
reports/onedoc/gates/G0.md	gate report (closes on acceptance)
tools/onedoc/traceability.py	catalogue generator + invariant checks
tools/onedoc/figures.py	figure generator
qa/onedoc/testcases.csv	generated catalogue (142 cases)
design/onedoc-figma/tokens.json · README.md	Figma contract (§7)

Nothing else changes: the predecessor program's `reports/`, `qa/`, and workflows remain untouched read-only history.